STPM Physics · Semester 2
Electricity & Magnetism

FULL NOTES

Cikgu Suhaili

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PhysiQ · ellyelias.github.io/physiq · Worked examples, exercises with answers and the simulations from the app.

Chapter 12 · Semester 2

ELECTROSTATICS

Notes by Cikgu Suhaili

23 worked examples 3 simulations Exercises with answers

Electrostatics is the study of **electric charges at rest**, the forces between them and the electric fields around them.

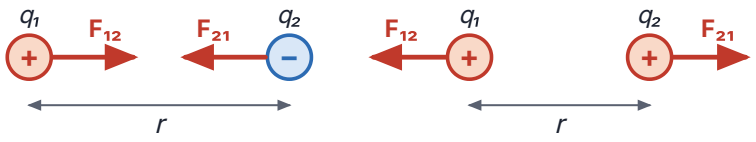
12.1 Coulomb's law

Coulomb's law

The electric force between two point charges is **directly proportional to the product of their charges** and **inversely proportional to the square of the distance** between them.

$$F = \frac{Qq}{4\pi\varepsilon_0 r^2} = \frac{kQq}{r^2}$$

$k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9\ \text{N m}^2\ \text{C}^{-2}$ (Coulomb's constant); $\varepsilon_0 = 8.85 \times 10^{-12}\ \text{C}^2\ \text{N}^{-1}\ \text{m}^{-2}$ (permittivity of free space)



unlike charges attract like charges repel

F_{12} = force on q_1 due to q_2; F_{21} = force on q_2 due to q_1

$F_{12} = -F_{21}$ (Newton's third law)

Figure 12.1: the forces between two point charges are equal in size and opposite in direction.

- Like charges repel; unlike charges attract.
- The two forces form an action–reaction pair: $F_{12} = F_{21} = kq_1q_2/r^2$, even if the charges are different in size.
- Doubling r makes the force four times smaller (inverse-square law).



Figure 12.2: F against r (a curve) and F against $1/r^2$ (a straight line through the origin).

Teacher's note

Use only the **size** of each charge in Coulomb's law. Decide the **direction** separately from "like repel, unlike attract", then draw the force on the diagram.

Relative permittivity

When the charges are in a medium (not a vacuum), the force is smaller. Replace ε_0 by the permittivity of the medium, ε.

Relative permittivity, ε_r

the **ratio of the permittivity of a medium to the permittivity of free space**. It has no unit.

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} \qquad F_{\text{medium}} = \frac{Qq}{4\pi\varepsilon_r\varepsilon_0 r^2} = \frac{F_{\text{vacuum}}}{\varepsilon_r}$$

ε_r of air ≈ 1.0006 (taken as 1); water ≈ 80; paper ≈ 3.7. The bigger ε_r, the weaker the force between the charges.

More than two charges

The resultant force on one charge is the **vector sum** of the forces from each of the other charges. Work out each force separately, then add them as vectors (along a line: add with signs; in two dimensions: resolve into x and y components).

Example 12.1.1

Two point charges $q_1 = -20$ nC and $q_2 = +90$ nC are 4.0 cm apart. Find the magnitude and direction of (a) the electric force on q_1 due to q_2, (b) the electric force on q_2 due to q_1. ($k = 9.0 \times 10^9$ N m^2 C^{-2})



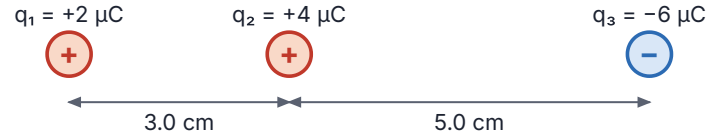
Solution

(a) $F_{12} = \dfrac{kq_1q_2}{r^2} = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-8})(9.0 \times 10^{-8})}{(4.0 \times 10^{-2})^2} =$ **1.0×10^{-2} N**, towards q_2 (attraction)

(b) $F_{21} =$ **1.0×10^{-2} N**, towards q_1: equal in size and opposite in direction (Newton's third law)

Example 12.1.2

Three point charges lie on the x-axis: $q_1 = +2$ µC, $q_2 = +4$ µC (3.0 cm to the right of q_1) and $q_3 = -6$ µC (5.0 cm to the right of q_2). Find the net electric force on q_2.



Solution

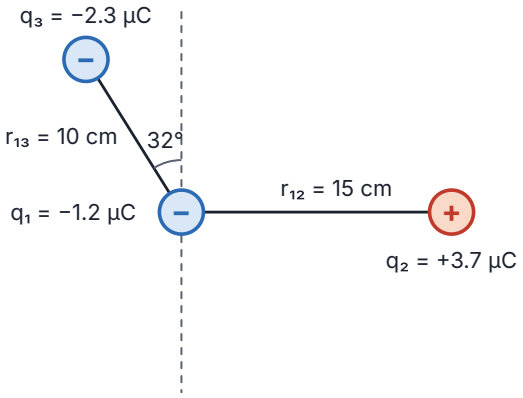
$F_{21} = \dfrac{(9.0 \times 10^9)(4 \times 10^{-6})(2 \times 10^{-6})}{(3.0 \times 10^{-2})^2} = 80$ N, to the right (q_1 repels q_2)

$F_{23} = \dfrac{(9.0 \times 10^9)(4 \times 10^{-6})(6 \times 10^{-6})}{(5.0 \times 10^{-2})^2} = 86.4$ N, to the right (q_3 attracts q_2)

Both forces point the same way: $F_2 = 80 + 86.4 =$ **166 N, to the right** (towards q_3)

Example 12.1.3

Three point charges are placed as shown. q_1 = −1.2 μC, q_2 = +3.7 μC, q_3 = −2.3 μC, r_{12} = 15 cm, r_{13} = 10 cm and the line q_1q_3 makes 32° with the vertical. Find the magnitude and direction of the resultant force on q_1.



Solution

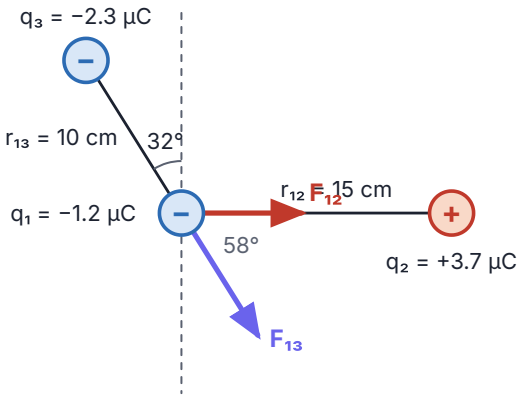
$F_{12} = \dfrac{(9.0 \times 10^9)(1.2 \times 10^{-6})(3.7 \times 10^{-6})}{(0.15)^2}$ = 1.78 N, along $+x$ (attracted towards q_2)

$F_{13} = \dfrac{(9.0 \times 10^9)(1.2 \times 10^{-6})(2.3 \times 10^{-6})}{(0.10)^2}$ = 2.48 N, away from q_3 (repulsion): down and to the right at 58° below the horizontal

ΣF_x = 1.78 + 2.48 cos 58° = 1.78 + 1.31 = 3.09 N

ΣF_y = −2.48 sin 58° = −2.10 N

$F_1 = \sqrt{3.09^2 + 2.10^2}$ = **3.74 N**; θ = $\tan^{-1}(2.10/3.09)$ = **34.2° below the $+x$ direction**



Example 12.1.4

Two charges +3.0 μC and −2.0 μC are 5.0 cm apart in oil of relative permittivity 2.5. Find the force between them.

Solution

$$F = \frac{Qq}{4\pi\varepsilon_r\varepsilon_0 r^2} = \frac{(9.0 \times 10^9)(3.0 \times 10^{-6})(2.0 \times 10^{-6})}{2.5 \times (0.050)^2} = \mathbf{8.6\ N}\text{, attractive}$$

In air the force would be 21.6 N; the oil reduces it by a factor of $\varepsilon_r = 2.5$.

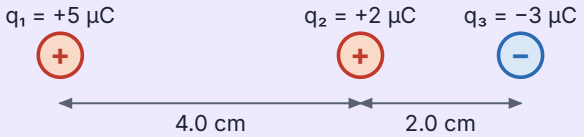
Exercise: Coulomb's law

1. Two point charges of -1.0×10^{-6} C and $+2.0 \times 10^{-6}$ C are 0.30 m apart. What is the electrostatic force on each charge?

   Answer

   0.20 N, each directed towards the other

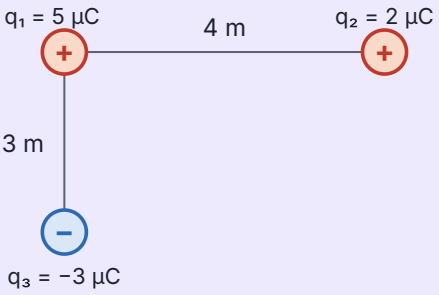
2. Calculate the net electrostatic force on (i) q_1, (ii) q_2, (iii) q_3.



   Answer

   (i) 18.75 N to the left (ii) 191.25 N to the right (iii) 172.5 N to the left

3. Calculate the net electrostatic force on q_2.



Answer

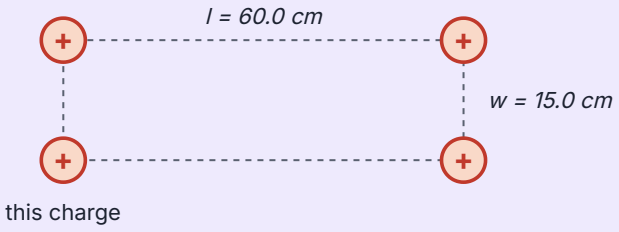
4.11×10^{-3} N, 18.4° below the $+x$ direction

4. q_1 = +2.5 μC is at (0, +0.30 m), q_2 = +2.5 μC is at (0, −0.30 m) and q_3 = +3 μC is at (+0.40 m, 0). What is the electrostatic force on q_3?

Answer

0.43 N in the $+x$ direction (each force is 0.27 N at 0.50 m; the y components cancel)

5. Four identical point charges (q = +10.0 μC) are at the corners of a rectangle, l = 60.0 cm and w = 15.0 cm. Find the magnitude and direction of the resultant force on the charge at the lower left corner.



Answer

40.9 N at 263° (measured anticlockwise from the $+x$ direction): almost straight down

12.2 Electric field

Electric field

a **region around a charge where another charge experiences an electric force**.

An electric field is shown with **electric field lines** (lines of force). The direction of the field at a point is the tangent to the line at that point: the direction of the force on a **positive** test charge.

Field patterns

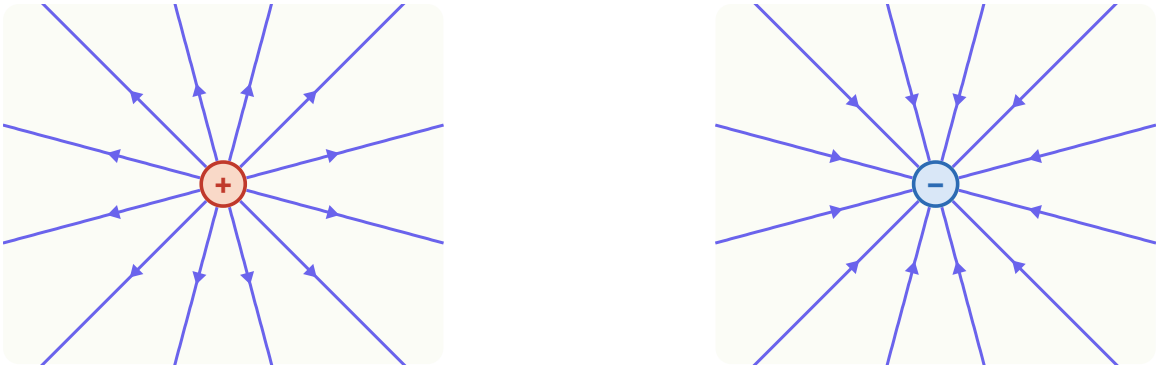


Figure 12.3: an isolated point charge. Positive: the lines point radially outwards. Negative: the lines point radially inwards.

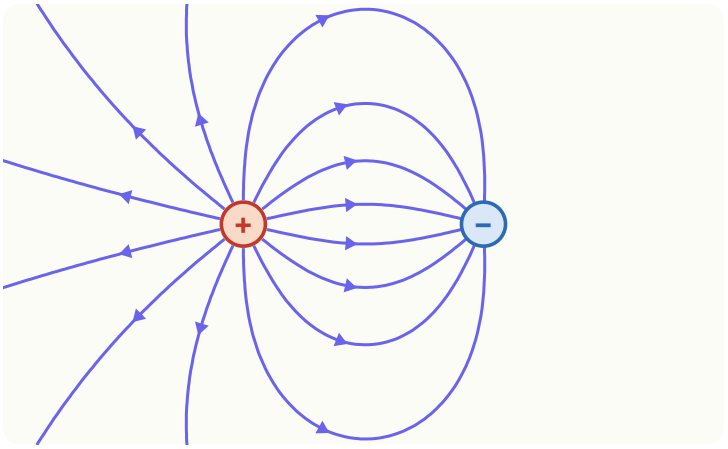


Figure 12.4: an electric dipole ($+q$ and $-q$). The lines curve from the positive charge to the negative charge.

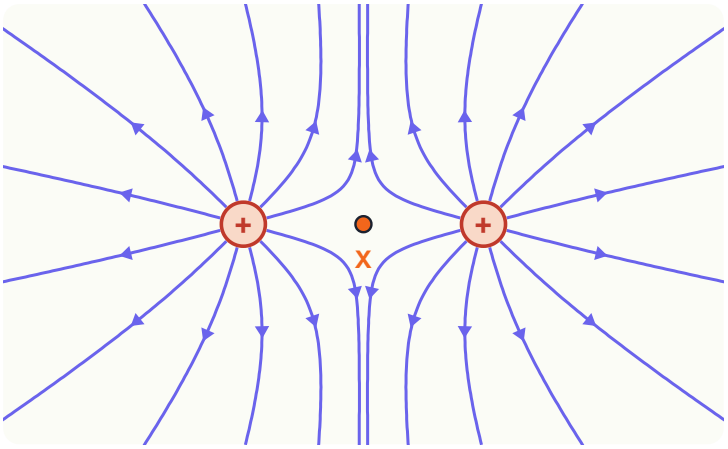


Figure 12.5: two equal positive charges. X is the **neutral point**: the resultant field (and force on a charge) there is zero.

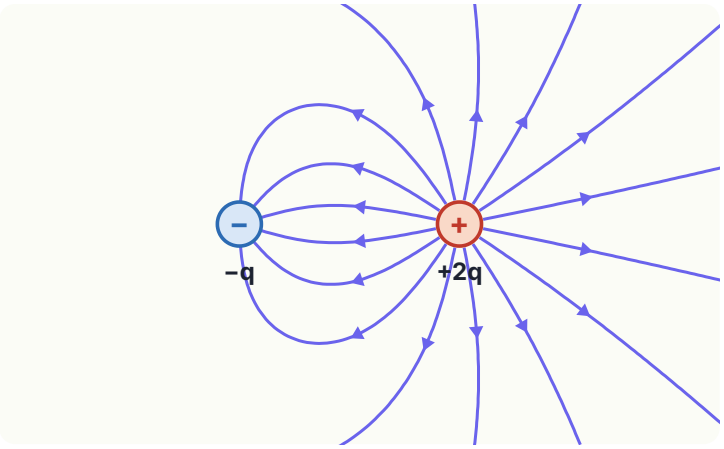
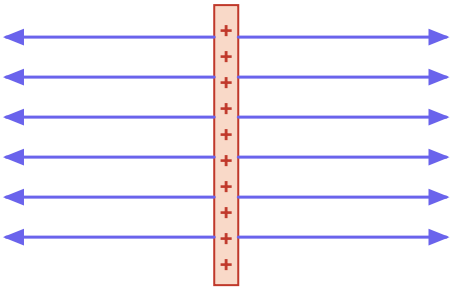
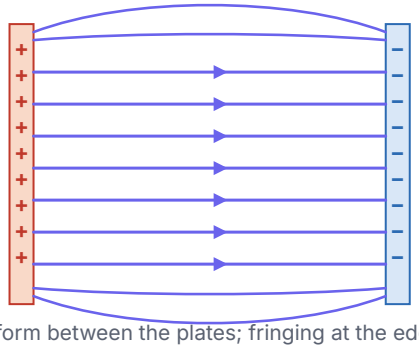


Figure 12.6: $+2q$ and $-q$. Twice as many lines leave $+2q$ as enter $-q$; the rest go off to infinity.



field lines perpendicular to the surface, on both sides

Figure 12.7: a uniformly charged surface (plate).



uniform between the plates; fringing at the edges

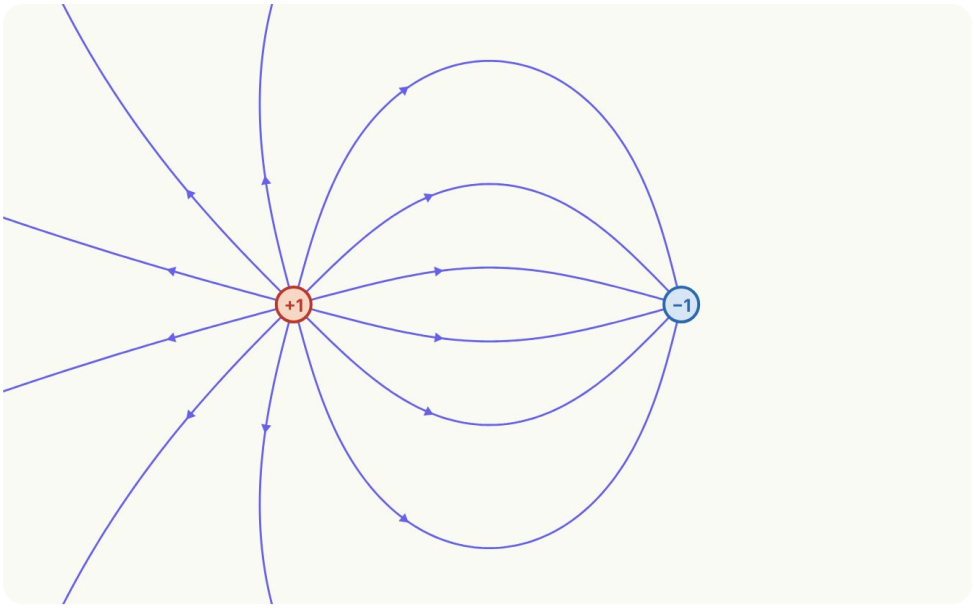
Figure 12.8: two oppositely charged parallel plates.

- Field lines start on positive charges and end on negative charges. The number of lines is proportional to the size of the charge.
- The closer the lines, the stronger the field.
- Field lines **never cross**: the field cannot have two directions at one point.
- At a charged conducting surface, the lines are **perpendicular** to the surface.
- Between two parallel plates the lines are parallel and equally spaced: the field is **uniform**. The fringing at the edges is ignored when the separation is small compared with the size of the plates.

Simulation

Electric field lines and the field at a point

Choose a set of charges and change the second charge. Tap anywhere to place a test point and read the field strength and potential there. Turn on the equipotentials to see that they always cross the field lines at right angles.



tap the picture	scale	charges
to place a test point	**picture width = 40 cm**	**q1 = +1 μC, q2 = −1 μC**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 12 to use the simulation.

Electric field strength

Electric field strength, E

the **electric force per unit positive charge** acting on a small test charge placed at that point.

$$E = \frac{F}{q} \qquad \text{for a point charge: } E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{kQ}{r^2}$$

vector; unit N C^{-1} (= V m^{-1}). Points away from a positive charge and towards a negative charge.

Derivation for a point charge: a test charge q at distance r from Q feels $F = kQq/r^2$, so $E = F/q = kQ/r^2$. It does not depend on the test charge.

Teacher's note

For several charges, find E from each charge using only its size, mark its direction on the diagram (away from +, towards −), then add the fields as **vectors**. The force on a charge q placed there is $F = qE$ (opposite to E if q is negative).

Example 12.2.1

An electric field of magnitude 0.370 N C^{-1} exerts a force of 6.20 × 10^{-4} N on a test charge. What is the size of the test charge?

Solution

$$q = \frac{F}{E} = \frac{6.20 \times 10^{-4}}{0.370} = \mathbf{1.68 \times 10^{-3}\ C}\ (1.68\ \text{mC})$$

Example 12.2.2

Find the electric field strength at point A, 2.0 cm from a point charge of (a) +5.0 μC, (b) −5.0 μC.



Solution

$$E = \frac{kq}{r^2} = \frac{(9.0 \times 10^{9})(5.0 \times 10^{-6})}{(2.0 \times 10^{-2})^2} = \mathbf{1.1 \times 10^{8}\ N\ C^{-1}}$$

(a) directed **away from** the charge; (b) the same size, directed **towards** the charge

Example 12.2.3

Point A is 2.0 cm from q_1 = +1.0 μC and 3.0 cm from q_2 = −4.0 μC, on the line between them. Find (a) the electric field at A, (b) the force on a charge q_0 = +4.0 μC placed at A.



Solution

$E_1 = \dfrac{(9.0 \times 10^9)(1.0 \times 10^{-6})}{(0.020)^2} = 2.25 \times 10^7$ N C^{-1}, to the right (away from $+q_1$)

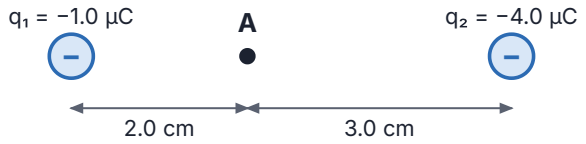
$E_2 = \dfrac{(9.0 \times 10^9)(4.0 \times 10^{-6})}{(0.030)^2} = 4.0 \times 10^7$ N C^{-1}, to the right (towards $-q_2$)

(a) $E = 2.25 \times 10^7 + 4.0 \times 10^7$ = **6.3×10^7 N C^{-1}, to the right**

(b) $F = q_0 E = (4.0 \times 10^{-6})(6.25 \times 10^7)$ = **250 N, to the right**

Example 12.2.4

Repeat Example 12.2.3 with q_1 = −1.0 μC.



Solution

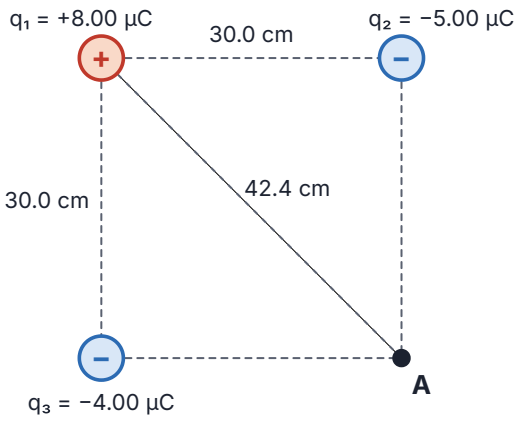
E_1 = 2.25 × 10^7 N C^{-1}, now to the **left** (towards $-q_1$); E_2 = 4.0 × 10^7 N C^{-1}, to the right

(a) $E = 4.0 \times 10^7 - 2.25 \times 10^7$ = **1.75×10^7 N C^{-1}, to the right**

(b) $F = (4.0 \times 10^{-6})(1.75 \times 10^7)$ = **70 N, to the right**

Example 12.2.5

Three charges are at three corners of a square of side 30.0 cm. Find the electric field at the fourth corner A, and the force on a +6.00 μC charge placed at A.



Solution

$E_1 = \dfrac{(9.0 \times 10^9)(8.00 \times 10^{-6})}{(0.424)^2} = 4.00 \times 10^5$ N C^{-1}, away from q_1 (down-right, 45°)

$E_2 = \dfrac{(9.0 \times 10^9)(5.00 \times 10^{-6})}{(0.300)^2} = 5.00 \times 10^5$ N C^{-1}, up (towards q_2)

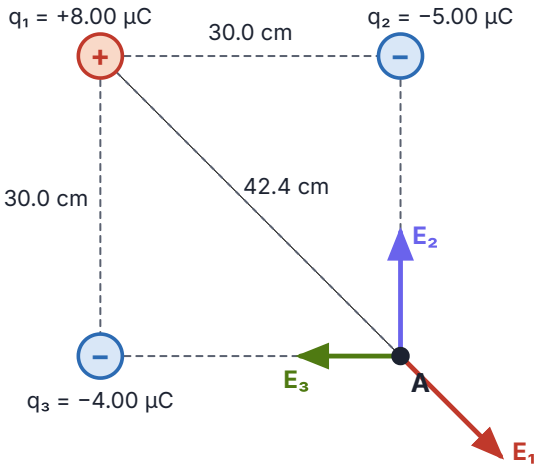
$E_3 = \dfrac{(9.0 \times 10^9)(4.00 \times 10^{-6})}{(0.300)^2} = 4.00 \times 10^5$ N C^{-1}, left (towards q_3)

$\Sigma E_x = 4.00 \times 10^5 \cos 45° - 4.00 \times 10^5 = -1.17 \times 10^5$ N C^{-1}

$\Sigma E_y = 5.00 \times 10^5 - 4.00 \times 10^5 \sin 45° = +2.17 \times 10^5$ N C^{-1}

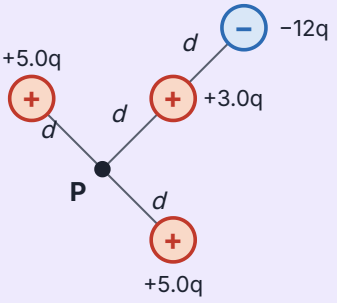
$E = \sqrt{(1.17 \times 10^5)^2 + (2.17 \times 10^5)^2} =$ **2.47×10^5 N C^{-1}**, at $\tan^{-1}(2.17/1.17) =$ **61.7° above the $-x$ direction** (up and to the left)

$F = qE = (6.00 \times 10^{-6})(2.47 \times 10^5) =$ **1.48 N**, in the same direction as E



Exercise: electric field

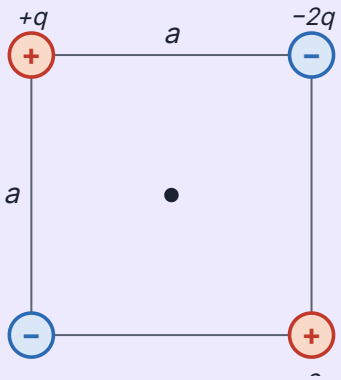
1. Find the magnitude of the electric field at P due to the four point charges shown, if q = 1 nC and d = 1 cm. ($\varepsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$)



Answer

Zero. The two +5.0q fields cancel. On the other line, +3.0q at d gives $3kq/d^2$ away from it and −12q at 2d gives $12kq/4d^2 = 3kq/d^2$ towards it: they cancel too.

2. Find the magnitude and direction of the electric field at the centre of the square if $q = 1.0 \times 10^{-8}$ C and $a = 5.0$ cm.



Answer

1.02×10^{5} N C^{-1}, upwards

Motion of a charge in a uniform electric field

In a uniform field the only force on the charge is $F = qE$ (its weight is negligible). The force is constant, so the acceleration is constant:

$$a = \frac{qE}{m} \qquad E = \frac{V}{d} \text{ between parallel plates}$$

a positive charge accelerates along E; a negative charge (electron) accelerates opposite to E

Moving along the field lines

- The charge speeds up (or slows down) in a straight line, like a body falling under gravity.
- Work done by the field = qV = gain in kinetic energy: $\frac{1}{2}mv^2 = qV$ (from rest)

Entering at right angles to the field

- Horizontal: no force, so $v_x = v_0$ stays constant.
- Vertical: constant acceleration $a = qE/m$.
- The path between the plates is a **parabola** (like a projectile). After leaving the plates it moves in a straight line.

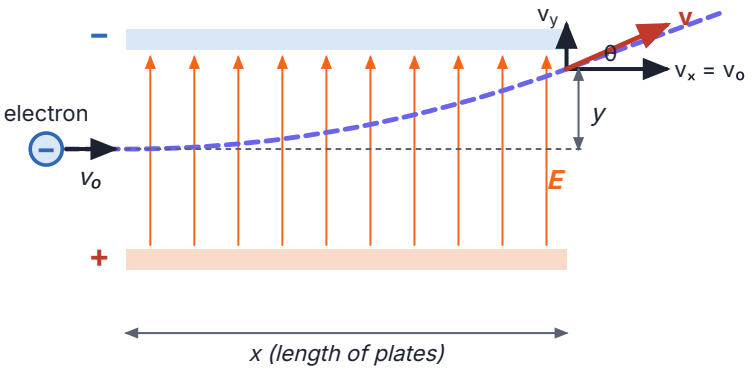


Figure 12.9: an electron enters the field between parallel plates at right angles to the field.

time between the plates: $t = \frac{x}{v_0}$

vertical velocity on leaving: $v_y = at = \frac{qEx}{mv_0}$

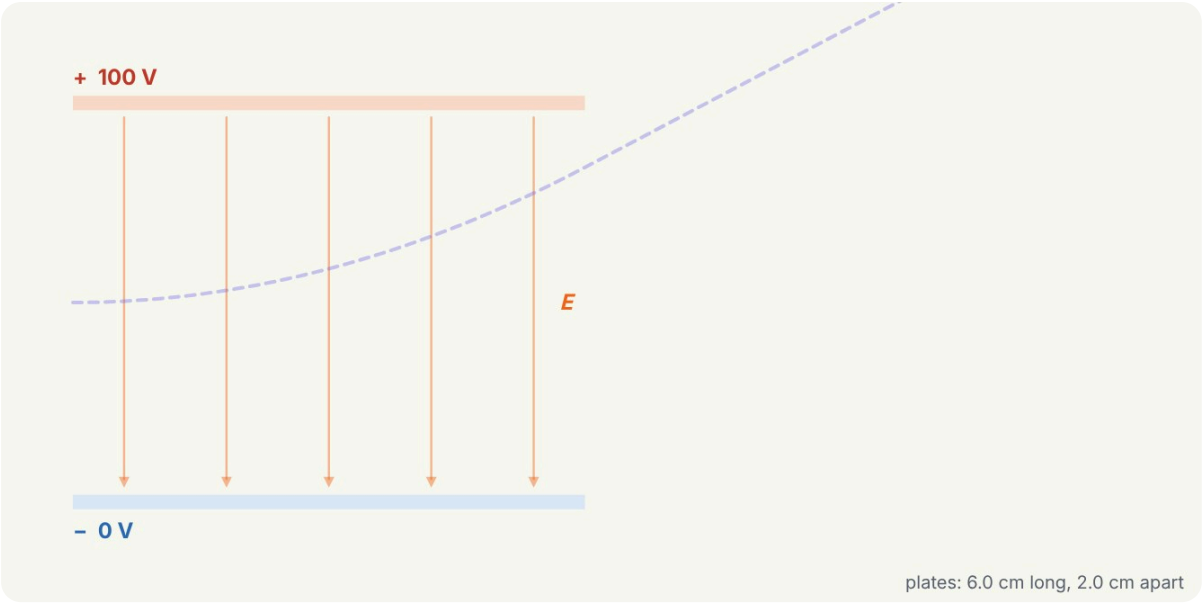
vertical displacement: $y = \frac{1}{2}at^2 = \frac{qEx^2}{2mv_0^2}$ ($y \propto x^2$: a parabola)

speed and direction on leaving: $v = \sqrt{v_x^2 + v_y^2}$, $\theta = \tan^{-1}\frac{v_y}{v_x}$

Simulation

A charge between parallel plates

Choose the particle, then change the voltage across the plates and the entry speed. See when the particle leaves the plates and when it hits one of them.



plates: 6.0 cm long, 2.0 cm apart

field E = V/d	acceleration qE/m	time between plates	vertical deflection y
5.00×10^{3} V m^{-1}	8.78×10^{14} m s^{-2}	4.00×10^{-9} s	7.03 mm

v_y on leaving	angle of deflection θ
3.51×10^{6} m s^{-1}	13.2°

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 12 to use the simulation.

Example 12.2.6

An electron travelling at 1.5×10^7 m s^{-1} enters midway between two parallel plates 60 mm long and 20 mm apart. The field between the plates is 4.0×10^3 V m^{-1}, with the upper plate positive. (a) Sketch the path of the electron between and after the plates. (b) Find the acceleration of the electron. (c) Find the components of its velocity as it leaves the plates. (d) Find the angle of deflection. ($e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg)

Solution

(a) a parabola curving towards the positive plate between the plates, then a straight line after it leaves

(b) $a = \dfrac{eE}{m} = \dfrac{(1.60 \times 10^{-19})(4.0 \times 10^3)}{9.11 \times 10^{-31}} = \mathbf{7.03 \times 10^{14}\ m\ s^{-2}}$, towards the positive plate (opposite to E)

(c) $t = 0.060/1.5 \times 10^7 = 4.0 \times 10^{-9}$ s; $v_x = \mathbf{1.5 \times 10^7\ m\ s^{-1}}$; $v_y = at = (7.03 \times 10^{14})(4.0 \times 10^{-9}) = \mathbf{2.8 \times 10^6\ m\ s^{-1}}$

Check: $y = \tfrac{1}{2}at^2 = 5.6$ mm, less than 10 mm, so it does leave the plates.

(d) $\theta = \tan^{-1}\dfrac{2.8 \times 10^6}{1.5 \times 10^7} = \mathbf{10.6°}$

Example 12.2.7

An electron starts from rest at the negative plate and is accelerated across a potential difference of 500 V. Find its speed when it reaches the positive plate.

Solution

$\tfrac{1}{2}mv^2 = eV \Rightarrow v = \sqrt{\dfrac{2eV}{m}} = \sqrt{\dfrac{2(1.60 \times 10^{-19})(500)}{9.11 \times 10^{-31}}} = \mathbf{1.33 \times 10^7\ m\ s^{-1}}$

12.3 Gauss's law

Electric flux, Φ

through a surface of area A in a uniform field E: **the scalar product of E and the area vector A**. It measures how many field lines pass through the surface.

$$\Phi = \boldsymbol{E} \cdot \boldsymbol{A} = EA\cos\theta$$

θ = angle between E and the **normal** to the surface; unit N m^2 C^{-1} (= V m); scalar

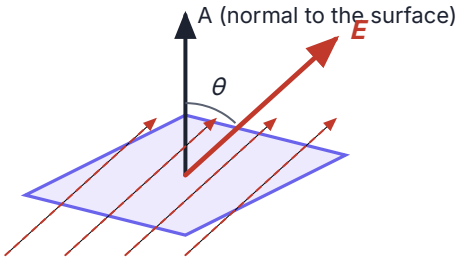


Figure 12.10: the area vector A is normal to the surface.

- θ = 0 (field perpendicular to the surface): Φ = EA, the maximum.
- θ = 90° (field parallel to the surface): Φ = 0, no lines pass through.
- For a **closed** surface the normal points outwards: flux leaving is positive, flux entering is negative.

Gauss's law

The **total electric flux through a closed surface is equal to the net charge enclosed by the surface divided by ε_0**.

$$\Phi = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$$

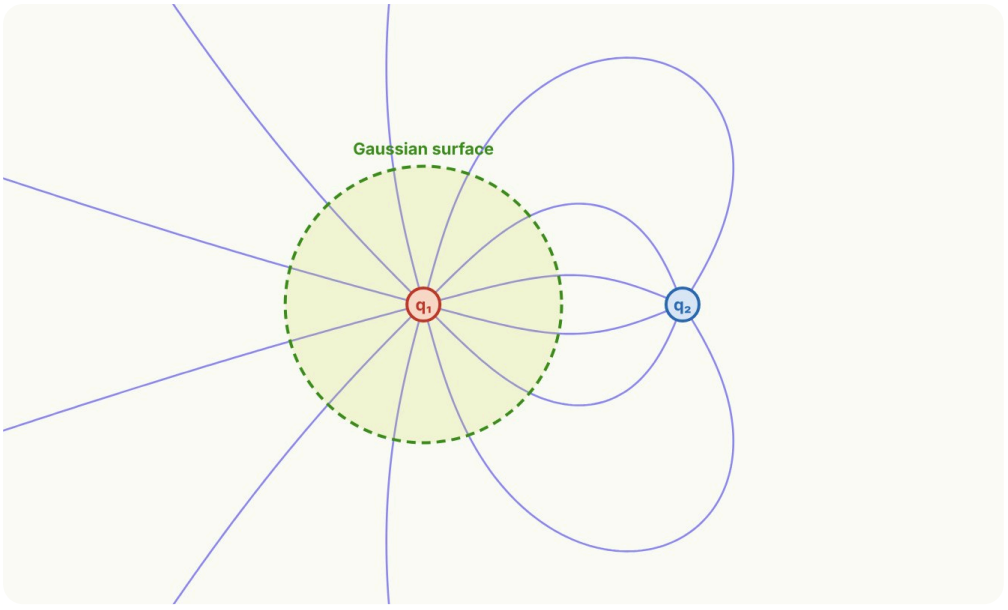
the closed surface is called a Gaussian surface; its shape and size do not matter, only the charge inside

- A charge **outside** the closed surface gives zero net flux: every line that enters also leaves.
- If the net charge inside is zero, the net flux is zero (but the field on the surface need not be zero).

Simulation

Gauss's law

Change the size of the Gaussian sphere and move the second charge in and out. The flux depends only on the charge inside. Then look at a charged conducting sphere: E and V inside and outside.



charge enclosed
+2 nC

flux $\Phi = Q/\varepsilon_o$
226 N m² C⁻¹

q_2 is
outside: its lines go in and out, net flux 0

size of the surface
does not change Φ

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 12 to use the simulation.

Example 12.3.1

A uniform field of 2.0×10^3 N C^{-1} passes through a flat square of side 10 cm. Find the flux through the square when its normal makes (a) 0°, (b) 60°, (c) 90° with the field.

Solution

$A = (0.10)^2 = 0.010$ m^2

(a) $\Phi = (2.0 \times 10^3)(0.010) \cos 0° =$ **20 N m² C⁻¹**

(b) $\Phi = 20 \cos 60° =$ **10 N m² C⁻¹**

(c) $\Phi = 20 \cos 90° =$ **0**

Example 12.3.2

A point charge of +5.0 nC is at the centre of a cube. Find (a) the total flux through the cube, (b) the flux through one face. (c) What is the flux through the cube if the charge is moved outside it?

Solution

(a) $\Phi = \dfrac{Q}{\varepsilon_0} = \dfrac{5.0 \times 10^{-9}}{8.85 \times 10^{-12}} = \mathbf{565\ N\ m^2\ C^{-1}}$

(b) by symmetry the six faces share it equally: 565/6 = **94.2 N m² C⁻¹**

(c) **zero**: no charge is enclosed

Using Gauss's law to find E

Choose a Gaussian surface on which E has the same size everywhere and is perpendicular to the surface. Then $\oint \boldsymbol{E}\cdot d\boldsymbol{A} = E \times$ (area).



Figure 12.11: Gaussian spheres around a point charge and around a charged conducting sphere.

(a) Isolated point charge Q. Take a sphere of radius r centred on the charge. E is radial and the same size at every point on it:

$$E(4\pi r^2) = \frac{Q}{\varepsilon_0} \quad \Rightarrow \quad E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

This is Coulomb's law again.

(b) Isolated charged conducting sphere (radius R, charge Q). The charge spreads over the **outer surface**.

- Outside ($r \geq R$): a Gaussian sphere of radius r encloses Q, so $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$: the sphere acts like a point charge at its centre.
- Inside ($r < R$): a Gaussian sphere inside encloses no charge, so $E = 0$.
- The potential is the same everywhere inside, equal to the surface value $Q/4\pi\varepsilon_0 R$.



Figure 12.12: E and V against r for a charged conducting sphere.

(c) Uniformly charged conducting plate. The surface charge density is σ = Q/A. Take a small cylinder (pillbox) with one end inside the conductor and the other end, of area A, just outside. Inside the conductor E = 0, and no flux passes through the curved side (E is perpendicular to the surface), so only the outer end counts:

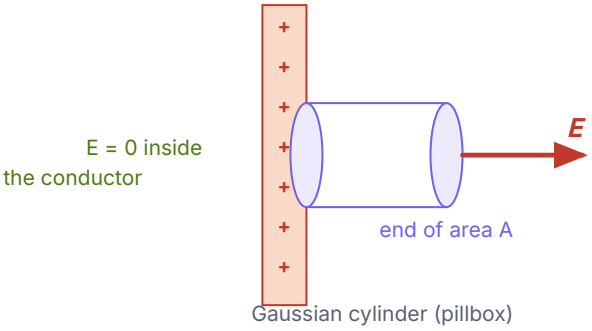


Figure 12.13: a Gaussian pillbox at the surface of a charged conducting plate.

$$EA = \frac{\sigma A}{\varepsilon_0} \quad \Rightarrow \quad E = \frac{\sigma}{\varepsilon_0}$$

The same result, $E = \sigma/\varepsilon_0$, gives the uniform field between two parallel plates with charges $+Q$ and $-Q$. It does not depend on the distance from the plate.

Teacher's note

Why is a car a safe place in a thunderstorm? The metal body is a conductor: the charge stays on its outer surface, and Gauss's law gives $E = 0$ inside.

Example 12.3.3

A conducting sphere of radius 10 cm carries a charge of +2.0 μC. Find the electric field and the potential at (a) 5.0 cm, (b) 10 cm, (c) 20 cm from its centre.

Solution

(a) inside: $E = 0$; $V = V_{surface} = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-6})}{0.10} = \mathbf{1.8 \times 10^5\ V}$

(b) $E = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-6})}{(0.10)^2} = \mathbf{1.8 \times 10^6\ N\ C^{-1}}$; $V = \mathbf{1.8 \times 10^5\ V}$

(c) $E = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-6})}{(0.20)^2} = \mathbf{4.5 \times 10^5\ N\ C^{-1}}$; $V = \dfrac{(9.0 \times 10^9)(2.0 \times 10^{-6})}{0.20} = \mathbf{9.0 \times 10^4\ V}$

Example 12.3.4

A charge of 4.0 nC is spread evenly over one face of a conducting plate of area 0.020 m^2. Find the electric field just outside the plate.

Solution

$\sigma = Q/A = 4.0 \times 10^{-9}/0.020 = 2.0 \times 10^{-7}\ C\ m^{-2}$

$E = \sigma/\varepsilon_0 = 2.0 \times 10^{-7}/8.85 \times 10^{-12} = \mathbf{2.3 \times 10^4\ N\ C^{-1}}$, perpendicular to the plate

12.4 Electric potential

Electric potential, V

at a point: the **work done to bring a unit positive charge from infinity to that point**. The potential at infinity is zero.

$$V = \frac{Q}{4\pi\varepsilon_0 r} = \frac{kQ}{r}$$

scalar; unit volt, V (= J C^{-1}). Put in the sign of Q: positive near a positive charge, negative near a negative charge.

Derivation

Bring a test charge q_0 from infinity to a point at distance r from a positive charge Q, slowly (no acceleration). The external force must balance the electric force, $F_{ext} = kQq_0/x^2$, pointing towards Q:

$$W = -\int_{\infty}^{r} \frac{kQq_0}{x^2}\,dx = kQq_0\left[\frac{1}{x}\right]_{\infty}^{r} = \frac{kQq_0}{r} \quad \Rightarrow \quad V = \frac{W}{q_0} = \frac{kQ}{r}$$

- For a positive Q, W is positive: work is done **by the external force** against the repulsion.
- For a negative Q, $W = -kQq_0/r$ is negative: the work is done **by the electric force** (the test charge is attracted).



Figure 12.14: the potential against distance for a positive and a negative point charge.

Potential due to several charges

Potential is a scalar: add the potentials from each charge **with their signs**. $V = k(Q_1/r_1 + Q_2/r_2 + \ldots)$

Example 12.4.1

Find the electric potential at a point A, 10 m from a point charge of +5.0 μC, and explain what the answer means.

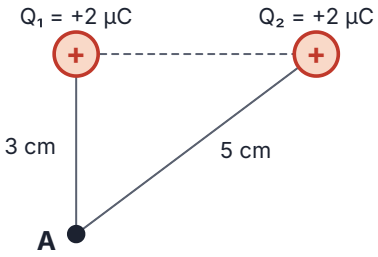
Solution

$$V = \frac{kq}{r} = \frac{(9.0 \times 10^9)(5.0 \times 10^{-6})}{10} = \mathbf{4.5 \times 10^3\ V}$$

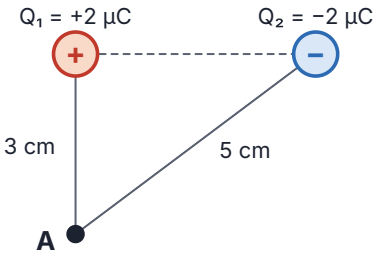
Meaning: 4.5×10^3 J of work is done to bring each coulomb of positive charge from infinity to A.

Example 12.4.2

Find the potential at A in each arrangement.



(a)



(b)

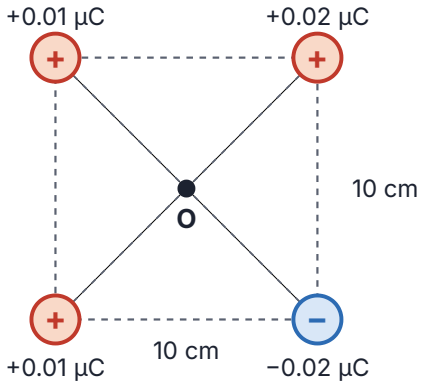
Solution

(a) $V_A = k\left(\frac{Q_1}{r_1} + \frac{Q_2}{r_2}\right) = (9.0 \times 10^9)\left(\frac{2 \times 10^{-6}}{0.03} + \frac{2 \times 10^{-6}}{0.05}\right) = \mathbf{9.6 \times 10^5\ V}$

(b) $V_A = (9.0 \times 10^9)\left(\frac{2 \times 10^{-6}}{0.03} - \frac{2 \times 10^{-6}}{0.05}\right) = \mathbf{2.4 \times 10^5\ V}$ (put in the minus sign of Q_2)

Example 12.4.3

Four charges are at the corners of a square of side 10 cm. Find the potential at the centre O.



Solution

Each charge is $\frac{10}{\sqrt{2}} = 7.07$ cm from O.

$V_O = \frac{k}{r}(\Sigma q) = \frac{9.0 \times 10^9}{0.0707}(0.01 + 0.02 + 0.01 - 0.02) \times 10^{-6} = \mathbf{2.55 \times 10^3\ V}$

Potential difference and work done

Potential difference between A and B

the work done per unit positive charge in moving a charge from B to A (against the field, with no acceleration).

$$V_{AB} = V_A - V_B = \frac{W_{B\to A}}{q} \qquad W_{A\to B} = q(V_B - V_A)$$

W is the work done by an **external force**. The work done by the **electric field** is the negative of this.

- If q and $(V_B - V_A)$ have the same sign, the external force does positive work (the charge is pushed against the field).
- If they have opposite signs, the work done by the external force is negative: the field does the work.

Example 12.4.4

(a) Find the potential at (i) 10 cm, (ii) 50 cm from a point charge of +2 μC. (b) Find the work done to move a charge of +0.05 μC from 50 cm to 10 cm from the point charge. Is the work done by an external force or by the electric field?

Solution

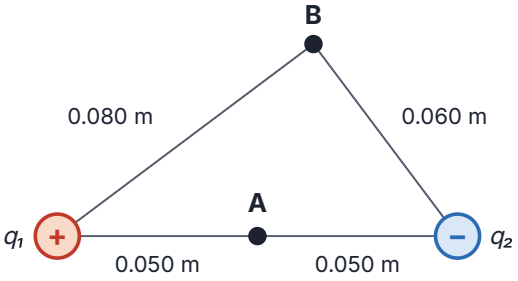
(a)(i) $V = (9.0 \times 10^9)(2 \times 10^{-6})/0.10 =$ **1.8×10^5 V** (ii) $V = (9.0 \times 10^9)(2 \times 10^{-6})/0.50 =$ **3.6×10^4 V**

(b) $W = q(V_{10} - V_{50}) = (0.05 \times 10^{-6})(1.8 \times 10^5 - 3.6 \times 10^4) =$ **7.2×10^{-3} J**

W is positive: the work is done **by an external force against the electric field** (a positive charge is pushed towards a positive charge). The field itself does -7.2×10^{-3} J.

Example 12.4.5

q_1 = +2.40 nC and q_2 = −6.50 nC are 0.100 m apart. A is midway between them; B is 0.080 m from q_1 and 0.060 m from q_2. Find (a) V_A, (b) V_B, (c) the work done by the electric field on a charge of +2.5 nC that moves from B to A.



Solution

(a) $V_A = (9.0 \times 10^9)\left(\frac{2.40 \times 10^{-9}}{0.050} - \frac{6.50 \times 10^{-9}}{0.050}\right) = 432 - 1170 = \mathbf{-738\ V}$

(b) $V_B = (9.0 \times 10^9)\left(\frac{2.40 \times 10^{-9}}{0.080} - \frac{6.50 \times 10^{-9}}{0.060}\right) = 270 - 975 = \mathbf{-705\ V}$

(c) work done by the field = $q(V_B - V_A) = (2.5 \times 10^{-9})(-705 + 738) = \mathbf{+8.3 \times 10^{-8}\ J}$

(The positive charge moves to a lower potential, so the field does positive work. An external force would do -8.3×10^{-8} J.)

Equipotential surfaces

Equipotential surface

a surface on which every point has the same potential.

- **No work** is done in moving a charge along an equipotential surface ($\Delta V = 0$).
- Equipotential surfaces are always **perpendicular to the field lines**.
- Where the equipotentials are closer together, the field is stronger.

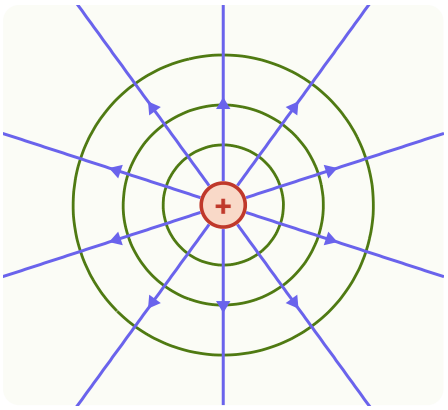


Figure 12.15: point charge: concentric spheres (circles in 2D).

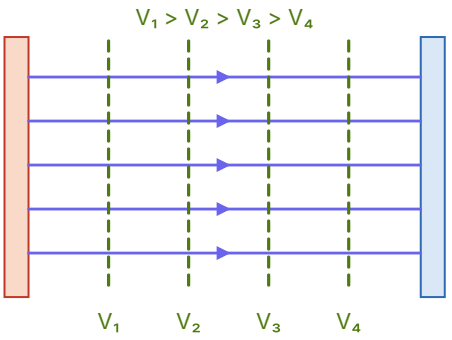


Figure 12.16: uniform field: equally spaced parallel planes.

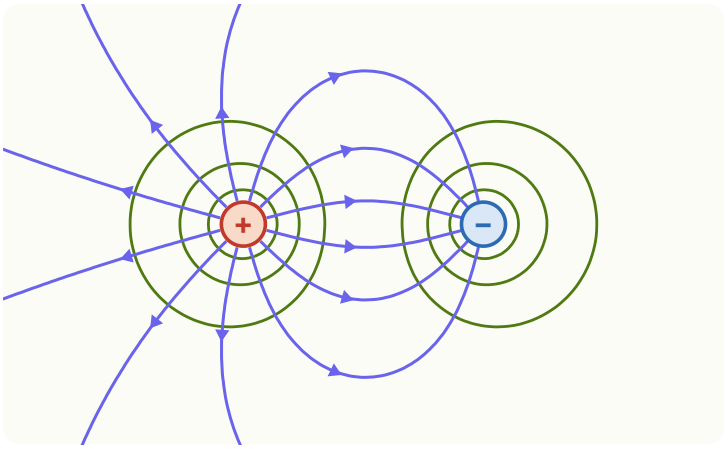


Figure 12.17: electric dipole. Dashed green lines are equipotentials; the middle one (V = 0) is a straight line.

Relation between E and V

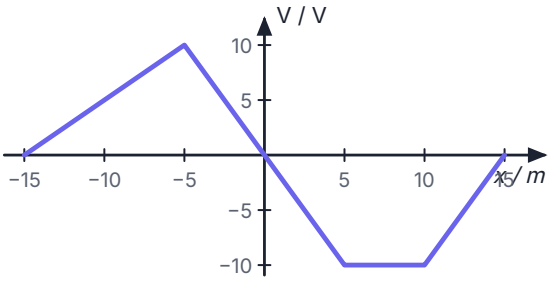
$$E = -\frac{\mathrm{d}V}{\mathrm{d}r} \qquad \text{uniform field: } E = \frac{V}{d}$$

the field strength is the negative of the potential gradient; E points from high potential to low potential

- The steeper the V–r graph, the stronger the field.
- Between parallel plates with p.d. V and separation d, V falls steadily from one plate to the other, so $E = V/d$ is uniform.

Example 12.4.6

The graph shows how the potential V varies with distance x along the x-axis. Sketch the graph of the electric field E against x.



Solution

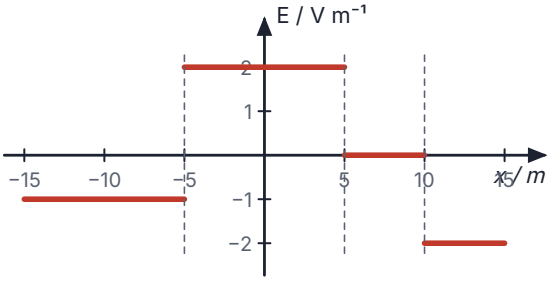
Use $E = -\mathrm{d}V/\mathrm{d}x$ for each straight section:

−15 m to −5 m: gradient = 10/10 = +1 ⇒ E = **−1 V m^{-1}**

−5 m to 5 m: gradient = −20/10 = −2 ⇒ E = **+2 V m^{-1}**

5 m to 10 m: gradient = 0 ⇒ E = **0**

10 m to 15 m: gradient = 10/5 = +2 ⇒ E = **−2 V m^{-1}**



Electric potential energy

Electric potential energy, U

of a charge at a point: the **work done to bring the charge from infinity to that point**.

$$U = qV = \frac{Qq}{4\pi\varepsilon_0 r} \qquad \Delta U = q\Delta V$$

scalar; unit joule. Put in the signs of both charges: $U > 0$ for like charges (repel), $U < 0$ for unlike charges (attract).

For a system of charges, add the potential energy of every pair:

$$U = k\left(\frac{q_1q_2}{r_{12}} + \frac{q_1q_3}{r_{13}} + \frac{q_2q_3}{r_{23}}\right)$$

Example 12.4.7

(a) Find the electric potential energy of a +0.05 μC charge at (i) 10 cm, (ii) 50 cm from a +2 μC point charge. (b) Find the change in potential energy between (i) and (ii).

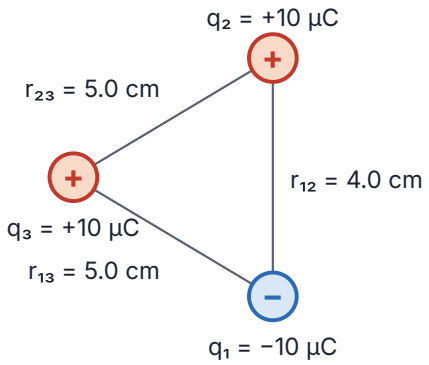
Solution

(a)(i) $U = qV = (0.05 \times 10^{-6})(1.8 \times 10^{5}) =$ **9.0×10^{-3} J** (ii) $U = (0.05 \times 10^{-6})(3.6 \times 10^{4}) =$ **1.8×10^{-3} J**

(b) $\Delta U = 9.0 \times 10^{-3} - 1.8 \times 10^{-3} =$ **7.2×10^{-3} J**: the same as the work done in Example 12.4.4

Example 12.4.8

Find the potential energy of the system of charges shown.



Solution

$$U = (9.0 \times 10^9)\left[\frac{(-10)(10)}{0.040} + \frac{(-10)(10)}{0.050} + \frac{(10)(10)}{0.050}\right] \times 10^{-12}$$

$$= (9.0 \times 10^{-3})(-2500 - 2000 + 2000) = \mathbf{-22.5\ J}$$

Negative: work would have to be done to pull the charges apart to infinity.

Exercise: electric potential

1. Two parallel plates 5.0 mm apart have a potential difference of 100 V. Find the electric field between them and the force on an electron there.

   Answer

   $E = 100/5.0 \times 10^{-3} = 2.0 \times 10^4$ V m^{-1}; $F = eE = 3.2 \times 10^{-15}$ N, towards the positive plate

2. How much work must an external force do to bring two protons from infinity to 1.0×10^{-14} m apart?

   Answer

   $W = ke^2/r = (9.0 \times 10^9)(1.60 \times 10^{-19})^2/1.0 \times 10^{-14} = 2.3 \times 10^{-14}$ J

3. Explain why no work is done when a charge moves along an equipotential surface, and why the field lines must cross the equipotentials at right angles.

   Answer

   $W = q\Delta V$ and $\Delta V = 0$ along the surface. If E had a component along the surface, a force would act along it and work would be done, so E must be perpendicular to the surface.

Summary

Quantity	Point charge Q	Scalar or vector	Unit
Force on q	$F = kQq/r^2$	vector	N
Field strength	$E = kQ/r^2 = F/q$	vector	N C^{-1} or V m^{-1}
Potential	$V = kQ/r$	scalar	V (J C^{-1})
Potential energy of q	$U = qV = kQq/r$	scalar	J
Flux through a closed surface	$\Phi = Q_{enclosed}/\varepsilon_0$	scalar	N m^2 C^{-1}
Link	$E = -\mathrm{d}V/\mathrm{d}r$; uniform field $E = V/d$		

Chapter 13 · Semester 2

CAPACITORS

Notes by Cikgu Suhaili

15 worked examples 5 simulations Exercises with answers

13.1 Capacitance

- A **capacitor** is a device that stores electric charge (and electrical energy).
- It is made of two conducting plates separated by a small gap of air or a thin insulator called a **dielectric** (mica, ceramic, paper, plastic, oil).

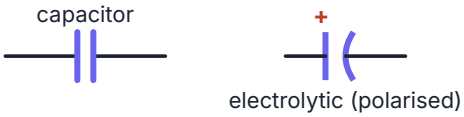


Figure 13.1: circuit symbols for a capacitor.

Capacitance, C

the **ratio of the charge on either plate to the potential difference between the plates**.

$$C = \frac{Q}{V}$$

Q = charge on one plate (the other has $-Q$); V = p.d. across the plates. Unit: farad (F) = C V^{-1}

- 1 F is the capacitance when a charge of 1 C on each plate gives a p.d. of 1 V.
- The farad is a very large unit: 1 μF = 10^{-6} F, 1 nF = 10^{-9} F, 1 pF = 10^{-12} F.
- For a given capacitor C is constant, so $Q \propto V$.
- The net charge on a capacitor is zero: it holds $+Q$ on one plate and $-Q$ on the other. "The charge on a capacitor" means Q.

13.2 Parallel plate capacitors

How a parallel plate capacitor is charged

- When the capacitor is connected to a battery, electrons flow from the negative terminal of the battery to one plate, which becomes negatively charged ($-Q$).
- At the same time, electrons are pulled off the other plate towards the positive terminal, leaving it positively charged ($+Q$). No charge crosses the gap.
- As charge builds up, the p.d. across the plates rises. The flow of electrons stops when the p.d. across the capacitor equals the e.m.f. of the battery.

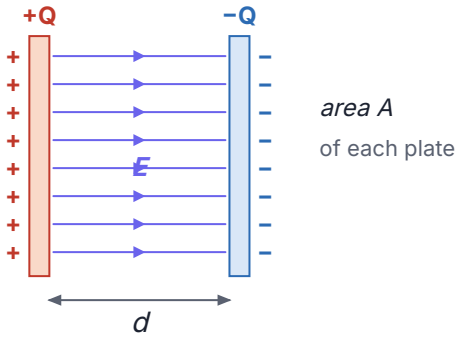


Figure 13.2: a charged parallel plate capacitor. The field between the plates is uniform.

Deriving C = ε₀A/d

Gauss's law gives the field between the plates: $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$

The field is uniform, so the p.d. is $V = Ed = \frac{Qd}{\varepsilon_0 A}$

$$C = \frac{Q}{V} = \frac{Q}{Qd/\varepsilon_0 A} \Rightarrow C_0 = \frac{\varepsilon_0 A}{d}$$

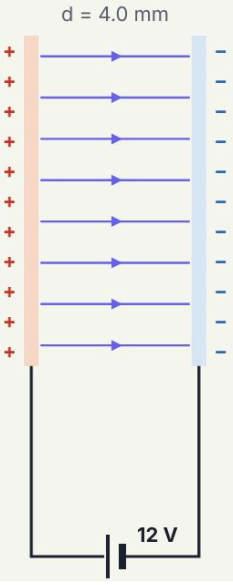
$$C_0 = \frac{\varepsilon_0 A}{d}$$

A = area of overlap of the plates; d = separation; $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}. The capacitance is proportional to A and inversely proportional to d.

Simulation

Parallel plate capacitor

Change the area, the separation and the dielectric. Compare a capacitor that stays connected to the battery (V fixed) with one that is disconnected first (Q fixed).



capacitance C	charge Q	p.d. V	field E = V/d	energy U = ½QV
4.43×10^{-11} F	**5.31×10^{-10} C**	**12.00 V**	**3.00×10^{3} V m^{-1}**	**3.19×10^{-9} J**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 13 to use the simulation.

Example 13.2.1

(a) Find the capacitance of a capacitor whose plates are 20 cm × 3.0 cm, separated by a 1.0 mm air gap. (b) What is the charge on each plate when it is connected to a 12 V battery? (c) What is the electric field between the plates?

Solution

(a) $C = \dfrac{\varepsilon_0 A}{d} = \dfrac{(8.85 \times 10^{-12})(0.20 \times 0.030)}{1.0 \times 10^{-3}} = \mathbf{5.3 \times 10^{-11}\ F}$ (53 pF)

(b) $Q = CV = (5.3 \times 10^{-11})(12) = \mathbf{6.4 \times 10^{-10}\ C}$

(c) $E = \dfrac{V}{d} = \dfrac{12}{1.0 \times 10^{-3}} = \mathbf{1.2 \times 10^{4}\ V\ m^{-1}}$

Example 13.2.2

An electric field of 2.80×10^5 V m^{-1} is needed between two parallel plates, each of area 21.0 cm^2 and 0.250 cm apart in air. Find the charge on each plate.

Solution

$$Q = CV = \frac{\varepsilon_0 A}{d}(Ed) = \varepsilon_0 AE$$

$Q = (8.85 \times 10^{-12})(21.0 \times 10^{-4})(2.80 \times 10^5) =$ **5.20×10^{-9} C** (the separation is not needed)

Exercise: parallel plate capacitors

1. The plates of a parallel plate capacitor in vacuum are 8.0 mm apart and each has an area of 4.0 cm^2. The p.d. across the plates is 2.0 kV. Find (a) the capacitance, (b) the charge on each plate, (c) the electric field between the plates.

   Answer

   (a) 4.4×10^{-13} F (0.44 pF) (b) 8.9×10^{-10} C (c) 2.5×10^5 V m^{-1}

2. The separation of the plates of an isolated charged capacitor is doubled. What happens to (a) C, (b) Q, (c) V?

   Answer

   (a) C halves (b) Q is unchanged (isolated) (c) $V = Q/C$ doubles

13.3 Dielectrics

- A **dielectric** is an insulator: it has no free electrons.
- When a dielectric fills the space between the plates, the capacitance increases by a factor ε_r, the **dielectric constant** (relative permittivity) of the material.

Dielectric constant, ε_r

the **ratio of the capacitance of a capacitor with the dielectric between the plates to its capacitance with a vacuum between the plates**.

$$\varepsilon_r = \frac{C}{C_0} \qquad C = \frac{\varepsilon A}{d} = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

$\varepsilon = \varepsilon_r \varepsilon_0$ is the permittivity of the dielectric; ε_r has no unit and is always greater than 1

Why a dielectric increases the capacitance

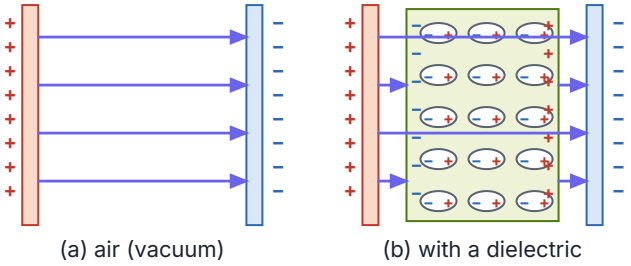


Figure 13.3: the molecules of the dielectric are polarised; induced surface charges appear on its faces.

The electric field between the plates **polarises** the molecules of the dielectric. In a **polar** dielectric (e.g. water) the permanent dipoles line up with the field; in a **non-polar** dielectric the field pulls the centres of positive and negative charge slightly apart, making induced dipoles.

The negative ends face the positive plate and the positive ends face the negative plate, so **induced surface charges** appear: negative on the face next to the + plate, positive on the face next to the − plate.

These surface charges produce a field opposite to the field of the plates, so the resultant field in the dielectric is smaller: $E = E_0/\varepsilon_r$.

For the same charge Q, the p.d. $V = Ed$ is smaller, so $C = Q/V$ is larger.

$$\varepsilon_r = \frac{C}{C_0} = \frac{V_0}{V} = \frac{E_0}{E} \quad \text{(same charge } Q\text{)}$$

Quantity	Isolated capacitor (Q fixed), dielectric inserted	Capacitor kept connected to the battery (V fixed)
Capacitance	$C = \varepsilon_r C_0$ (increases)	$C = \varepsilon_r C_0$ (increases)
Charge	unchanged	$Q = \varepsilon_r Q_0$ (the battery supplies more)
p.d.	$V = V_0/\varepsilon_r$ (decreases)	unchanged
Field	$E = E_0/\varepsilon_r$ (decreases)	unchanged ($E = V/d$)
Energy stored	$U = Q^2/2C$: decreases	$U = \frac{1}{2}CV^2$: increases

Material	Dielectric constant ε_r	Dielectric strength / 10^6 V m^{-1}
Air	1.0006	3
Teflon	2.1	60
Silicone oil	2.5	15
Mylar	3.2	7
Paper	3.7	16
Water	80	—

The **dielectric strength** is the largest electric field a material can stand before it breaks down and conducts. A dielectric also lets the plates be very close together without touching.

Example 13.3.1

A parallel plate capacitor has plates of area $2.0 \times 10^{-2}\ \text{m}^2$, 1.0 cm apart in air. It is charged to 3000 V and the battery is disconnected. A dielectric sheet is then slid in to fill the gap, and the p.d. falls to 1000 V. Find (a) the capacitance in air, (b) the charge on each plate, (c) the capacitance with the dielectric, (d) the dielectric constant, (e) the permittivity of the dielectric, (f) the field before, (g) the field after the dielectric is inserted.

Solution

(a) $C_0 = \dfrac{(8.85 \times 10^{-12})(2.0 \times 10^{-2})}{0.010} = \mathbf{1.77 \times 10^{-11}\ F}$

(b) $Q_0 = C_0 V_0 = (1.77 \times 10^{-11})(3000) = \mathbf{5.31 \times 10^{-8}\ C}$

(c) the charge stays the same: $C = Q_0/V = 5.31 \times 10^{-8}/1000 = \mathbf{5.31 \times 10^{-11}\ F}$

(d) $\varepsilon_r = C/C_0 = \mathbf{3.0}$

(e) $\varepsilon = \varepsilon_r \varepsilon_0 = \mathbf{2.66 \times 10^{-11}\ F\ m^{-1}}$

(f) $E_0 = V_0/d = 3000/0.010 = \mathbf{3.0 \times 10^{5}\ V\ m^{-1}}$

(g) $E = E_0/\varepsilon_r = \mathbf{1.0 \times 10^{5}\ V\ m^{-1}}$

Teacher's note

First ask: is the capacitor still connected to the battery? If yes, V stays the same. If it was disconnected first, Q stays the same. Everything else follows from $C = Q/V$.

A natural capacitor: the thundercloud and the ground

The base of a thundercloud and the ground below it behave like the two plates of a huge parallel plate capacitor, with the **air between them as the dielectric**.

Inside the cloud, collisions between ice crystals and soft hail separate charge. The base of the cloud usually becomes **negative**.

This negative charge repels electrons in the ground below, so the ground surface becomes **positive** (charging by induction): the two "plates" carry $+Q$ and $-Q$.

As more charge builds up, the electric field in the air between them grows: $E = V/d = Q/(\varepsilon_0 A)$.

When E reaches the **dielectric strength of air** (about 3×10^6 V m^{-1}), the air is ionised and becomes a conductor. The dielectric breaks down and the capacitor discharges through it: this is **lightning**.

The huge current heats the air channel to about 30 000 K. The air expands suddenly, and we hear this as **thunder**. We see the flash almost at once (light travels at 3×10^8 m s^{-1}) but hear the thunder later (sound travels at about 340 m s^{-1}): distance ≈ 340 × delay, roughly 1 km for every 3 s.

Simulation

Thundercloud and ground: a natural capacitor

Let the storm charge the cloud. Watch the charges on the cloud base and the ground, and the field in the air between them. When the field reaches the dielectric strength of air, the air breaks down and the capacitor discharges as lightning. Try a higher cloud: does it flash at a different charge? Turn on the thunder sound and count the seconds between the flash and the thunder. Use slow motion to see the stepped leader come down before the bright return stroke.



charge on the cloud Q	field E = Q/(ε₀A)	capacitance C = ε₀A/d	p.d. V = Q/C
11.0 C	**1.25 × 10⁶ V m⁻¹**	**8.8 nF**	**1.25 × 10⁹ V**

energy stored ½QV	flashes	thunder heard after d/340
6.9 × 10⁹ J	**0**	**2.9 s**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 13 to use the simulation.

Example 13.3.2

The base of a thundercloud is 1.0 km above flat ground and has an area of 1.0 km^2. Treat the cloud and the ground as a parallel plate capacitor with air between them (dielectric strength 3.0×10^6 V m^{-1}). Find (a) the capacitance, (b) the largest charge the cloud can hold before the air breaks down, (c) the p.d. between the cloud and the ground at that moment, (d) the energy released if the whole charge is discharged.

Solution

(a) $C = \dfrac{\varepsilon_0 A}{d} = \dfrac{(8.85 \times 10^{-12})(1.0 \times 10^6)}{1.0 \times 10^3} = \mathbf{8.9 \times 10^{-9}\ F}$ (8.9 nF)

(b) $E = \dfrac{Q}{\varepsilon_0 A} \Rightarrow Q = \varepsilon_0 A E = (8.85 \times 10^{-12})(1.0 \times 10^6)(3.0 \times 10^6) = \mathbf{27\ C}$

(c) $V = Ed = (3.0 \times 10^6)(1.0 \times 10^3) = \mathbf{3.0 \times 10^9\ V}$

(d) $U = \tfrac{1}{2}QV = \tfrac{1}{2}(26.6)(3.0 \times 10^9) = \mathbf{4.0 \times 10^{10}\ J}$

This is an idealised model. In a real storm the flash starts at a lower average field, because the field is much stronger near sharp points, raindrops and ice, and the charge is spread through the cloud. Still, the answers are the right size: a real flash carries tens of coulombs.

13.4 Capacitors in series and in parallel

Series

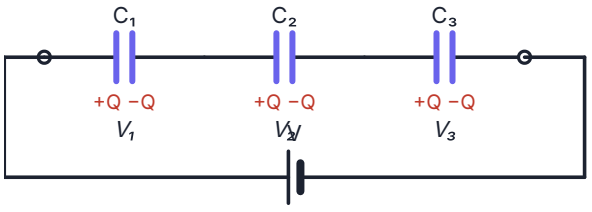


Figure 13.4: three capacitors in series.

Electrons from the battery charge the end plate $-Q$. This repels electrons from the facing plate, which becomes $+Q$, and so on along the chain: **every capacitor carries the same charge Q.**

The p.d.s add up: $V = V_1 + V_2 + V_3 = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$

Divide by Q, with $C_{eq} = Q/V$: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

Parallel

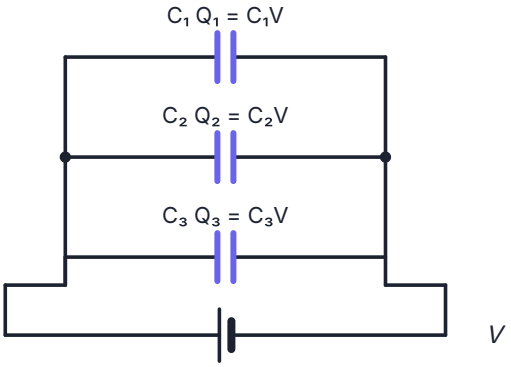


Figure 13.5: three capacitors in parallel.

Each capacitor is connected across the same two points: **every capacitor has the same p.d.** V.

The charges add up: $Q = Q_1 + Q_2 + Q_3 = C_1V + C_2V + C_3V$

Divide by V: $C_{eq} = C_1 + C_2 + C_3$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

series: same Q; C_{eq} is smaller than the smallest

$$C_{eq} = C_1 + C_2 + \dots$$

parallel: same V; C_{eq} is larger than the largest

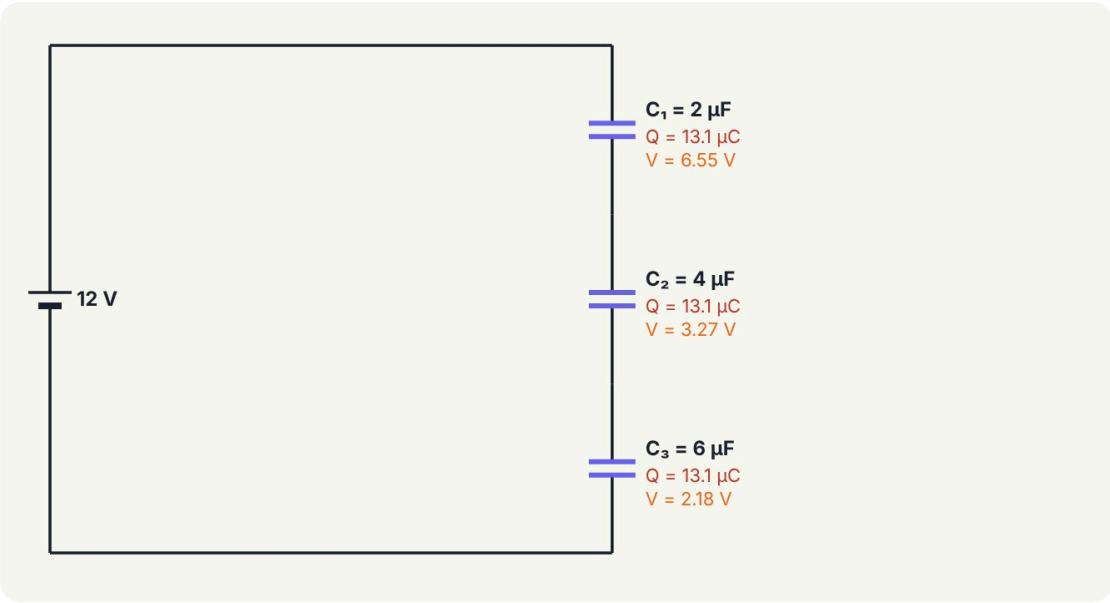
Teacher's note

Two in series: $C_{eq} = \frac{C_1C_2}{C_1 + C_2}$ ("product over sum"). Capacitors combine the opposite way to resistors.

Simulation

Capacitors in series and parallel

Pick an arrangement and change the capacitances and the supply voltage. Check which quantity is shared (Q in series, V in parallel) and how the charge and p.d. split.

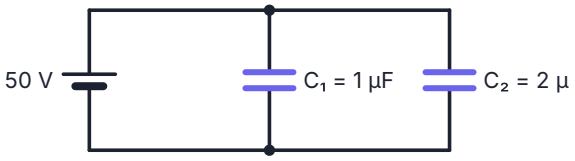


equivalent capacitance	total charge from the supply	energy stored	shared quantity
1.09 µF	**13.1 µC**	**78.5 µJ**	**same Q on each**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 13 to use the simulation.

Example 13.4.1

A 1 µF and a 2 µF capacitor are connected in parallel to a 50 V supply. Find (a) the charge on each capacitor, (b) the equivalent capacitance.



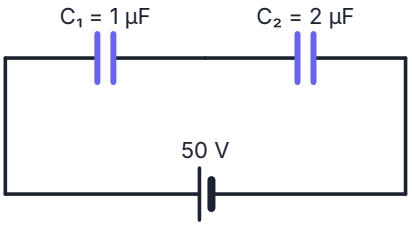
Solution

(a) $Q_1 = C_1 V = (1\ \mu F)(50) =$ **50 µC**; $Q_2 = (2\ \mu F)(50) =$ **100 µC**

(b) $C_{eq} = 1 + 2 =$ **3 µF**

Example 13.4.2

The same capacitors are connected in series to the 50 V supply. Find (a) the equivalent capacitance, (b) the charge on each capacitor, (c) the p.d. across each.



Solution

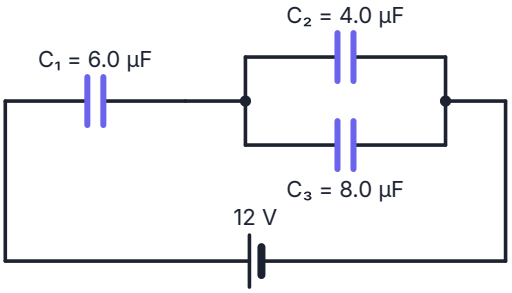
(a) $\frac{1}{C_{eq}} = \frac{1}{1} + \frac{1}{2} \Rightarrow C_{eq} =$ **0.667 μF**

(b) $Q = C_{eq}V$ = (0.667 μF)(50) = **33.3 μC** on each capacitor

(c) $V_1 = Q/C_1$ = 33.3/1 = **33.3 V**; V_2 = 33.3/2 = **16.7 V** (check: 33.3 + 16.7 = 50 V)

Example 13.4.3

Find (a) the equivalent capacitance, (b) the charge on each capacitor, (c) the p.d. across each capacitor.



Solution

(a) $C_{23} = 4.0 + 8.0 = 12.0\ \mu F$; $\frac{1}{C_{eq}} = \frac{1}{6.0} + \frac{1}{12.0} \Rightarrow C_{eq} = \mathbf{4.0\ \mu F}$

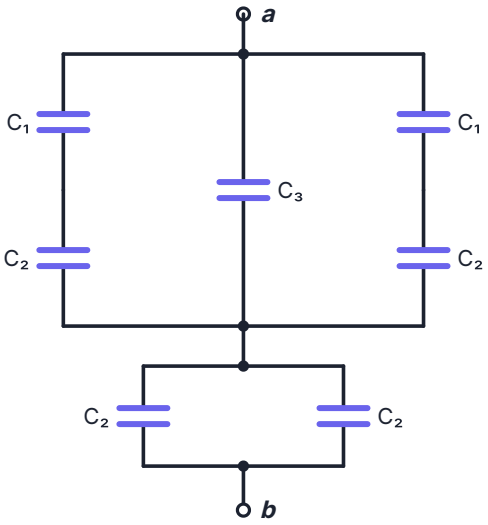
(b) $Q_1 = Q_{total} = (4.0\ \mu F)(12) = \mathbf{48\ \mu C}$

(c) $V_1 = 48/6.0 = \mathbf{8.0\ V}$; $V_2 = V_3 = 12 - 8.0 = \mathbf{4.0\ V}$

(b) $Q_2 = (4.0\ \mu F)(4.0) = \mathbf{16\ \mu C}$; $Q_3 = (8.0\ \mu F)(4.0) = \mathbf{32\ \mu C}$ (check: 16 + 32 = 48 μC)

Example 13.4.4

Find the equivalent capacitance between a and b. C_1 = 5.00 μF, C_2 = 10.0 μF, C_3 = 2.00 μF.



Solution

Each side branch: C_1 and C_2 in series: $C_{12} = \dfrac{(5.00)(10.0)}{15.0} = 3.33\ \mu F$

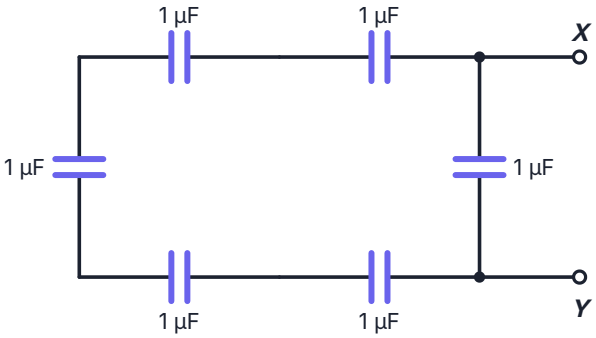
Upper group in parallel: $C_a = 3.33 + 2.00 + 3.33 = 8.67\ \mu F$

Lower pair in parallel: $C_{22} = 10.0 + 10.0 = 20.0\ \mu F$

In series: $\dfrac{1}{C_{eq}} = \dfrac{1}{8.67} + \dfrac{1}{20.0} \Rightarrow C_{eq} = \mathbf{6.05\ \mu F}$

Example 13.4.5

All the capacitors are identical, each 1 μF. Find the equivalent capacitance between X and Y.



Solution

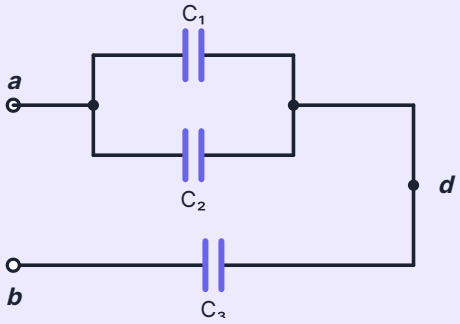
The three capacitors on the left (top-left, left side, bottom-left) are in series: C_a = 1/3 = 0.33 μF

C_a is in series with the top-right and bottom-right capacitors: $\frac{1}{C_b}$ = 1 + 1 + 3 ⇒ C_b = 0.20 μF

C_b is in parallel with the capacitor across XY: C_{eq} = 0.20 + 1 = **1.20 μF**

Exercise: combinations

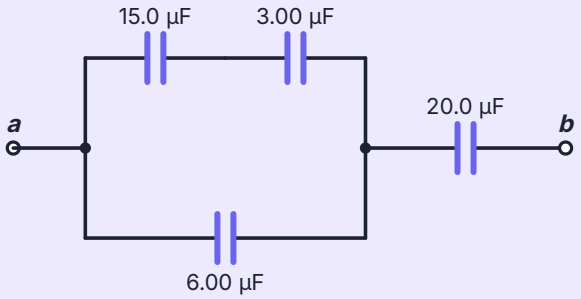
1. C_1 = 2.00 μF, C_2 = 4.00 μF and C_3 = 9.00 μF; V_{ab} = 61.5 V. Find (a) the charge on each capacitor, (b) the p.d. across each, (c) the p.d. between a and d.



Answer

(a) Q_1 = 73.8 μC, Q_2 = 148 μC, Q_3 = 221 μC (b) V_1 = V_2 = 36.9 V, V_3 = 24.6 V (c) V_{ad} = 36.9 V

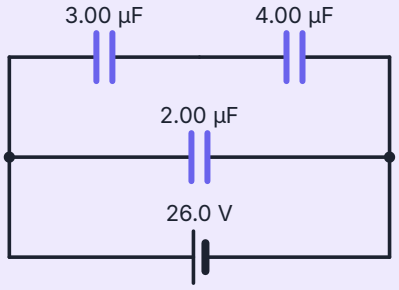
2. Find (a) the equivalent capacitance between a and b, (b) the charge on each capacitor if V_{ab} = 15.0 V.



Answer

(a) 5.96 μF (b) 89.5 μC on 20 μF, 63.2 μC on 6 μF, 26.3 μC on both 15 μF and 3 μF

3. A 3.00 μF and a 4.00 μF capacitor in series are connected in parallel with a 2.00 μF capacitor, across 26.0 V. Find (a) the equivalent capacitance, (b) the p.d. across each capacitor.



Answer

(a) 3.71 μF (b) 2.00 μF: 26.0 V; 3.00 μF: 14.9 V; 4.00 μF: 11.1 V

13.5 Energy stored in a charged capacitor

- Work must be done to charge a capacitor: once some charge is on the plates, adding more charge of the same sign needs work against the repulsion. This work is stored as electric potential energy in the field between the plates.
- On a V–Q graph (a straight line, since $V = Q/C$), the energy stored is the **area under the graph**.

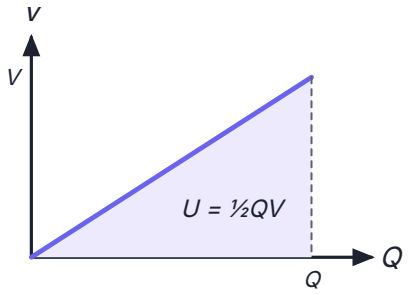


Figure 13.6: the energy stored is the area of the triangle under the V–Q graph.

$$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$$

unit: joule. Derivations are not required by the syllabus.

For understanding: moving a small charge $\mathrm{d}q$ through the p.d. $V = q/C$ needs $\mathrm{d}W = V\,\mathrm{d}q$, so $W = \int_0^Q (q/C)\,\mathrm{d}q = Q^2/2C$.

Teacher's note

A battery that charges a capacitor supplies energy QV, but the capacitor stores only $\frac{1}{2}QV$. The other half is changed into heat in the resistance of the circuit (and some electromagnetic radiation), whatever the value of the resistance.

Example 13.5.1

A camera flash unit stores energy in a 150 μF capacitor at 200 V. How much energy is stored?

Solution

$U = \frac{1}{2}CV^2 = \frac{1}{2}(150 \times 10^{-6})(200)^2 =$ **3.0 J**

Example 13.5.2

A 2 μF capacitor is charged to 200 V by a battery. Find (a) the charge delivered by the battery, (b) the energy supplied by the battery, (c) the energy stored in the capacitor.

Solution

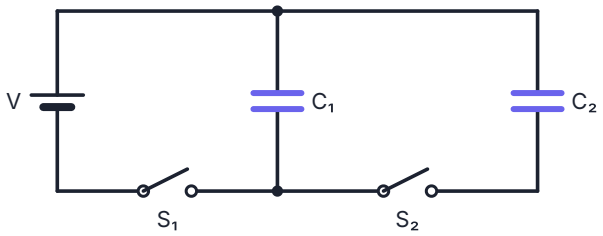
(a) $Q = CV = (2 \times 10^{-6})(200) =$ **4.0×10^{-4} C**

(b) $E = QV = (4.0 \times 10^{-4})(200) =$ **0.080 J**

(c) $U = \frac{1}{2}QV =$ **0.040 J**: half of the energy supplied

Example 13.5.3

C_1 = 6.00 μF is charged by closing S_1 (V = 20.0 V). S_1 is then opened and S_2 closed, connecting C_1 to the uncharged C_2 = 3.00 μF. Find (a) the initial charge on C_1, (b) the final p.d., (c) the final charge on each capacitor, (d) the energy lost.



Solution

(a) $Q_1 = C_1V$ = (6.00 μF)(20.0) = **120 μC**

(b) After S_2 is closed the capacitors are in parallel and share the charge until their p.d.s are equal. Charge is conserved: $V = \dfrac{Q_{\text{total}}}{C_1 + C_2} = \dfrac{120}{9.00}$ = **13.3 V**

(c) Q_1' = (6.00)(13.33) = **80.0 μC**; Q_2' = (3.00)(13.33) = **40.0 μC**

(d) before: ½(120 × 10^{-6})(20.0) = 1.20 × 10^{-3} J; after: ½(120 × 10^{-6})(13.33) = 0.80 × 10^{-3} J; lost = **4.0 × 10^{-4} J** (as heat in the connecting wires)

Store slowly, release quickly: camera flash and defibrillator

A capacitor can be charged slowly by a small battery and then discharged in a fraction of a millisecond. The **same energy** released in a much **shorter time** gives a huge **power**. This is how a camera flash and a heart defibrillator work.

- **Charging** goes through a large resistance (the charging circuit), so the time constant RC is about a second or more: you wait for the "ready" light.
- **Discharging** goes through a small resistance (the flash tube, or the patient's chest, about 50 Ω), so the time constant is about a millisecond.
- Energy stored: $U = \tfrac{1}{2}CV^2$. Power ≈ energy ÷ time, so releasing it in about a millisecond gives tens of kilowatts.

Simulation Camera flash and defibrillator

Choose the device and press Charge: watch the capacitor charge slowly (seconds). Then press Fire: the energy comes out in about a millisecond. A defibrillator shock stops the chaotic heartbeat so that a normal rhythm can restart.







capacitor	energy stored ½CV²	charging time constant RC
32 µF charged to 3.5 kV	**196 J**	**3.2 s**

discharge time constant RC	peak current V/R	power ≈ U ÷ 2RC
1.60 ms	**70 A**	**61 kW**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 13 to use the simulation.

Example 13.5.4

A defibrillator has a 32 μF capacitor charged to 3.5 kV. It discharges through the patient's chest, of resistance 50 Ω. Find (a) the energy stored, (b) the time constant of the discharge, (c) the initial current, (d) the average power if the energy is delivered in 3.0 ms.

Solution

(a) $U = \tfrac{1}{2}CV^2 = \tfrac{1}{2}(32 \times 10^{-6})(3500)^2 =$ **196 J**

(b) $\tau = RC = (50)(32 \times 10^{-6}) =$ **1.6×10^{-3} s**

(c) $I_0 = \dfrac{V_0}{R} = \dfrac{3500}{50} =$ **70 A**

(d) $P \approx \dfrac{196}{3.0 \times 10^{-3}} =$ **6.5×10^4 W** (65 kW): far more than its battery could give directly

Exercise: energy

1. C_1 = 3.00 μF and C_2 = 6.00 μF are connected in series to a 4.00 V battery. Find (a) the equivalent capacitance, (b) the charge on each capacitor, (c) the p.d. across each, (d) the energy stored in each, (e) the area of each plate of C_1 if its plates are 0.010 mm apart in vacuum.

Answer

(a) 2.00 μF (b) 8.00 μC (c) $V_1 = 2.67$ V, $V_2 = 1.33$ V (d) $U_1 = 1.07 \times 10^{-5}$ J, $U_2 = 5.33 \times 10^{-6}$ J (e) 3.39 m²

13.6 Charging and discharging through a resistor

Charging

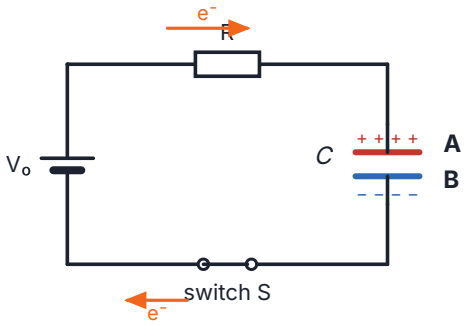


Figure 13.7: charging a capacitor through a resistor.

- When S is closed, electrons flow from the negative terminal of the battery, through R, onto plate B. Electrons leave plate A for the positive terminal, so A becomes positive.
- At the start the capacitor is uncharged, so the whole e.m.f. is across R: the current is largest, $I_0 = V_0/R$.
- As charge builds up, the p.d. across C rises, so the p.d. across R (and the current) falls.
- Charging stops when the p.d. across C equals V_0: then $I = 0$ and the charge is maximum, $Q_0 = CV_0$.

Derivation (charging)

Kirchhoff's loop rule at any time: $V_0 = IR + \frac{Q}{C}$, with $I = \frac{dQ}{dt}$

Rearrange: $\frac{dQ}{dt} = \frac{CV_0 - Q}{RC} = \frac{Q_0 - Q}{RC} \Rightarrow \int_0^Q \frac{dQ}{Q_0 - Q} = \int_0^t \frac{dt}{RC}$

$-\ln \frac{Q_0 - Q}{Q_0} = \frac{t}{RC} \Rightarrow Q = Q_0(1 - e^{-t/RC})$

Divide by C: $V = V_0(1 - e^{-t/RC})$. Differentiate: $I = \frac{dQ}{dt} = \frac{Q_0}{RC} e^{-t/RC} = I_0 e^{-t/RC}$

$$Q = Q_0(1 - e^{-t/RC}) \quad V = V_0(1 - e^{-t/RC}) \quad I = I_0 e^{-t/RC}$$

charging: $Q_0 = CV_0$, $I_0 = V_0/R$



Figure 13.8: charging. Q and V rise to their maximum; the current falls exponentially.

Discharging

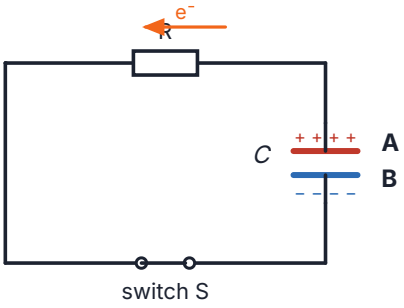


Figure 13.9: discharging a charged capacitor through a resistor.

- When S is closed, electrons flow from plate B through R to plate A and neutralise its positive charge.
- At the start the p.d. is V_0, so the current is largest, $I_0 = V_0/R$.
- As the charge falls, the p.d. falls, so the current falls. Discharging ends when $Q = 0$, $V = 0$ and $I = 0$.
- The current flows in the opposite direction to the charging current.

Derivation (discharging)

Loop rule with no battery: $0 = IR + \frac{Q}{C}$, with $I = \frac{dQ}{dt}$ (the charge decreases)

$$\frac{dQ}{Q} = -\frac{dt}{RC} \Rightarrow \ln\frac{Q}{Q_0} = -\frac{t}{RC} \Rightarrow Q = Q_0 e^{-t/RC}$$

$V = Q/C = V_0 e^{-t/RC}$; $I = \frac{dQ}{dt} = -I_0 e^{-t/RC}$ (minus: opposite direction; use the size in calculations)

$$Q = Q_0 e^{-t/RC} \quad V = V_0 e^{-t/RC} \quad I = I_0 e^{-t/RC}$$

discharging



Figure 13.10: discharging. Q, V and I all fall exponentially.

Time constant

Time constant, τ

of an RC circuit: the **time for the charge (or current) to fall to 1/e (37%) of its initial value** during discharging. During charging, it is the time for the charge to rise to (1 – 1/e), 63% of its final value.

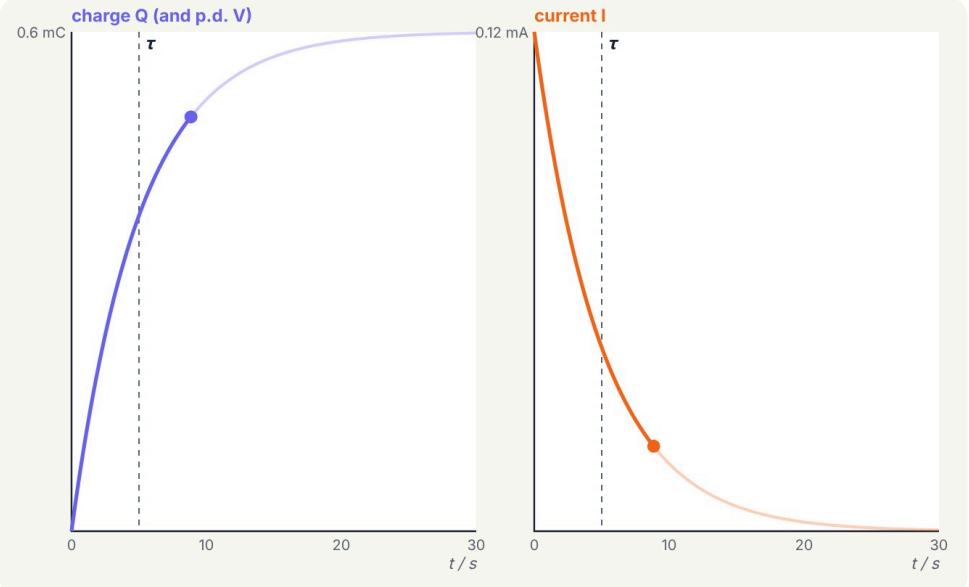
$$\tau = RC$$

unit: second (Ω × F = s). A large τ means slow charging and discharging.

- At t = τ: e^{-1} = 0.37, so discharging gives Q = 0.37Q_0; charging gives Q = 0.63Q_0 and I = 0.37I_0.
- After about 5τ the capacitor is practically fully charged (99.3%) or fully discharged.
- Half-life: the time for Q to halve is $t_{½}$ = RC ln 2 = 0.693RC.

Simulation — Charging and discharging through a resistor

Choose charging or discharging, then change R and C. Watch how the time constant τ = RC sets how fast the curves rise and fall. The dashed line marks t = τ.



time constant τ = RC	t	charge Q	p.d. across C
5.0 s	**8.9 s (= 1.77τ)**	**0.50 mC (83% of Q_0)**	**9.96 V**

current I	supply
0.020 mA (17% of I_0)	**12 V battery**

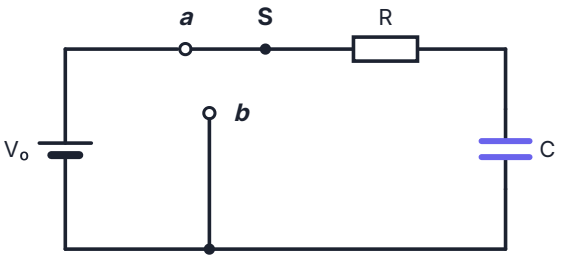
This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 13 to use the simulation.

Teacher's note

To find a time, take natural logs: e.g. discharging, $\frac{Q}{Q_0} = e^{-t/RC} \Rightarrow t = -RC \ln \frac{Q}{Q_0} = RC \ln \frac{Q_0}{Q}$.

Example 13.6.1

A capacitor is fully charged by a 20.0 V battery, then at t = 0 the switch is moved from a to b. C = 1.02 μF and the current falls to half its initial value in 40 μs. Find (a) R, (b) the time constant, (c) the charge at t = 0, (d) the charge at t = 60 μs.



Solution

(a) $0.50 = e^{-40 \times 10^{-6}/RC} \Rightarrow RC = \frac{40 \times 10^{-6}}{\ln 2} = 5.77 \times 10^{-5}$ s $\Rightarrow R = \frac{5.77 \times 10^{-5}}{1.02 \times 10^{-6}} =$ **57 Ω**

(b) τ = RC = **5.8×10^{-5} s**

(c) $Q_0 = CV_0 = (1.02 \times 10^{-6})(20.0) =$ **2.04×10^{-5} C**

(d) $Q = (2.04 \times 10^{-5})e^{-60/57.7} =$ **7.2×10^{-6} C**

Example 13.6.2

A 470 μF capacitor is charged through a 100 kΩ resistor from a 9.0 V supply. Find (a) the time constant, (b) the initial current, (c) the p.d. across the capacitor after 30 s, (d) the time for the capacitor to reach 90% of its final charge.

Solution

(a) $\tau = RC = (100 \times 10^3)(470 \times 10^{-6}) =$ **47 s**

(b) $I_0 = V_0/R = 9.0/100 \times 10^3 =$ **9.0×10^{-5} A**

(c) $V = 9.0(1 - e^{-30/47}) = 9.0(1 - 0.528) =$ **4.2 V**

(d) $0.90 = 1 - e^{-t/47} \Rightarrow t = 47 \ln 10 =$ **108 s** (2.3τ)

Exercise: RC circuits

1. A 1000 μF capacitor charged to 12 V is discharged through a 10 kΩ resistor. Find (a) the time constant, (b) the initial current, (c) the p.d. after 5.0 s, (d) the time for the charge to halve.

   Answer

   (a) 10 s (b) 1.2 mA (c) $12e^{-0.5} = 7.3$ V (d) $10 \ln 2 = 6.9$ s

2. Sketch, on the same axes, the current–time graphs for charging the same capacitor through (a) a resistor R, (b) a resistor $2R$, from the same supply.

   Answer

   (a) starts at $I_0 = V_0/R$ and falls with $\tau = RC$. (b) starts at half the value, $I_0/2$, and falls more slowly ($\tau = 2RC$). The areas under both graphs are equal (the same final charge $Q_0 = CV_0$).

Chapter 14 · Semester 2

ELECTRIC CURRENT

Notes by Cikgu Suhaili

9 worked examples 2 simulations Exercises with answers

14.1 Conduction of electricity

Electric current, I

the **rate of flow of electric charge** through a cross-section of a conductor.

$$I = \frac{\mathrm{d}Q}{\mathrm{d}t} \qquad \text{steady current: } Q = It$$

unit: ampere (A) = C s^{-1}. 1 A is a flow of 1 C of charge per second.

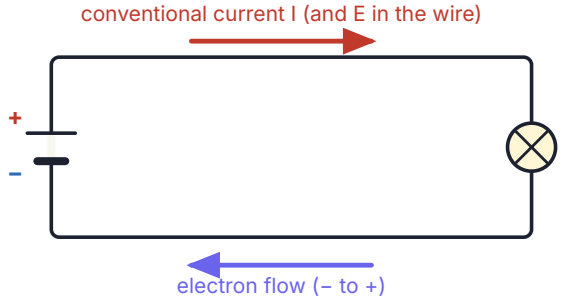


Figure 14.1: the conventional current flows from the positive to the negative terminal outside the cell; electrons flow the other way.

Charge carriers

Material	Charge carriers
Metals	free (conduction) electrons
Semiconductors	electrons and holes (a hole behaves like a positive charge)
Electrolytes and gases (for comparison)	positive and negative ions (and electrons in gases)

How a metal conducts

A metal is a lattice of positive ions with many **free electrons** that are not bound to any atom.

Without a p.d., the free electrons move randomly at high speed (about 10^6 m s^{-1}), colliding with the vibrating ions. Their average velocity is zero, so there is no current.

When a p.d. is applied, an electric field E is set up in the metal. Each electron feels a force eE opposite to the field and accelerates between collisions.

At each collision with an ion the electron loses the velocity it gained. The result is a slow average **drift** opposite to the field, added to the random motion. This drift is the current.

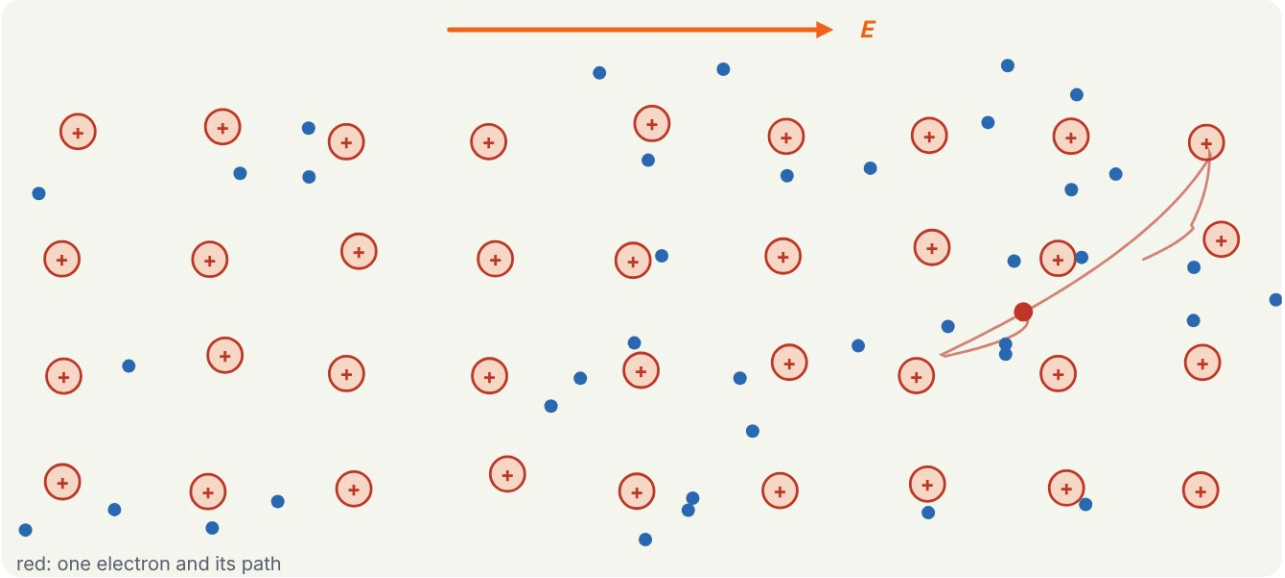


Figure 14.2: the path of a free electron without a field and with a field.

Simulation

Free electrons in a metal

Turn the field on and watch the electrons drift slowly against the field while still moving randomly. Raise the temperature: the ions vibrate more, collisions are more frequent, and the drift becomes slower.



red: one electron and its path

drift (average velocity)
36 units to the left

collisions per second
45

mean time between collisions t
892 ms (simulated)

what to notice
drift is tiny compared with the random speed

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 14 to use the simulation.

Example 14.1.1

A silver wire carries a current of 3.0 A. Find (a) the number of electrons passing a cross-section each second, (b) the charge that flows through a cross-section in 55 s. (e = 1.60 × 10^{-19} C)

Solution

(a) $N = \dfrac{I}{e} = \dfrac{3.0}{1.60 \times 10^{-19}}$ = **1.9 × 10^{19} electrons per second**

(b) $Q = It$ = (3.0)(55) = **165 C**

14.2 Drift velocity

Drift velocity, v

the **average velocity of the charge carriers along the conductor** when an electric field is applied.

- For electrons it is opposite to the field (and to the conventional current).
- It is very small (about 10^{-4} m s^{-1}) compared with the random speed of the electrons. A lamp lights at once because the field is set up along the whole wire almost immediately and all the electrons start drifting together.

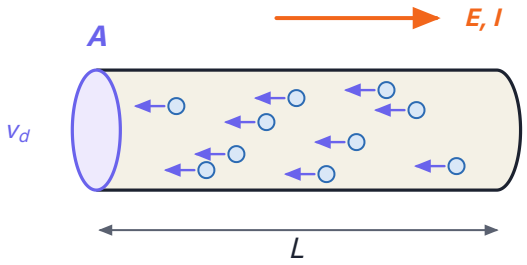


Figure 14.3: a conductor of cross-sectional area A and length L.

Deriving I = Anev

Let n = number of free electrons per unit volume. The volume of the length L is AL, so it holds $N = nAL$ free electrons.

Total charge of these electrons: $Q = Ne = nALe$

They all pass through the end face in the time $t = \frac{L}{v}$

$$I = \frac{Q}{t} = \frac{nALe}{L/v} \Rightarrow I = Anev$$

$$I = Anev$$

n = number density of free electrons (m^{-3}); copper: $n \approx 8.5 \times 10^{28}$ m^{-3}

- For the same current, a thinner wire (smaller A) has a larger drift velocity.
- A semiconductor has a much smaller n, so for the same current its carriers drift much faster.

Example 14.2.1

A copper wire of cross-sectional area 1.0 mm^2 carries a current of 2.0 A. Copper has 8.5×10^{28} free electrons per m^3. Find the drift velocity.

Solution

$$v = \frac{I}{Ane} = \frac{2.0}{(1.0 \times 10^{-6})(8.5 \times 10^{28})(1.60 \times 10^{-19})} = \mathbf{1.5 \times 10^{-4}\ m\,s^{-1}}$$

Example 14.2.2

A transmission line of diameter 3.00 cm and length 100 km carries a steady current of 1500 A. The copper has 8.49×10^{28} free electrons per m^3. How long does one electron take to travel the full length of the line?

Solution

$$A = \pi(0.0150)^2 = 7.07 \times 10^{-4}\ m^2$$

$$v = \frac{1500}{(7.07 \times 10^{-4})(8.49 \times 10^{28})(1.60 \times 10^{-19})} = 1.56 \times 10^{-4}\ m\,s^{-1}$$

$$t = \frac{L}{v} = \frac{1.00 \times 10^{5}}{1.56 \times 10^{-4}} = \mathbf{6.4 \times 10^{8}\ s}\ \text{(about 20 years)}$$

14.3 Current density

Current density, J

the **current per unit cross-sectional area** of a conductor, the area being perpendicular to the current.

$$J = \frac{I}{A} = nev$$

vector, in the direction of the conventional current; unit A m^{-2}

Example 14.3.1

A copper wire carries 10.0 A. Its cross-section is a square of side 2.0 mm. Copper has 8.49×10^{28} free electrons per m^3. Find (a) the current density, (b) the drift velocity.

Solution

(a) $J = \dfrac{I}{A} = \dfrac{10.0}{(2.0 \times 10^{-3})^2} = \mathbf{2.5 \times 10^{6}\ A\ m^{-2}}$

(b) $v = \dfrac{J}{ne} = \dfrac{2.5 \times 10^{6}}{(8.49 \times 10^{28})(1.60 \times 10^{-19})} = \mathbf{1.8 \times 10^{-4}\ m\,s^{-1}}$

14.4 Electrical conductivity and resistivity

Resistance and Ohm's law

Resistance, R

the ratio of the potential difference across a conductor to the current in it: $R = V/I$. Unit: ohm (Ω) = V A^{-1}.

Ohm's law

The current in a metallic conductor is directly proportional to the potential difference across it, provided the temperature and other physical conditions are constant.

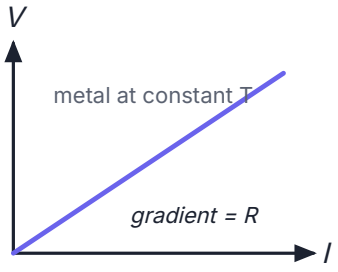


Figure 14.4: an ohmic conductor gives a straight line through the origin.

Resistivity and conductivity

Resistivity, ρ

of a material: the resistance of a sample of unit length and unit cross-sectional area. It depends on the material and its temperature, not on its shape.

$$\rho = \frac{RA}{l} \qquad \left(R = \frac{\rho l}{A}\right)$$

unit: Ω m. Good conductors have a small ρ (copper 1.7×10^{-8} Ω m); insulators a very large ρ.

Electrical conductivity, σ

the **ratio of the current density to the electric field strength** in a conductor.

$$J = \sigma E \qquad \sigma = \frac{1}{\rho}$$

unit of σ: $\Omega^{-1}\,m^{-1}$ ($S\,m^{-1}$). Conductivity is the reciprocal of resistivity.

Ohm's law and J = σE are the same law

For a uniform conductor of length l and area A: $V = El$ and $I = JA$.

Ohm's law $V = IR$ with $R = \rho l/A$: $El = (JA)\frac{\rho l}{A} = J\rho l$

So $E = \rho J$, i.e. $J = \frac{1}{\rho}E = \sigma E$. A constant σ (at constant temperature) means $J \propto E$, which is Ohm's law for the material.

Deriving σ = ne²t/m

Between collisions, a free electron in the field E has acceleration $a = \frac{eE}{m}$.

Let t be the **mean time between collisions**. Each collision wipes out the velocity gained, so the average (drift) velocity gained is $v = at = \frac{eEt}{m}$.

$$J = nev = ne\frac{eEt}{m} = \frac{ne^2t}{m}E$$

Compare with $J = \sigma E$: $\sigma = \frac{ne^2t}{m}$ and $\rho = \frac{m}{ne^2t}$

$$\sigma = \frac{ne^2t}{m} \qquad \rho = \frac{1}{\sigma} = \frac{m}{ne^2t}$$

n = free electrons per unit volume; t = mean time between collisions; m, e = mass and charge of an electron

Temperature and the resistivity of a metal

In a metal, n does **not** change with temperature: every atom already gives its free electrons.

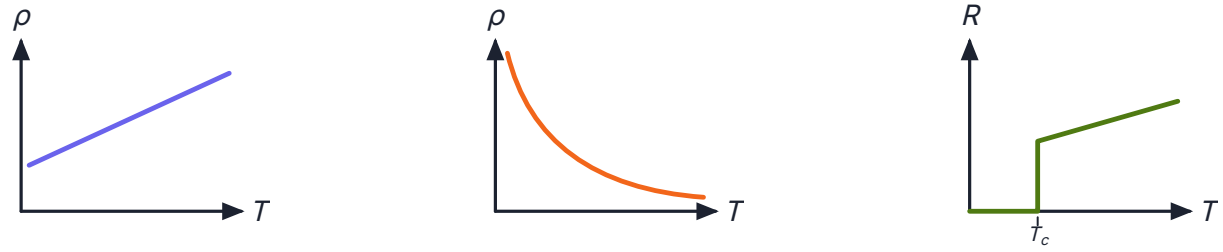
When the temperature rises, the ions vibrate with larger amplitude, so the free electrons collide with them more often: the mean time between collisions t **decreases**.

From $\sigma = ne^2t/m$, σ decreases, so the resistivity $\rho = m/ne^2t$ **increases** (roughly linearly) with temperature.

Superconductivity

the **property of some materials whose resistivity falls to zero when they are cooled below a critical temperature, T_c.**

Material	When the temperature rises	Reason
Metal	ρ increases	n constant; more collisions with the vibrating ions, so t decreases
Semiconductor	ρ decreases (quickly)	more electrons and holes are released, so n increases greatly; this outweighs the fall in t
Superconductor	ρ = 0 below T_c; above T_c it behaves like a normal conductor	mechanism not required



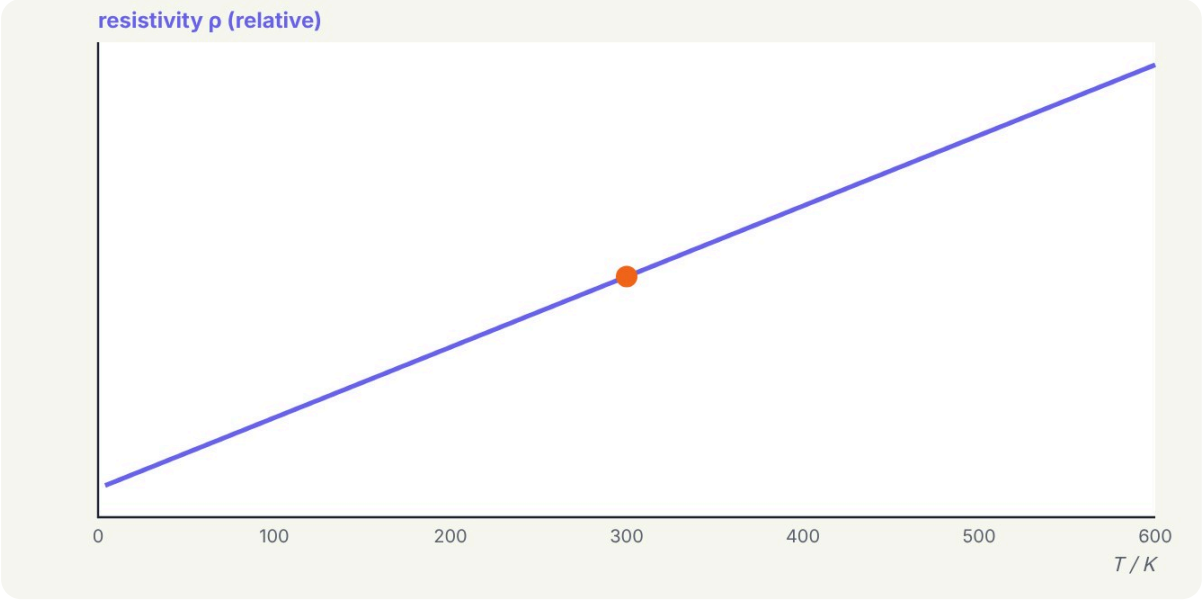
metal semiconductor superconductor

Figure 14.5: how resistivity (or resistance) varies with temperature.

Simulation

Resistivity and temperature

Choose a material and change the temperature. Watch the number of carriers n and the mean time between collisions t, and see how they decide the resistivity.



carriers per unit volume n
constant

mean time between collisions t
1.00 × the value at 300 K

resistivity ρ = m/ne²t
1.00 × the value at 300 K

note
curves are sketches, not data

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 14 to use the simulation.

Example 14.4.1

A wire 2.0 m long with a diameter of 0.50 mm has a resistance of 0.171 Ω. Find (a) the resistivity, (b) the conductivity of the material.

Solution

$A = \pi(0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7}\ \text{m}^2$

(a) $\rho = \dfrac{RA}{l} = \dfrac{(0.171)(1.96 \times 10^{-7})}{2.0} = \mathbf{1.68 \times 10^{-8}\ \Omega\,m}$

(b) $\sigma = 1/\rho = \mathbf{5.95 \times 10^{7}\ \Omega^{-1}\,m^{-1}}$

Example 14.4.2

A p.d. of 240 V is applied across a wire 200 cm long. The current density in the wire is 7.14×10^9 A m^{-2}. Find (a) the resistivity, (b) the conductivity of the wire.

Solution

$E = V/l = 240/2.00 = 120$ V m^{-1}

(a) $\rho = \dfrac{E}{J} = \dfrac{120}{7.14 \times 10^9} = \mathbf{1.68 \times 10^{-8}\ \Omega\ m}$

(b) $\sigma = 1/\rho = \mathbf{5.95 \times 10^7\ \Omega^{-1}\ m^{-1}}$

Example 14.4.3

Copper has 8.5×10^{28} free electrons per m^3 and a resistivity of 1.7×10^{-8} Ω m. Find the mean time between collisions of the free electrons. ($m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C)

Solution

$$t = \frac{m}{ne^2\rho} = \frac{9.11 \times 10^{-31}}{(8.5 \times 10^{28})(1.60 \times 10^{-19})^2(1.7 \times 10^{-8})} = \mathbf{2.5 \times 10^{-14}\ s}$$

Example 14.4.4

The copper wire in Example 14.3.1 (10.0 A, square cross-section of side 2.0 mm) is 50 m long. ρ = 1.68×10^{-8} Ω m. Find (a) the electric field in the wire, (b) the p.d. across it, (c) its resistance.

Solution

(a) $E = \rho J = (1.68 \times 10^{-8})(2.5 \times 10^6) = \mathbf{0.042\ V\ m^{-1}}$

(b) $V = El = (0.042)(50) = \mathbf{2.1\ V}$

(c) $R = V/I = 2.1/10.0 = \mathbf{0.21\ \Omega}$ (check: $\rho l/A = (1.68 \times 10^{-8})(50)/(4.0 \times 10^{-6}) = 0.21\ \Omega$)

Section A practice

The resistivity of a wire depends on its

A. length

B. material

C. cross-sectional area

D. shape

Answer

Answer: B

Resistivity is a property of the material (and its temperature). Changing the length or area changes the resistance, not the resistivity.

Exercise: electric current

1. Two wires A and B of the same material and length carry the same current. The diameter of A is twice that of B. Find the ratio of the drift velocities $v_A : v_B$.

   Answer

   $v \propto 1/A \propto 1/d^2$, so $v_A : v_B = 1 : 4$

2. Explain why the resistance of a metal wire increases when it gets hotter, but the resistance of a semiconductor decreases.

   Answer

   Metal: n stays the same but the ions vibrate more, so the mean time between collisions t falls; $\sigma = ne^2t/m$ falls and ρ rises. Semiconductor: heating frees many more charge carriers, so n rises greatly; this outweighs the small fall in t, so ρ falls.

3. An aluminium wire of resistivity 2.8×10^{-8} Ω m and cross-sectional area 2.0 mm^2 carries 4.0 A. Find the electric field in the wire.

   Answer

   $J = 4.0/2.0 \times 10^{-6} = 2.0 \times 10^6$ A m^{-2}; $E = \rho J = 0.056$ V m^{-1}

Chapter 15 · Semester 2

DIRECT CURRENT CIRCUITS

Notes by Cikgu Suhaili

20 worked examples 3 simulations Exercises with answers

15.1 Electromotive force, potential difference and internal resistance

Electromotive force (e.m.f.), ε

of a source: the **energy converted from other forms (e.g. chemical) into electrical energy per unit charge** that passes through the source. It equals the p.d. across the terminals when no current flows (open circuit).

Potential difference, V

between two points: the **electrical energy converted into other forms per unit charge** passing from one point to the other.

$$\varepsilon = \frac{W}{Q} \qquad V = \frac{W}{Q}$$

both measured in volts (J C^{-1}); the difference is the direction of the energy change

	e.m.f.	potential difference
Energy change	other forms → electrical energy (in the source)	electrical energy → other forms (in a component)
Where	across a source (cell, generator)	between any two points in a circuit
Depends on the current?	no: a property of the source	yes (e.g. $V = IR$)

Internal resistance

Every real source has some resistance of its own, its **internal resistance** r (in a chemical cell: the electrolyte and the electrodes oppose the flow of charge). Part of the e.m.f. is used to drive the current through r, so the p.d. at the terminals is less than ε.

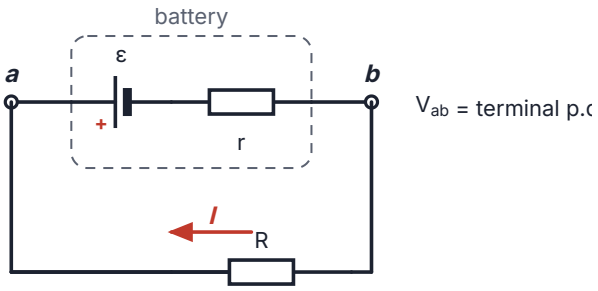


Figure 15.1: a cell of e.m.f. ε and internal resistance r connected to an external resistor R.

$$\varepsilon = I(R + r) \qquad V = \varepsilon - Ir$$

$V = IR$ = terminal p.d.; Ir = the "lost volts" across the internal resistance

- The larger the current drawn, the larger Ir and the lower the terminal p.d. (headlights dim when a car engine is started).
- $V = \varepsilon$ only when $I = 0$ (open circuit) or when $r = 0$ (an ideal source).
- When a cell is being **charged** (current forced into its + terminal), $V = \varepsilon + Ir$, which is larger than ε.

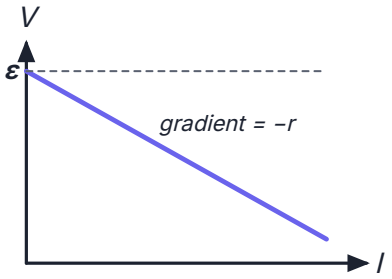
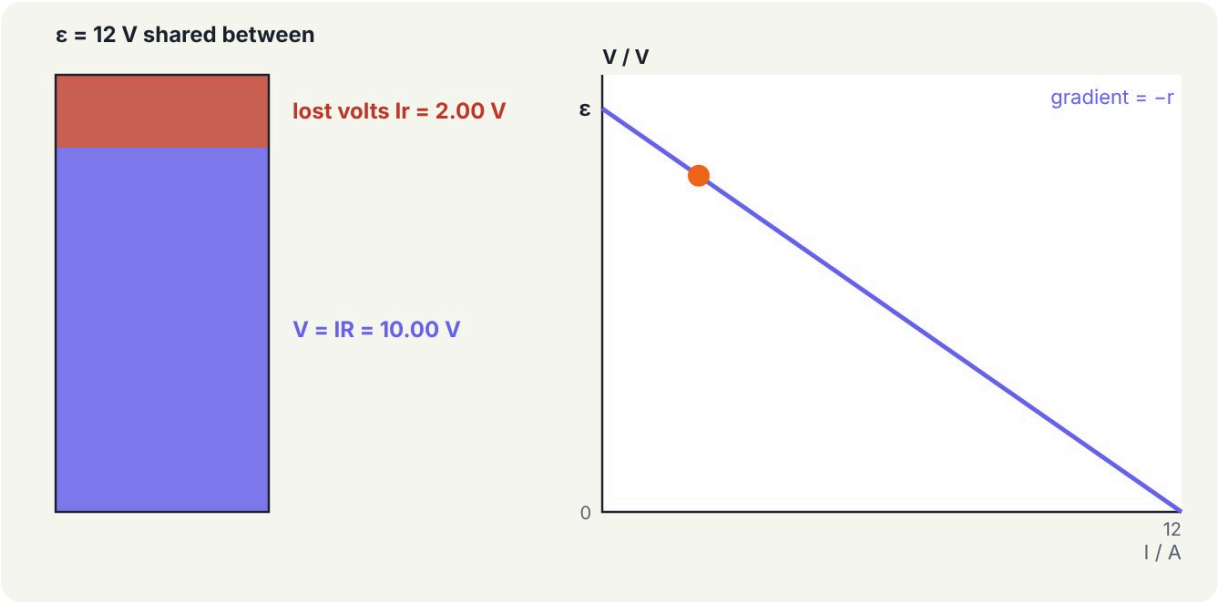


Figure 15.2: terminal p.d. against current: the intercept is ε and the gradient is −r.

Simulation

E.m.f. and internal resistance

Change the load resistance R and the internal resistance r. Watch the terminal p.d. fall as the current rises, and where the point lies on the V–I line.



current I = ε/(R + r)	terminal p.d. V	lost volts Ir	power to R	power wasted in r
2.00 A	**10.00 V**	**2.00 V**	**20.0 W**	**4.0 W**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 15 to use the simulation.

Example 15.1.1

A battery delivering 0.50 A has a terminal p.d. of 11.95 V and an internal resistance of 0.10 Ω. What is its e.m.f.?

Solution

ε = $V + Ir$ = 11.95 + (0.50)(0.10) = **12.0 V**

Example 15.1.2

A battery of e.m.f. 9.0 V is connected to a resistor. The current is 0.10 A and a voltmeter across the terminals reads 8.9 V. Find the internal resistance.

Solution

$$r = \frac{\varepsilon - V}{I} = \frac{9.0 - 8.9}{0.10} = \mathbf{1.0\ \Omega}$$

Example 15.1.3

When a cell gives a current of 0.20 A its terminal p.d. is 1.40 V; when it gives 0.60 A the terminal p.d. is 1.20 V. Find the e.m.f. and the internal resistance.

Solution

$1.40 = \varepsilon - 0.20r$ and $1.20 = \varepsilon - 0.60r$

Subtract: $0.20 = 0.40r \Rightarrow r = \mathbf{0.50\ \Omega}$; $\varepsilon = 1.40 + 0.10 = \mathbf{1.50\ V}$

Electrical energy and power

$$W = QV = VIt \qquad P = IV = I^2R = \frac{V^2}{R}$$

$P = IV$ for any device; I^2R and V^2/R only for a resistor (all the energy becomes heat).

Power supplied by the source = $I\varepsilon$.

Example 15.1.4

(a) Find the resistance of a 40 W, 12 V car headlamp. (b) A heater element of resistance 12 Ω carries 13 A. Find its power. (c) An iron with a 15 Ω element works at 120 V. How much heat does it produce in 1.0 h?

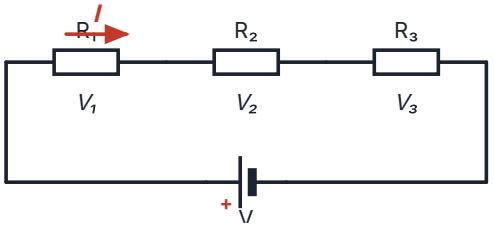
Solution

(a) $R = V^2/P = 12^2/40 =$ **3.6 Ω**

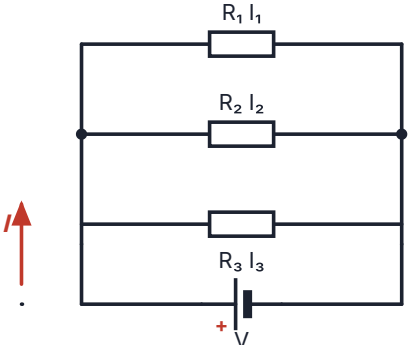
(b) $P = I^2R = (13)^2(12) =$ **2.0×10^3 W**

(c) $W = \dfrac{V^2}{R}t = \dfrac{120^2}{15}(3600) =$ **3.5×10^6 J**

Resistors in series and in parallel



Series: same current



Parallel: same p.d.

$$R_{eq} = R_1 + R_2 + R_3$$

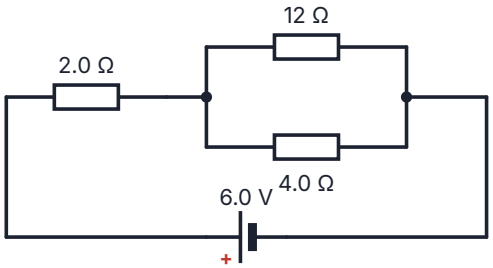
series: $V = V_1 + V_2 + V_3$ and $V = IR_{eq}$, so $IR_{eq} = IR_1 + IR_2 + IR_3$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

parallel: $I = I_1 + I_2 + I_3$ and each $I = V/R$, then divide by V

Example 15.1.5

Find (a) the total resistance, (b) the current from the 6.0 V supply, (c) the p.d. across the 4.0 Ω resistor.



Solution

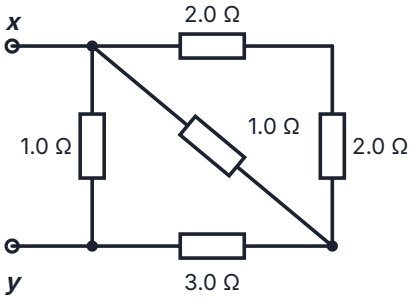
(a) 12 Ω ∥ 4.0 Ω: $\frac{(12)(4.0)}{16}$ = 3.0 Ω; R_{total} = 2.0 + 3.0 = **5.0 Ω**

(b) I = 6.0/5.0 = **1.2 A**

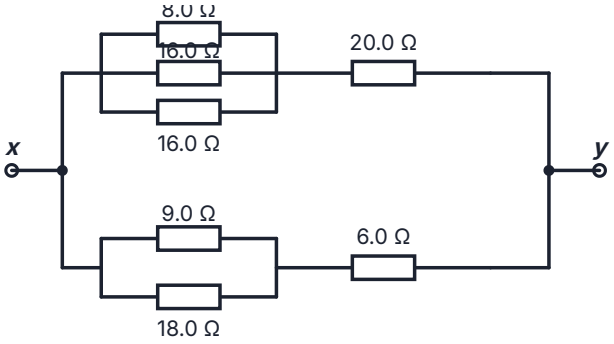
(c) $V_{2\Omega}$ = (1.2)(2.0) = 2.4 V, so $V_{4\Omega}$ = 6.0 − 2.4 = **3.6 V**

Example 15.1.6

Find the equivalent resistance between x and y in each network.



(a)



(b)

Solution

(a) 2.0 Ω + 2.0 Ω = 4.0 Ω (top and right side, in series) in parallel with the 1.0 Ω diagonal: 0.80 Ω; add the 3.0 Ω: 3.80 Ω; in parallel with the 1.0 Ω between x and y: $\frac{(3.80)(1.0)}{4.80}$ = **0.79 Ω**

(b) upper branch: 8.0 ‖ 16.0 ‖ 16.0 = 4.0 Ω, plus 20.0 Ω = 24.0 Ω. Lower branch: 9.0 ‖ 18.0 = 6.0 Ω, plus 6.0 Ω = 12.0 Ω. Together: $\frac{(24)(12)}{36}$ = **8.0 Ω**

15.2 Kirchhoff's laws

Kirchhoff's first law (junction rule)

The **sum of the currents entering a junction equals the sum of the currents leaving it** ($\Sigma I = 0$). It follows from the conservation of charge.

Kirchhoff's second law (loop rule)

In any closed loop, the **sum of the e.m.f.s equals the sum of the products IR** ($\Sigma \varepsilon = \Sigma IR$). It follows from the conservation of energy.



Figure 15.3: sign convention when going round a loop.

Mark a current (any direction you like) in each branch. Use the first law so the junctions balance.

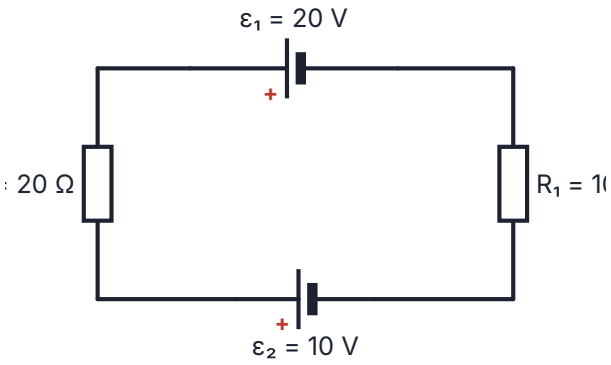
Choose a loop and a direction to go round it.

Apply $\Sigma\varepsilon = \Sigma IR$: an e.m.f. counts + if you pass through it from − to +; an IR counts + if you go the same way as the current you marked.

Solve the equations. A negative answer means that current really flows the other way; its size is still correct.

Example 15.2.1

Find the current in the circuit.



Solution

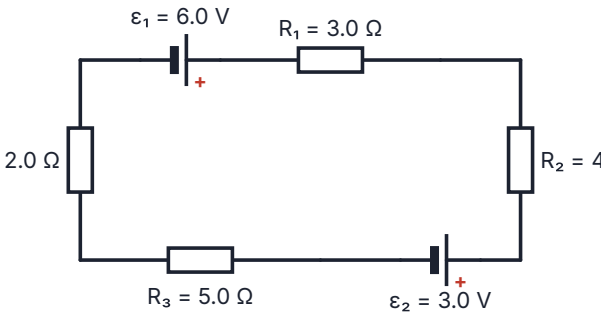
One loop, one current. The two cells oppose each other; ε_1 is larger, so take I anticlockwise (driven by ε_1).

$\Sigma\varepsilon = \Sigma IR$: $20 - 10 = I(20 + 10)$

$I = 10/30 =$ **0.33 A** in each resistor

Example 15.2.2

Find the current in each resistor.



Solution

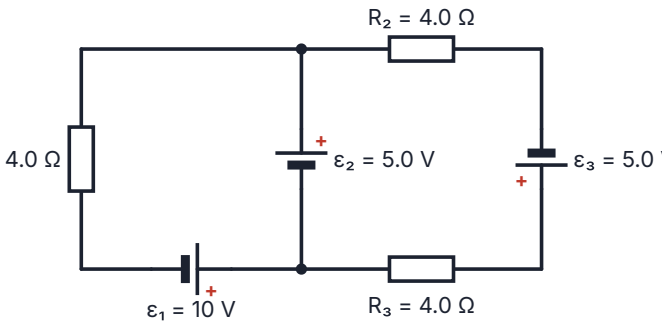
One loop. Take I clockwise. ε_1 is passed from − to + (+6.0 V); ε_2 is passed from + to − (−3.0 V).

$6.0 - 3.0 = I(3.0 + 4.0 + 5.0 + 2.0) = 14I$

I = **0.21 A**, clockwise, the same in every resistor

Example 15.2.3

Find the current in each branch.



Solution

Let I_1 flow through R_1 and ε_1, I_3 through R_2, ε_3 and R_3, and I_2 up through ε_2. Junction: $I_2 = I_1 + I_3$.

Left loop (ε_1, ε_2, R_1): $10 + 5.0 = 4.0 I_1 \Rightarrow I_1 =$ **3.75 A**

Right loop (ε_2, ε_3, R_2, R_3): $5.0 + 5.0 = (4.0 + 4.0) I_3 \Rightarrow I_3 =$ **1.25 A**

$I_2 = 3.75 + 1.25 =$ **5.0 A**

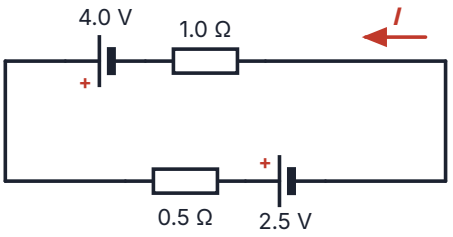
Example 15.2.4

A cell of e.m.f. 4.0 V and internal resistance 1.0 Ω is connected in series with a cell of e.m.f. 2.5 V and internal resistance 0.5 Ω in a closed loop, so that the current is as small as possible. Draw the circuit and find the current.

Solution

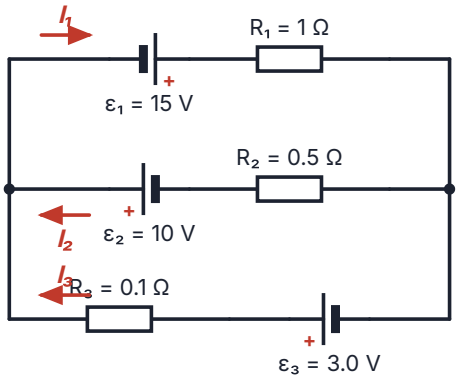
The current is smallest when the cells **oppose** each other: connect + to + (and − to −).

$$I = \frac{4.0 - 2.5}{1.0 + 0.5} = \mathbf{1.0\ A}$$



Example 15.2.5

Find I_1, I_2 and I_3. Ignore the internal resistances.



Solution

Junction (right-hand node): $I_1 = I_2 + I_3$ ... (1)

Top loop (branches 1 and 2): $15 + 10 = (1)I_1 + (0.5)I_2 \Rightarrow 25 = I_1 + 0.5I_2$ ... (2)

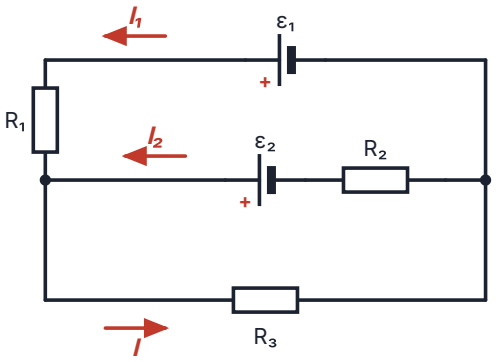
Bottom loop (branches 2 and 3): $3.0 - 10 = 0.1I_3 - 0.5I_2 \Rightarrow -7 = 0.1I_3 - 0.5I_2$ ... (3)

From (3): $I_3 = 5I_2 - 70$. Put into (1) and (2): $25 = 1.5I_2 + 5I_2 - 70 \Rightarrow I_2 =$ **14.6 A**

$I_3 = 5(14.62) - 70 =$ **3.1 A**; $I_1 = 14.62 + 3.08 =$ **17.7 A**

Example 15.2.6

ε_1 = 8 V, R_1 = 1 Ω, R_2 = 2 Ω, R_3 = 3 Ω and I = 3 A. Ignore the internal resistances. Find (a) I_1 and I_2, (b) ε_2.



Solution

Loop through ε_1, R_1 and R_3: with the directions marked, $0 = 1(I_1) - 8 + 3(3) \Rightarrow I_1 = -1$ A.

(a) I_1 **= 1 A**, flowing in the **opposite** direction to the arrow. Junction: $I = I_1 + I_2 \Rightarrow 3 = -1 + I_2 \Rightarrow I_2$ **= 4 A**

(b) Loop through ε_2, R_2 and R_3: $\varepsilon_2 = I_2R_2 + IR_3 = (4)(2) + (3)(3) =$ **17 V**

15.3 Potential divider

Two resistors in series across a supply share the supply voltage in the ratio of their resistances. Taking the output across one of them gives a smaller, chosen voltage.

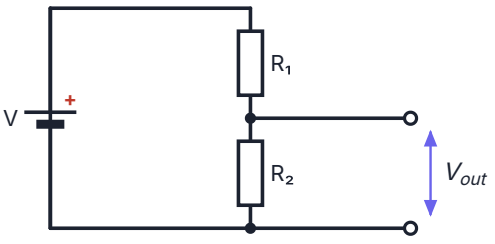


Figure 15.4: a potential divider.

$$V_{\text{out}} = \frac{R_2}{R_1 + R_2} V$$

the same current $I = V/(R_1 + R_2)$ flows through both resistors (no load connected)

A source of variable voltage

If the two resistors are replaced by one uniform resistance wire (or a rheostat) with a sliding contact, the output can be varied smoothly from 0 to the full supply voltage. For a uniform wire, $R \propto l$, so $V_{\text{out}} = \frac{l}{L} V$.

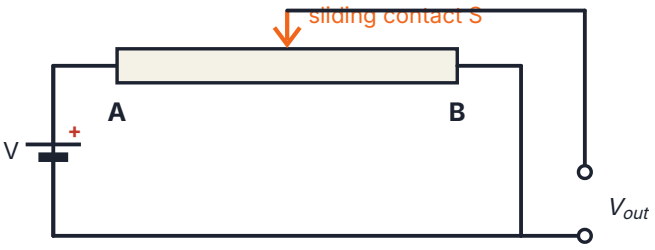


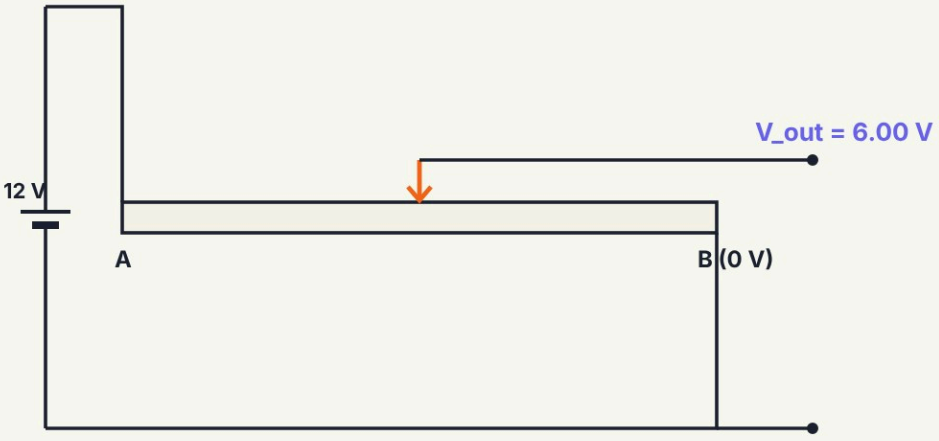
Figure 15.5: a rheostat used as a potential divider (potentiometer connection).

If a load is connected across the output, it is in parallel with R_2. The combined resistance is smaller, so V_{out} falls. The output only stays near the calculated value if the load resistance is much larger than R_2.

Simulation

Potential divider

Move the sliding contact to change the output voltage. Then connect a load and make it smaller: watch the output fall below the value without a load.



output (no load)	output now	lower part of the rheostat	load
6.00 V	6.00 V	0.50 kΩ (total 1 kΩ)	not connected

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 15 to use the simulation.

Example 15.3.1

Resistors of 3.0 Ω and 6.0 Ω are connected in series to a 12.0 V battery of negligible internal resistance. Find the p.d. across each resistor.

Solution

$V_3 = \frac{3.0}{9.0}(12.0) = \mathbf{4.0\ V}$; $V_6 = \frac{6.0}{9.0}(12.0) = \mathbf{8.0\ V}$

Example 15.3.2

A 12 V supply is connected across two 10 kΩ resistors in series. Find the output across the lower resistor (a) with no load, (b) with a 10 kΩ load connected across it.

Solution

(a) $V_{out} = ½(12) = \mathbf{6.0\ V}$

(b) 10 kΩ ∥ 10 kΩ = 5.0 kΩ: $V_{out} = \frac{5.0}{10 + 5.0}(12) = \mathbf{4.0\ V}$

Shunts and multipliers

A galvanometer has a resistance R_g and gives full-scale deflection for a small current I_g. It can be converted into an ammeter or a voltmeter.



Figure 15.6: a shunt in parallel makes an ammeter; a multiplier in series makes a voltmeter.

Ammeter: shunt

- A small resistance R_s is connected in **parallel** with the galvanometer so that most of the current bypasses it.
- Same p.d.: $I_g R_g = (I - I_g) R_s$

$$R_s = \frac{I_g R_g}{I - I_g}$$

an ammeter should have a very small resistance

Voltmeter: multiplier

- A large resistance R_m is connected in **series** with the galvanometer.
- At full scale: $V = I_g(R_g + R_m)$

$$R_m = \frac{V}{I_g} - R_g$$

a voltmeter should have a very large resistance

Example 15.3.3

A galvanometer of resistance 50 Ω gives full-scale deflection for 1.0 mA. How can it be converted into (a) an ammeter reading up to 1.0 A, (b) a voltmeter reading up to 10 V?

Solution

(a) $R_s = \frac{(1.0 \times 10^{-3})(50)}{1.0 - 0.0010}$ = **0.050 Ω**, connected in parallel

(b) $R_m = \frac{10}{1.0 \times 10^{-3}} - 50$ = **9950 Ω**, connected in series

15.4 Potentiometer and Wheatstone bridge

Potentiometer

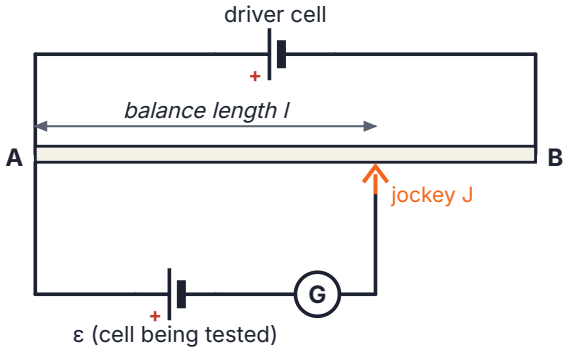


Figure 15.7: a potentiometer measuring the e.m.f. of a cell.

- A **driver cell** sends a steady current through a uniform resistance wire AB (often 100 cm long). The p.d. along the wire is proportional to length: $V_{AJ} \propto l$.

- The cell being tested is connected with its + terminal to A (the same side as the + of the driver cell), through a galvanometer to a sliding **jockey** J.
- The jockey is moved until the galvanometer shows **no deflection** (balance). Then no current flows through the test cell, so the p.d. across length l equals its **e.m.f.** exactly.

$$\varepsilon = \frac{l}{L} V_{AB} \qquad \frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2} \qquad r = R\frac{l_0 - l}{l}$$

measuring an e.m.f.; comparing two e.m.f.s; internal resistance (l_0 = balance length on open circuit, l = balance length with a resistor R across the cell)

- **Uses:** measuring an e.m.f. (no current is drawn, unlike a voltmeter), comparing e.m.f.s, measuring internal resistance, comparing resistances.
- **No balance point** (deflection one way all along the wire)? The e.m.f. being measured is larger than the p.d. across the whole wire, or the cells are connected + to −, or a connection is loose.

Wheatstone bridge

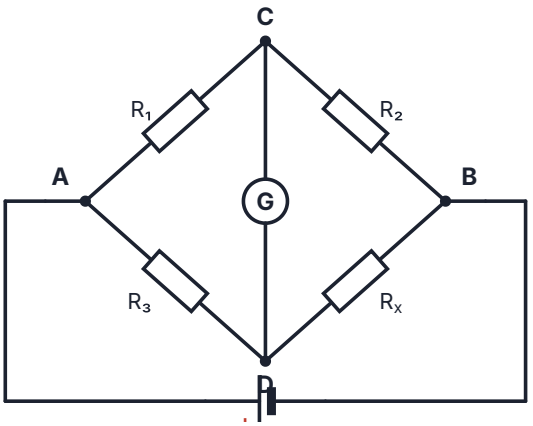


Figure 15.8: a Wheatstone bridge.

Four resistors form a bridge with a galvanometer between C and D. One of them, R_x, is unknown.

A known resistor is adjusted until the galvanometer reads zero: the bridge is **balanced**, so C and D are at the same potential.

No current in CD means R_1 and R_2 carry the same current I_1, and R_3 and R_x carry I_2.

$V_{AC} = V_{AD}$: $I_1R_1 = I_2R_3$; $V_{CB} = V_{DB}$: $I_1R_2 = I_2R_x$. Divide the two equations.

$$\frac{R_1}{R_2} = \frac{R_3}{R_x} \Rightarrow R_x = \frac{R_2 R_3}{R_1}$$

balance does not depend on the e.m.f. of the cell or on the galvanometer resistance

Metre bridge

A practical Wheatstone bridge: two arms are the two parts of a uniform wire 100 cm long. At balance, the ratio of resistances equals the ratio of lengths.

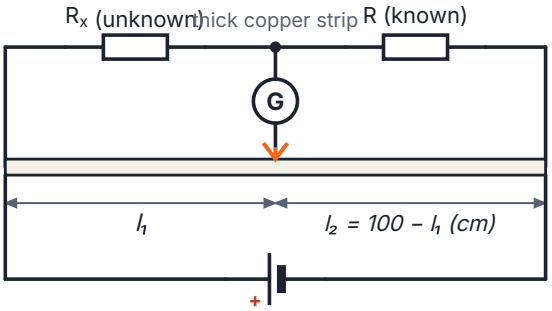


Figure 15.9: a metre bridge.

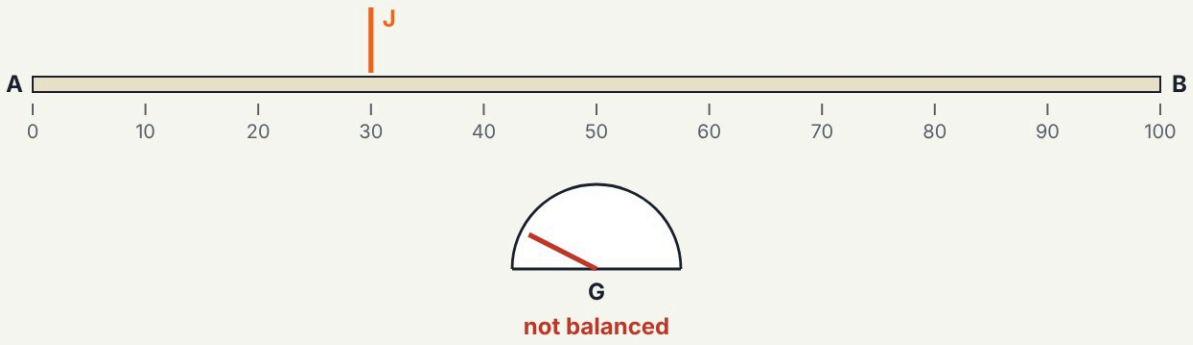
$$\frac{R_x}{R} = \frac{l_1}{l_2}$$

best accuracy when the balance point is near the middle of the wire

Simulation Potentiometer and metre bridge

Slide the jockey along the wire until the galvanometer reads zero. Read the balance length and use it to find the unknown e.m.f. or the unknown resistance.

p.d. across AB = 2.00 V; unknown cell ε (hidden)



balance length l	ε = (l/100) × 2.00 V	hint
—	**find the balance first**	**move J towards B**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 15 to use the simulation.

Example 15.4.1

A potentiometer wire 100.0 cm long has a p.d. of 2.00 V across it. A cell balances at 75.0 cm. Find its e.m.f.

Solution

$$\varepsilon = \frac{75.0}{100.0}(2.00) = \mathbf{1.50\ V}$$

Example 15.4.2

A standard cell of e.m.f. 1.018 V balances at 60.0 cm on a potentiometer. Another cell balances at 72.0 cm. Find its e.m.f.

Solution

$$\varepsilon_2 = \varepsilon_1 \frac{l_2}{l_1} = 1.018 \times \frac{72.0}{60.0} = \mathbf{1.22\ V}$$

Example 15.4.3

A cell balances at 80.0 cm on open circuit. When a 10 Ω resistor is connected across it, the balance length becomes 64.0 cm. Find the internal resistance of the cell.

Solution

$$r = R\frac{l_0 - l}{l} = 10 \times \frac{80.0 - 64.0}{64.0} = \mathbf{2.5\ \Omega}$$

(Open circuit: $\varepsilon \propto l_0$. With R: $V = IR \propto l$. Since $\varepsilon/V = (R + r)/R = l_0/l$.)

Example 15.4.4

A platinum wire of diameter 0.920 mm is the unknown resistance in a Wheatstone bridge. R_1 = 38.0 Ω and R_2 = 46.0 Ω, and balance is reached when R_3 = 3.48 Ω. Find the length of the wire. ($\rho = 10.6 \times 10^{-8}$ Ω m)

Solution

$$R_x = \frac{R_2 R_3}{R_1} = \frac{(46.0)(3.48)}{38.0} = 4.21\ \Omega$$

$$l = \frac{R_x A}{\rho} = \frac{(4.21)\pi(0.920 \times 10^{-3})^2/4}{10.6 \times 10^{-8}} = \mathbf{26.4\ m}$$

Example 15.4.5

In a metre bridge the unknown resistor is in the left gap and a 10.0 Ω resistor in the right gap. Balance is at 40.0 cm from the left end. Find the unknown resistance.

Solution

$$R_x = R\frac{l_1}{l_2} = 10.0 \times \frac{40.0}{60.0} = \mathbf{6.67\ \Omega}$$

Exercise: d.c. circuits

1. A battery of e.m.f. 12 V and internal resistance 0.50 Ω is connected to a 5.5 Ω resistor. Find (a) the current, (b) the terminal p.d., (c) the power lost in the battery.

   Answer

   (a) 2.0 A (b) 11 V (c) I^2r = 2.0 W

2. Why does a potentiometer measure the e.m.f. of a cell more accurately than a voltmeter?

   Answer

   At balance no current flows through the cell, so there is no p.d. across its internal resistance. A voltmeter draws a small current, so it reads ε − Ir, slightly less than ε.

3. A galvanometer of resistance 20 Ω gives full-scale deflection with 5.0 mA. Find the resistance needed to make (a) a 0–5.0 A ammeter, (b) a 0–15 V voltmeter, and say how each is connected.

   Answer

   (a) 0.020 Ω in parallel (b) 2980 Ω in series

Chapter 16 · Semester 2

MAGNETIC FIELDS

Notes by Cikgu Suhaili

19 worked examples 5 simulations Exercises with answers

16.1 Magnetic field

Magnetic field

a **region of space where a magnetic force is experienced** by a magnetic material, a current-carrying conductor or a moving charge. It is a field of force produced by **permanent magnets or by current-carrying conductors**.

- A magnet has two poles, north (N) and south (S). **Like poles repel, unlike poles attract.**
- The direction of the field at a point is the direction in which the **north pole of a compass needle** points when placed there.
- Field patterns are shown with iron filings or a plotting compass.

Magnetic field lines

- Field lines go **from N to S outside** a magnet (and S to N inside it), so they form closed loops.
- The tangent to a line at a point gives the direction of the field there.
- Field lines **never cross** (the field has only one direction at each point).
- The closer the lines, the stronger the field: the number of lines per unit area is proportional to the field strength.



Figure 16.1: a uniform field and a non-uniform field.



Figure 16.2: a field perpendicular to the page.

Field patterns

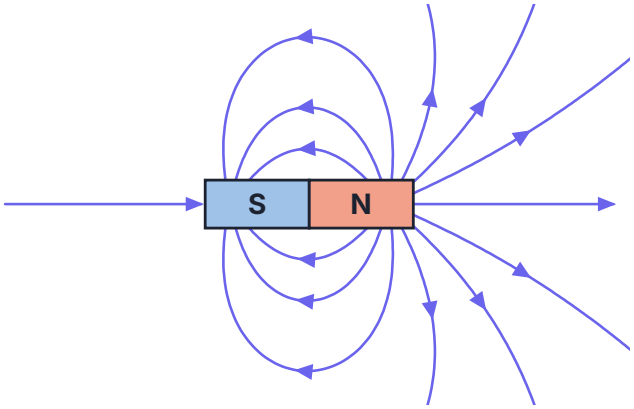


Figure 16.3: a bar magnet.

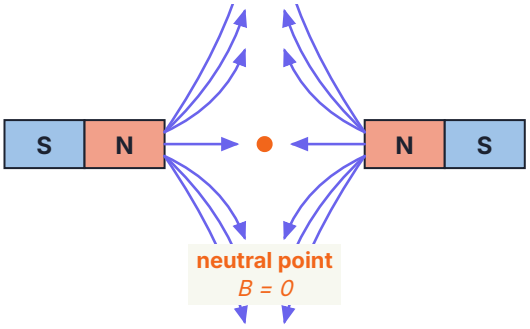


Figure 16.4: two like poles facing each other. At the neutral point the two fields cancel.

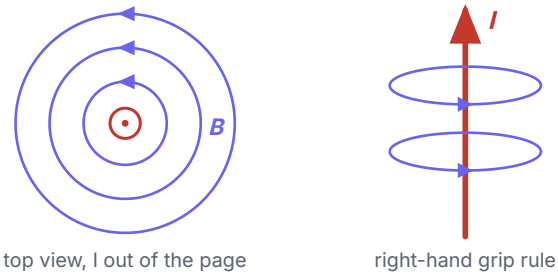


Figure 16.5: a long straight wire: concentric circles. Right-hand grip rule.

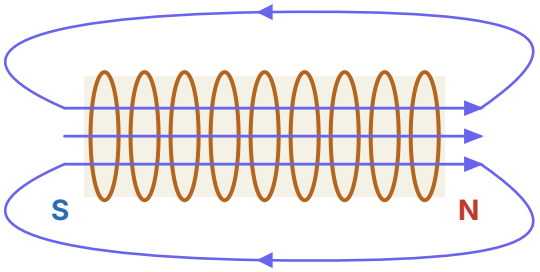


Figure 16.6: a solenoid: like a bar magnet outside, uniform inside.

- **Long straight wire**: concentric circles centred on the wire, further apart away from the wire (weaker field). Direction: **right-hand grip rule**, thumb along the current, fingers curl in the direction of B.
- **Flat circular coil**: near the centre the lines are almost straight and perpendicular to the coil; the coil behaves like a short magnet.
- **Solenoid**: outside, like a bar magnet; inside, the field is **uniform** and parallel to the axis. The end where the current flows anticlockwise (seen from outside) is a north pole.
- The **Earth** has a field like a large bar magnet whose south pole is near the geographic north pole, so a compass needle points north.

Teacher's note

Magnetic flux Φ = **B**·**A** is in Chapter 17 (electromagnetic induction), where the STPM syllabus puts it.

16.2 Force on a moving charge

A charge q moving with velocity **v** in a magnetic field **B** feels a force that is **perpendicular to both v and B**.

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B} \qquad F = qvB\sin\theta$$

θ = angle between v and B

Magnetic flux density, B

defined by $F = qvB\sin\theta$: $B = \dfrac{F}{qv\sin\theta}$. One tesla is the flux density in which **a charge of 1 C moving at 1 m s^{-1} perpendicular to the field feels a force of 1 N**.

- Unit: tesla (T) = N A^{-1} m^{-1} = Wb m^{-2}. B is a vector, along the field lines.
- $F = 0$ when the charge is at rest ($v = 0$) or moves parallel to B (θ = 0° or 180°).
- F is greatest, qvB, when the charge moves perpendicular to B.

Direction: Fleming's left-hand rule

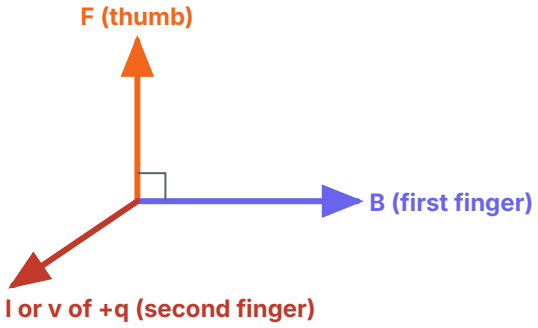


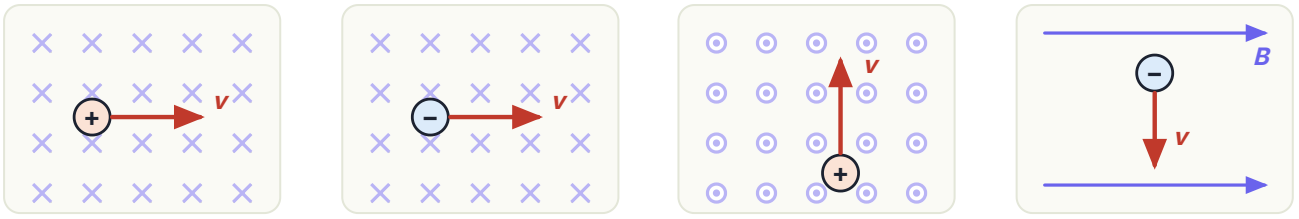
Figure 16.7: Fleming's left-hand rule. For a positive charge the second finger points along v; for a negative charge (electron) it points opposite to v.

Teacher's note

Some notes use the **right** hand for a negative charge. It gives the same answer, but it is easy to mix up with Fleming's right-hand rule for induction in Chapter 17. Safer: always use the left hand, with the second finger along the **conventional current** (opposite to the velocity of an electron).

Example 16.2.1

Find the direction of the magnetic force on the charge in each case.



(a) (b) (c) (d) electron

Solution

(a) $+q$ moving right, B into the page: **upwards** (towards the top of the page).

(b) $-q$ moving right, B into the page: the force is opposite to (a), **downwards**.

(c) $+q$ moving up, B out of the page: **to the right**.

(d) electron moving down, B to the right: for a positive charge it would be out of the page, so for the electron **into the page**.

Example 16.2.2

A proton travels at 3.1×10^7 m s^{-1} in a uniform field of 1.6 T. Find the force on it if its velocity (a) is perpendicular to the field, (b) makes 60° with the field. (e = 1.60×10^{-19} C)

Solution

(a) $F = qvB = (1.60 \times 10^{-19})(3.1 \times 10^7)(1.6) =$ **7.9×10^{-12} N**

(b) $F = qvB \sin 60° = (7.94 \times 10^{-12})(0.866) =$ **6.9×10^{-12} N**

Example 16.2.3

A charge q_1 = 25.0 μC moves at 4.5 × 10^3 m s^{-1} perpendicular to a uniform field and feels a force of 7.31 × 10^{-3} N. A second charge q_2 = 5.00 μC moves at 40.0° to the same field and feels 1.90 × 10^{-3} N. Find (a) B, (b) the speed of q_2.

Solution

(a) $B = \dfrac{F_1}{q_1 v_1} = \dfrac{7.31 \times 10^{-3}}{(25.0 \times 10^{-6})(4.5 \times 10^3)} = \mathbf{6.50 \times 10^{-2}\ T}$

(b) $v_2 = \dfrac{F_2}{q_2 B \sin 40.0°} = \dfrac{1.90 \times 10^{-3}}{(5.00 \times 10^{-6})(6.50 \times 10^{-2})(0.643)} = \mathbf{9.10 \times 10^3\ m\,s^{-1}}$

Example 16.2.4

An electron moves from left to right at 2.0 × 10^6 m s^{-1} into a field of 3.0 T directed vertically downwards (in the plane of the page). Find the magnitude and direction of the force on it.

Solution

$F = evB = (1.60 \times 10^{-19})(2.0 \times 10^6)(3.0) = \mathbf{9.6 \times 10^{-13}\ N}$

For a positive charge the force would be into the page, so on the electron it is **out of the page**.

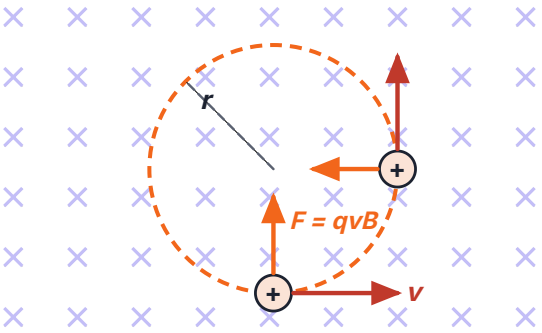
Motion parallel to the field

If **v** is parallel (or antiparallel) to **B**, sin θ = 0, so **there is no magnetic force**. The charge carries on in a **straight line at constant speed**.

Motion perpendicular to the field

The force is always perpendicular to the velocity, so it does **no work**: the speed and kinetic energy stay constant, only the direction changes.

A force of constant size always perpendicular to the velocity gives **uniform circular motion**. The magnetic force provides the centripetal force:



B into the page; a positive charge goes round anticlockwise

Figure 16.8: circular motion of a charge in a uniform magnetic field.

$$qvB = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB} \qquad T = \frac{2\pi r}{v} = \frac{2\pi m}{qB} \qquad f = \frac{qB}{2\pi m}$$

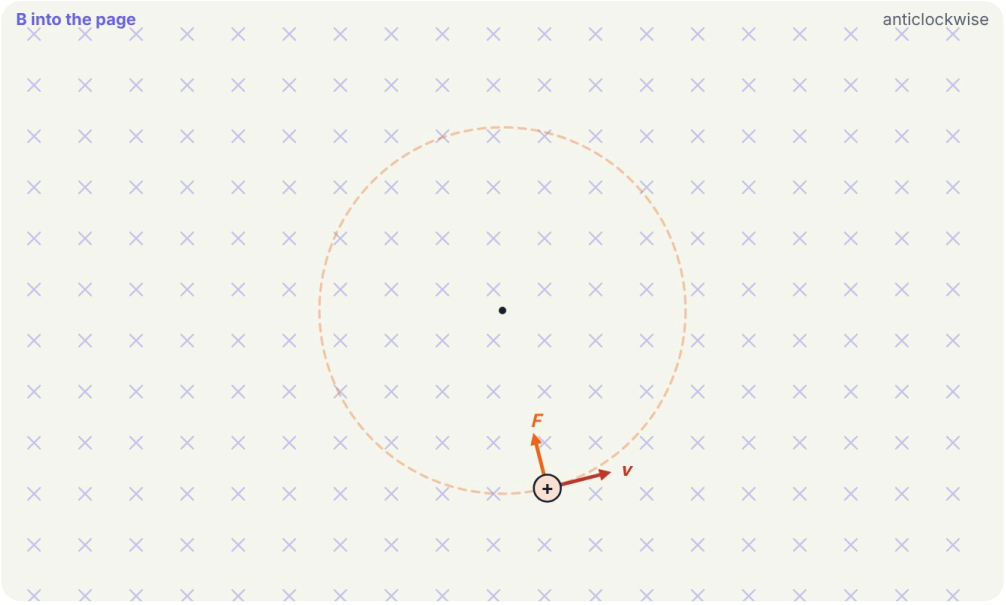
- A faster particle moves in a **larger** circle, but the **period does not depend on the speed**.
- A positive and a negative charge with the same speed curve in opposite directions.

Teacher's note

If v makes any other angle with B, the component of v along B is unchanged and the perpendicular component goes round in a circle, so the path is a helix (a spiral along B).

Simulation — A charge in a uniform magnetic field

Choose the sign of the charge, its speed and the flux density. Perpendicular to B it moves in a circle: watch the radius change with v and B while the time for one turn depends only on B. Then switch to "parallel to B".



radius r = mv/(qB)	period T = 2πm/(qB)	magnetic force qvB	speed
13.9 cm	**1.093 μs**	**7.68×10^{-15} N**	**constant (F ⊥ v)**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 16 to use the simulation.

Example 16.2.5

A proton moves at 3.0×10^5 m s^{-1} perpendicular to a field of 0.020 T. Find (a) its kinetic energy, (b) the magnetic force on it, (c) the radius of its path, (d) the period. ($m_p = 1.67 \times 10^{-27}$ kg)

Solution

(a) $K = \tfrac{1}{2}mv^2 = \tfrac{1}{2}(1.67 \times 10^{-27})(3.0 \times 10^5)^2 = \mathbf{7.5 \times 10^{-17}\ J}$

(b) $F = qvB = (1.60 \times 10^{-19})(3.0 \times 10^5)(0.020) = \mathbf{9.6 \times 10^{-16}\ N}$

(c) $$r = \frac{mv}{qB} = \frac{(1.67 \times 10^{-27})(3.0 \times 10^5)}{(1.60 \times 10^{-19})(0.020)} = \mathbf{0.16\ m}$$

(d) $$T = \frac{2\pi m}{qB} = \frac{2\pi(1.67 \times 10^{-27})}{(1.60 \times 10^{-19})(0.020)} = \mathbf{3.3 \times 10^{-6}\ s}$$

Exercise: force on a moving charge

1. An electron enters a uniform field of 0.50 mT at 4.0×10^6 m s^{-1}, perpendicular to the field. Find the radius of its path. ($m_e = 9.11 \times 10^{-31}$ kg)

   Answer

   $r = mv/(eB)$ = **4.6 cm**

2. Explain why a magnetic field cannot change the kinetic energy of a moving charge.

   Answer

   The force is always perpendicular to the velocity, so it does no work. It changes the direction of motion but not the speed.

3. A proton and an alpha particle (charge $2e$, mass $4m$) enter the same field perpendicular to it with the same speed. Find the ratio of the radii $r_\alpha : r_p$.

   Answer

   $r \propto m/q$: (4/2) : 1 = **2 : 1**

16.3 Force on a current-carrying conductor

A current is a stream of moving charges. Each charge feels a force qvB sin θ, and the forces on all the charges in the wire add up to a force on the wire.

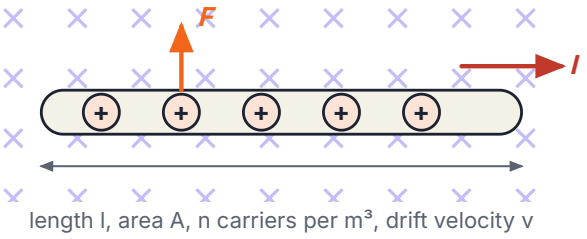


Figure 16.9: a length l of wire in a uniform field.

Deriving F = IlB sin θ

Number of carriers in the length l: $N = nAl$.

Force on each carrier: qvB sin θ, so the total force is $F = (nAl)qvB$ sin θ.

The current is $I = nAvq$ (Chapter 14), so $F = (nAvq)lB$ sin θ:

$$F = IlB\sin\theta \qquad \mathbf{F} = \boldsymbol{l} \times \mathbf{B}$$

θ = angle between the wire (current) and B. F = 0 when the wire is parallel to B; F = IlB when perpendicular.

- Direction: **Fleming's left-hand rule** with the second finger along the current.
- This is the principle of the electric motor and the moving-coil meter.

Example 16.3.1

A vertical wire is in a uniform field directed into the page. Find the direction of the force on it when the current flows (a) upwards, (b) downwards.



Solution

(a) **to the left** (b) **to the right** (reversing the current reverses the force)

Example 16.3.2

A wire 0.655 m long carries 21.0 A in a 0.470 T field and feels a force of 5.46 N. Find the angle (less than 90°) between the wire and the field.

Solution

$$\sin\theta = \frac{F}{IlB} = \frac{5.46}{(21.0)(0.655)(0.470)} = 0.845$$

$\theta = \mathbf{57.6°}$

Example 16.3.3

A square loop of side 0.32 m carries 12 A. It lies in the plane of a uniform 0.25 T field, with two sides perpendicular to the field and two sides parallel to it. Find the force on each side.

Solution

Sides perpendicular to B: $F = IlB$ = (12)(0.32)(0.25) = **0.96 N** each (in opposite directions)

Sides parallel to B: sin 0° = 0, so **no force**.

16.4 Magnetic fields due to currents

Ampère's law

the **line integral of B around any closed loop is equal to μ_0 times the current enclosed by the loop**.

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

μ_0 = permeability of free space = $4\pi \times 10^{-7}$ H m^{-1} (T m A^{-1})

Applying Ampère's law to a long straight wire

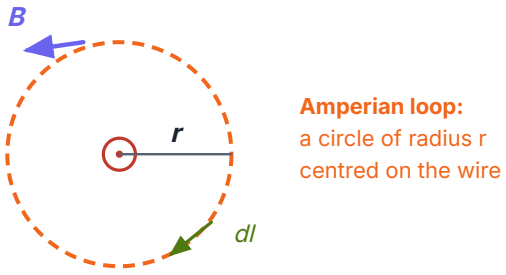


Figure 16.10: a circular Amperian loop around a long straight wire.

Choose a circle of radius r centred on the wire, in a plane perpendicular to it.

By symmetry B has the same size at every point on the circle and is tangential, so it is parallel to d**l** everywhere: **B**·d**l** = $B\,dl$.

$\oint B\,dl = B(2\pi r) = \mu_0 I$

$$B = \frac{\mu_0 I}{2\pi r}$$

long straight wire; B ∝ 1/r

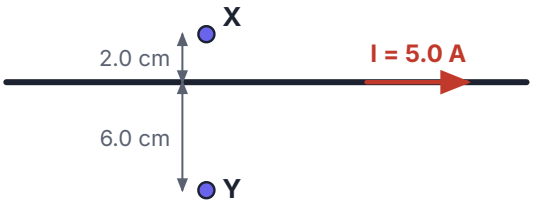
Circular coil and solenoid

Conductor	Flux density	Notes
Long straight wire	$B = \frac{\mu_0 I}{2\pi r}$	r = perpendicular distance from the wire
Flat circular coil, at its centre	$B = \frac{\mu_0 N I}{2r}$	N turns of radius r
Long solenoid, inside	$B = \mu_0 n I$	$n = \frac{N}{L}$ = turns per unit length

With a core of relative permeability μ_r, replace μ_0 by $\mu = \mu_r \mu_0$. A soft iron core makes the field of a solenoid much stronger.

Example 16.4.1

A long straight wire carries 5.0 A to the right. Point X is 2.0 cm above the wire and point Y is 6.0 cm below it. Find the flux density at X and at Y.



Solution

$$B_X = \frac{(4\pi \times 10^{-7})(5.0)}{2\pi(0.020)} = \mathbf{5.0 \times 10^{-5}\ T,\ out\ of\ the\ page}$$

$$B_Y = \frac{(4\pi \times 10^{-7})(5.0)}{2\pi(0.060)} = \mathbf{1.7 \times 10^{-5}\ T,\ into\ the\ page}\ \text{(right-hand grip rule)}$$

Example 16.4.2

A flat circular coil of 40 turns and radius 6.0 cm is in air. What current gives a flux density of 2.0 mT at its centre?

Solution

$$I = \frac{2rB}{\mu_0 N} = \frac{2(0.060)(2.0 \times 10^{-3})}{(4\pi \times 10^{-7})(40)} = \mathbf{4.8\ A}$$

Example 16.4.3

An air-cored solenoid has 2000 turns, length 60 cm and diameter 2.0 cm. Find the flux density inside it when the current is 5.0 A.

Solution

$$B = \mu_0 \frac{N}{L} I = (4\pi \times 10^{-7}) \frac{2000}{0.60} (5.0) = \mathbf{0.021\ T}$$ (the diameter is not needed)

Example 16.4.4

A solenoid of 400 turns is wound on an iron core 20 cm long with relative permeability 13 000. What current gives 0.50 T at its centre?

Solution

$$\mu = \mu_r \mu_0 = (13\,000)(4\pi \times 10^{-7}) = 1.63 \times 10^{-2}\ \text{H m}^{-1}$$

$$I = \frac{BL}{\mu N} = \frac{(0.50)(0.20)}{(1.63 \times 10^{-2})(400)} = \mathbf{0.015\ A}$$

Exercise: fields due to currents

1. Two long straight parallel wires are 30 cm apart and each carries 20 A. Find the flux density at a point between them, in their plane, 10 cm from one wire and 20 cm from the other, when the currents are (a) in the same direction, (b) in opposite directions.

Answer

The fields of the wires are 4.0×10^{-5} T and 2.0×10^{-5} T. (a) They are opposite: 2.0×10^{-5} T, in the direction of the nearer wire's field. (b) They add: 6.0×10^{-5} T. Both perpendicular to the plane of the wires.

2. How far from a long straight wire carrying 12 A is the flux density 3.0×10^{-5} T?

Answer

$r = \mu_0 I/(2\pi B)$ = **8.0 cm**

16.5 Force between two current-carrying conductors

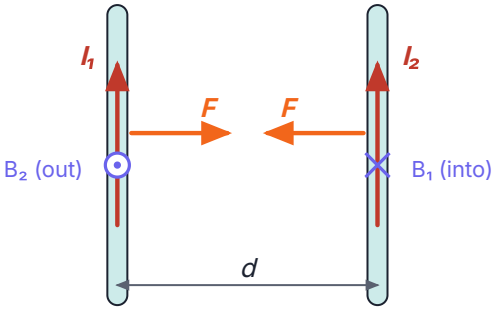


Figure 16.11: two long parallel wires a distance d apart.

Derivation

Wire 1 produces a field at wire 2: $B_1 = \dfrac{\mu_0 I_1}{2\pi d}$, perpendicular to wire 2.

A length l of wire 2 feels $F = B_1 I_2 l \sin 90°$.

So $F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}$. Working the other way (field of wire 2 at wire 1) gives a force of the same size on wire 1, in the opposite direction (Newton's third law).

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi d} \qquad \frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

currents in the **same direction attract**; in **opposite directions repel**

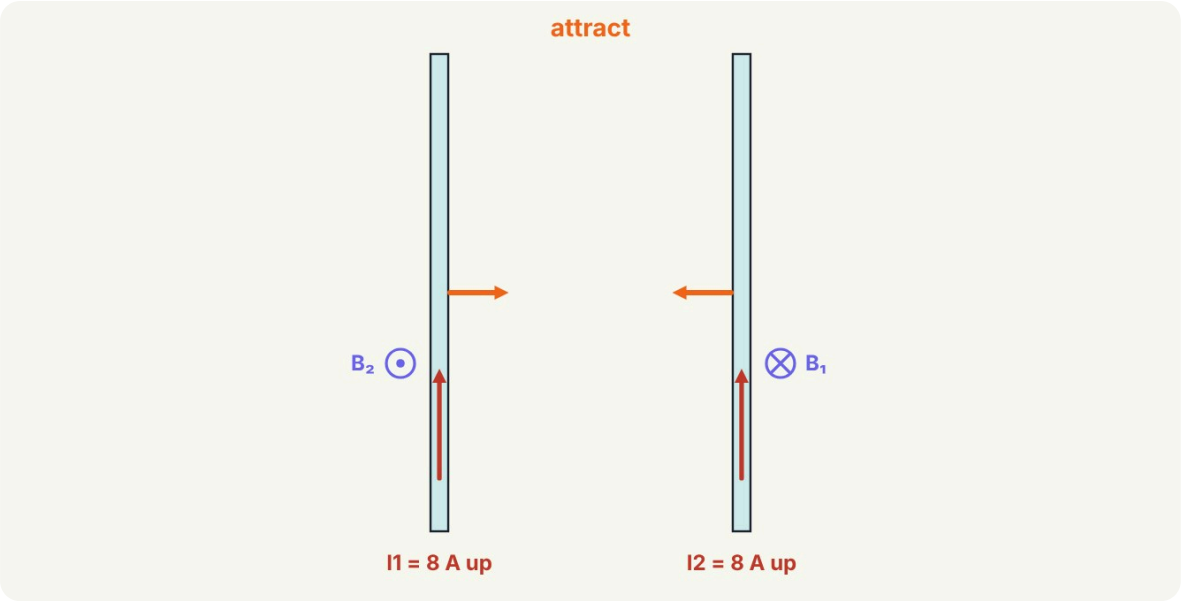
- With $I_1 = I_2 = 1$ A and $d = 1$ m: $F/l = 2 \times 10^{-7}$ N m^{-1}. This was used to define the ampere: the constant current which, in each of two infinitely long, straight, parallel conductors of negligible cross-section 1 m apart in a vacuum, gives a force of 2×10^{-7} N per metre of length.

Teacher's note

Since 2019 the SI defines the ampere by fixing the value of the elementary charge, but this result is still true and is a favourite exam question.

Simulation Two parallel currents

Change the size and direction of each current and the separation. See the field of each wire at the other one and the forces: same direction attract, opposite directions repel. The forces are always equal and opposite.



B_1 at wire 2 = $\mu_0 I_1/(2\pi d)$	force per metre $F/l = \mu_0 I_1 I_2/(2\pi d)$	B_2 at wire 1	forces on the wires
4.00×10^{-5} T	**3.20×10^{-4} N m^{-1}**	**4.00×10^{-5} T**	**equal and opposite**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 16 to use the simulation.

Example 16.5.1

Two long parallel wires 2.0 cm apart in air carry 8.0 A and 10 A. Find the force on each metre of wire.

Solution

$$\frac{F}{l} = \frac{(4\pi \times 10^{-7})(8.0)(10)}{2\pi(0.020)} = \mathbf{8.0 \times 10^{-4}\ N\ m^{-1}}$$

Example 16.5.2

Two long parallel wires 5.0 cm apart attract each other with a force of 6.0×10^{-7} N per metre. One carries 400 mA. Find (a) the current in the other, (b) its direction relative to the first.

Solution

(a) $I_2 = \dfrac{(F/l)(2\pi d)}{\mu_0 I_1} = \dfrac{(6.0 \times 10^{-7})(2\pi)(0.050)}{(4\pi \times 10^{-7})(0.400)} = \mathbf{0.375\ A}$

(b) They attract, so the currents are in the **same direction**.

16.6 Determination of the ratio e/m

Crossed electric and magnetic fields

When **E**, **B** and **v** are perpendicular to each other, the electric and magnetic forces on a charge are in opposite directions. If they are equal, the resultant force is zero and the charge goes **straight through, undeflected**.

$$qE = qvB \Rightarrow v = \frac{E}{B}$$

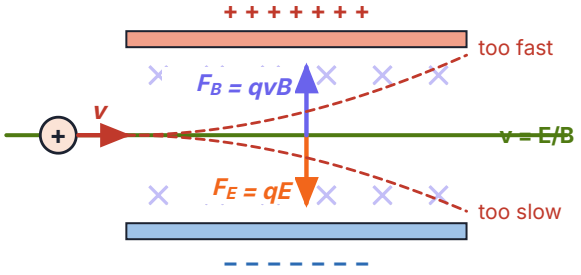


Figure 16.12: a velocity selector. Only particles with v = E/B pass straight through; the result does not depend on q or m.

- Faster particles: $qvB > qE$, they bend towards the magnetic force. Slower particles bend towards the electric force.
- This arrangement is a **velocity selector** (used, for example, before the magnetic field of a mass spectrometer).

Thomson's experiment

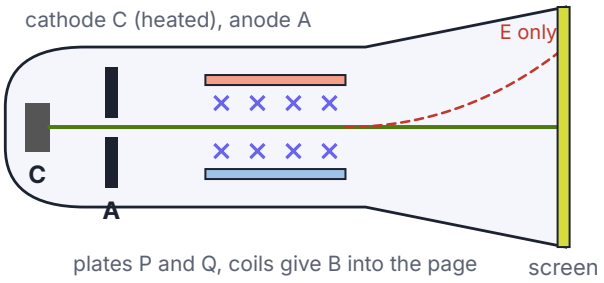


Figure 16.13: Thomson's apparatus (simplified).

Electrons from a heated cathode C are accelerated to the anode A and pass through it as a narrow beam.

Between plates P and Q there is an electric field E; coils outside the tube give a magnetic field B at right angles to it and to the beam.

E (or B) is adjusted until the beam is **undeflected**: then $eE = evB$, so the speed is $v = \dfrac{E}{B}$.

The electric field is switched off. In the magnetic field alone the beam moves in a circular arc of radius r, found from the deflection on the screen.

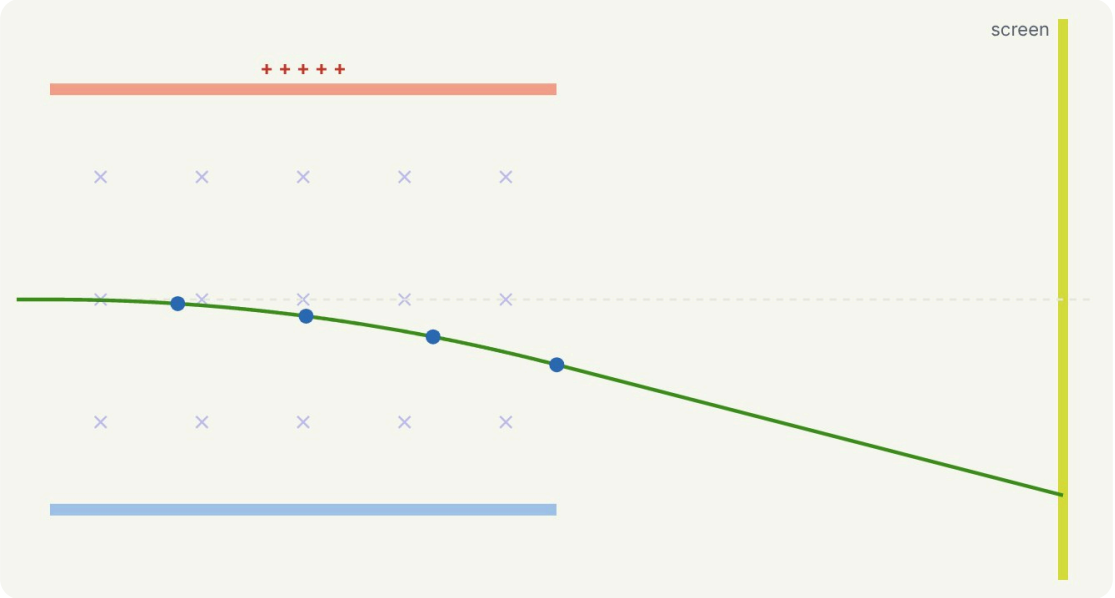
The magnetic force provides the centripetal force: $evB = \dfrac{mv^2}{r}$, so $\dfrac{e}{m} = \dfrac{v}{Br}$. Substituting $v = E/B$:

$$\frac{e}{m} = \frac{E}{B^2 r}$$

Thomson found the same e/m for any cathode material: electrons are part of every atom. Accepted value 1.76×10^{11} C kg^{-1}.

Simulation Crossed fields and Thomson's e/m

Fire electrons through the plates. Adjust E and B until the beam goes straight through (v = E/B). Then switch off E: the beam bends into a circle, and its radius gives e/m.



v needed = E/B
1.40 × 10⁷ m s⁻¹

beam
bends down: evB > eE (too fast)

electric force eE
2.24 × 10⁻¹⁵ N

magnetic force evB
3.20 × 10⁻¹⁵ N

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 16 to use the simulation.

Example 16.6.1

Protons pass undeflected through crossed fields with $E = 3.0 \times 10^5$ V m^{-1} and $B = 0.25$ T. Find their speed.

Solution

$$v = \frac{E}{B} = \frac{3.0 \times 10^5}{0.25} = \mathbf{1.2 \times 10^6\ m\,s^{-1}}$$

Example 16.6.2

A velocity selector must let positive ions moving to the right at 6.0 km s^{-1} through undeflected. The electric field is 300 V m^{-1} into the page. Find the size and direction of the magnetic field.

Solution

$$B = \frac{E}{v} = \frac{300}{6.0 \times 10^3} = \mathbf{0.050\ T}$$

The electric force on the ions is into the page, so the magnetic force must be out of the page. By Fleming's left-hand rule (second finger to the right, thumb out of the page) B points **up the page**.

Example 16.6.3

In a Thomson experiment the beam is undeflected with $E = 2.0 \times 10^4$ V m^{-1} and $B = 1.0$ mT. With the electric field switched off, the beam moves in an arc of radius 11.4 cm. Find (a) the speed of the electrons, (b) e/m.

Solution

(a) $$v = \frac{E}{B} = \frac{2.0 \times 10^4}{1.0 \times 10^{-3}} = \mathbf{2.0 \times 10^7\ m\,s^{-1}}$$

(b) $$\frac{e}{m} = \frac{E}{B^2 r} = \frac{2.0 \times 10^4}{(1.0 \times 10^{-3})^2 (0.114)} = \mathbf{1.75 \times 10^{11}\ C\ kg^{-1}}$$

A real use: the mass spectrometer

Atoms of the sample are **ionised** (given the same charge, usually $+e$) and accelerated into a beam.

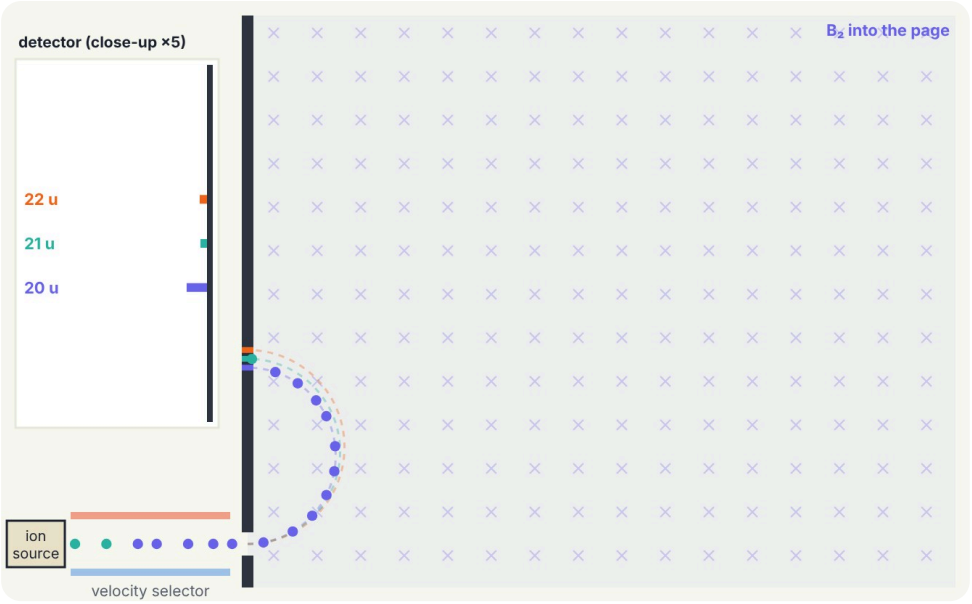
A **velocity selector** (crossed E and B_1) lets through only ions with $v = E/B_1$, so all ions leave it with the same speed.

In a second field B_2 alone, each ion moves in a semicircle of radius $r = mv/(qB_2) \propto m$: heavier ions make bigger circles and land further along the detector.

So the detector separates **isotopes** (same charge, different mass). It is used to date rocks and fossils (carbon-14), test for drugs in sport and check the purity of chemicals.

Simulation Mass spectrometer

Choose an element and watch its isotopes leave the velocity selector at the same speed and then separate in the magnetic field B_2. Change B_2: a stronger field makes smaller circles and brings the isotopes closer together on the detector.



speed from the selector v = E/B₁
1.0 × 10⁵ m s⁻¹

r for mass 20 u
4.15 cm

r for mass 21 u
4.36 cm

r for mass 22 u
4.56 cm

gap between the outer two on the detector
8.3 mm

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 16 to use the simulation.

Example 16.6.4

Singly charged neon ions leave a velocity selector at 1.0×10^5 m s^{-1} and enter a field $B_2 = 0.50$ T. Find the distance between the places where neon-20 and neon-22 ions hit the detector. (1 u = 1.66×10^{-27} kg)

Solution

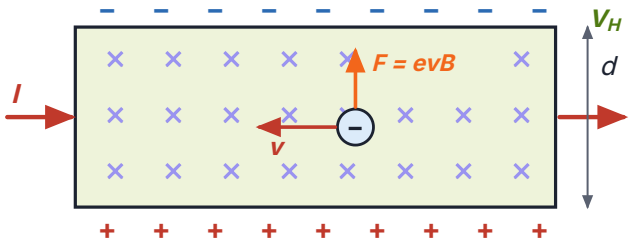
$$r_{20} = \frac{mv}{eB} = \frac{(20 \times 1.66 \times 10^{-27})(1.0 \times 10^5)}{(1.60 \times 10^{-19})(0.50)} = 4.15 \times 10^{-2}\ \text{m};\ \text{similarly } r_{22} = 4.57 \times 10^{-2}\ \text{m}$$

Each ion lands a diameter ($2r$) from the entrance slit, so the separation is $2(r_{22} - r_{20}) = 2(0.42 \times 10^{-2}) =$ **8.3×10^{-3} m** (8.3 mm)

16.7 Hall effect

Hall effect

when a current flows through a conductor or semiconductor in a magnetic field perpendicular to the current, **a p.d. (the Hall voltage) appears across the sides of the specimen**, perpendicular to both the current and the field.



top view: B into the page; t = thickness along B

Figure 16.14: the Hall effect in a slab with electrons as charge carriers.

How the Hall voltage builds up

The charge carriers drift along the slab with velocity v. The magnetic force evB pushes them towards one side.

Charges collect on that side and an equal opposite charge is left on the other side. This sets up an electric field E_H across the slab.

The electric force eE_H opposes the magnetic force. Charges keep collecting until the two forces balance; after that the carriers flow straight along the slab and the Hall voltage is steady.

Deriving the Hall voltage

At equilibrium: $eE_H = evB$, and $E_H = \frac{V_H}{d}$ (d = width across which V_H appears), so $V_H = Bvd$.

The current is $I = nevA$ with $A = td$ (t = thickness, measured along B), so $v = \frac{I}{netd}$.

$V_H = B\frac{I}{netd}d$:

$$V_H = \frac{BI}{net}$$

n = number of charge carriers per unit volume, t = thickness of the specimen in the direction of B

- $V_H \propto 1/n$. Metals have a very large n, so their Hall voltage is tiny. Semiconductors have a much smaller n, so V_H is large enough to measure: Hall probes use semiconductors.
- A thin specimen (small t) gives a larger V_H.

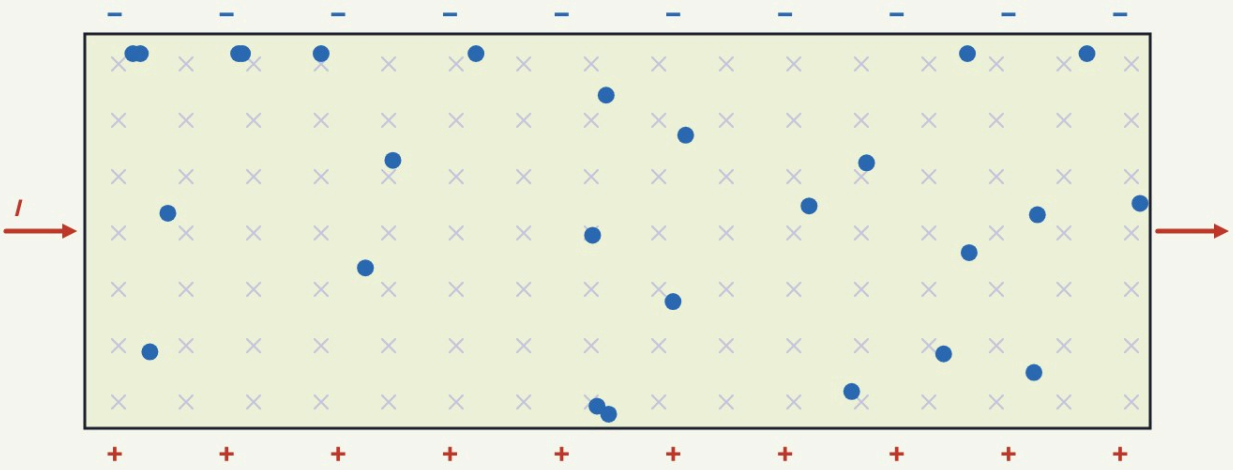
Applications of the Hall effect

- **Measuring magnetic flux density** with a Hall probe: with I constant, $V_H \propto B$. The probe is small, so it can measure B at a point, and it responds quickly to changing fields.
- **Finding the sign of the charge carriers**: the side that becomes positive shows whether the carriers are electrons (n-type) or holes (p-type).
- **Finding the number density** n of the carriers: $n = BI/(eV_H t)$.
- Hall sensors in everyday devices: measuring current without breaking the circuit, detecting the speed of a rotating wheel, and as contactless switches.

Simulation

Hall effect

Choose electrons or holes as the charge carriers and change B and the current. Watch the carriers pile up on one face until the Hall voltage stops further build-up. The sign of the top face tells you the sign of the carriers.



top face: negative

V_H = 3.12 mV

Hall voltage V_H = BI/(net)	carriers pushed to	n (this slab)	thickness t
3.12 mV	**the top face**	**1.0×10^{22} m^{-3}**	**1.0 mm**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 16 to use the simulation.

Example 16.7.1

A thin semiconductor slab is 10.0 mm long, 3.0 mm wide and 1.0 mm thick, with 4.0×10^{21} charge carriers per m^3. A current of 0.0025 mA flows along its length. A uniform field B is applied perpendicular to its largest face (along the 1.0 mm thickness), and a Hall voltage of 60 μV appears across its width. Find (a) the drift velocity of the carriers, (b) B. (c) The top face (as in Figure 16.14) becomes negative. What are the charge carriers?

Solution

(a) $$v = \frac{I}{neA} = \frac{2.5 \times 10^{-6}}{(4.0 \times 10^{21})(1.60 \times 10^{-19})(3.0 \times 10^{-3} \times 1.0 \times 10^{-3})} = \mathbf{1.3 \times 10^{-3}\ m\,s^{-1}}$$

(b) $$B = \frac{V_H net}{I} = \frac{(60 \times 10^{-6})(4.0 \times 10^{21})(1.60 \times 10^{-19})(1.0 \times 10^{-3})}{2.5 \times 10^{-6}} = \mathbf{15\ T}$$

(c) With the current to the right and B into the page, positive carriers moving right and negative carriers moving left are both pushed to the top face. The top face is negative, so the carriers are **electrons** (n-type).

Exercise: crossed fields and the Hall effect

1. Why does a Hall probe use a semiconductor and not a metal?

   Answer

   $V_H = BI/(net)$ is inversely proportional to n. A metal has so many free electrons that V_H is too small to measure; a semiconductor has far fewer carriers, so V_H is much larger.

2. Electrons accelerated to 3.0×10^7 m s^{-1} enter crossed fields with B = 2.0 mT. What electric field lets them pass undeflected?

   Answer

   $E = vB = 6.0 \times 10^4$ V m^{-1}

3. Derive $e/m = E/(B^2 r)$ for Thomson's experiment.

   Answer

   Undeflected: $eE = evB$ so $v = E/B$. Magnetic field only: $evB = mv^2/r$, so $e/m = v/(Br) = E/(B^2 r)$.

Chapter 17 · Semester 2

ELECTROMAGNETIC INDUCTION

Notes by Cikgu Suhaili

21 worked examples 6 simulations Exercises with answers

17.1 Magnetic flux

Magnetic flux, Φ

through a surface: the **scalar product of the magnetic flux density B and the area vector A** of the surface. It measures the number of field lines passing through the area.

$$\Phi = \mathbf{B} \cdot \mathbf{A} = BA\cos\theta$$

θ = angle between B and the normal to the surface (the direction of the area vector A).
Unit: weber (Wb) = T m^2. Scalar.

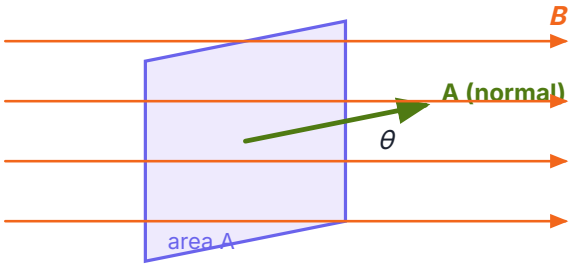


Figure 17.1: the area vector A is normal to the surface.



θ = 0°: Φ = BA (maximum) θ = 90°: Φ = 0

Figure 17.2: B along the normal gives the largest flux; B in the plane of the surface gives none.

- Flux linkage of a coil of N turns: $N\Phi$.
- Rearranged, $B = \Phi/A_\perp$: magnetic flux density is the flux per unit area perpendicular to the field (1 T = 1 Wb m^{-2}).

Teacher's note

Read the question carefully: "the **plane** of the loop makes θ with B" means the normal makes (90° − θ) with B, so Φ = BA sin θ.

Example 17.1.1

A rectangular loop of area 200 cm^2 has its plane at 41° to a 0.28 T field. Find the flux through it.

Solution

Angle between B and the normal = 90° − 41° = 49°

Φ = BA cos 49° = (0.28)(200 × 10^{-4})(0.656) = **3.7 × 10^{-3} Wb**

Example 17.1.2

A 25 cm × 15 cm loop is in a 0.60 T field and the flux through it is 0.015 Wb. Find the angle between the plane of the loop and the field.

Solution

$$\cos\theta = \frac{\Phi}{BA} = \frac{0.015}{(0.60)(0.25 \times 0.15)} = 0.667,\ \theta = 48.2°$$ (between B and the normal)

Angle between the plane and B = 90° − 48.2° = **41.8°**

Example 17.1.3

A flux of 50 μWb passes perpendicularly through a loop of area 0.78 m^2. Find the flux density.

Solution

$$B = \frac{\Phi}{A} = \frac{50 \times 10^{-6}}{0.78} = \mathbf{6.4 \times 10^{-5}\ T}$$

17.2 Faraday's law and Lenz's law

Faraday's experiments

- A steady current in one coil produces **no** current in a second coil. A current is induced only at the moment the first current is switched on or off.
- Moving a magnet towards a coil induces a current; moving it away induces a current in the opposite direction; no current flows when the magnet and coil are at rest relative to each other.
- The induced current is larger when the magnet moves faster, when the coil has more turns and when the magnet is stronger.

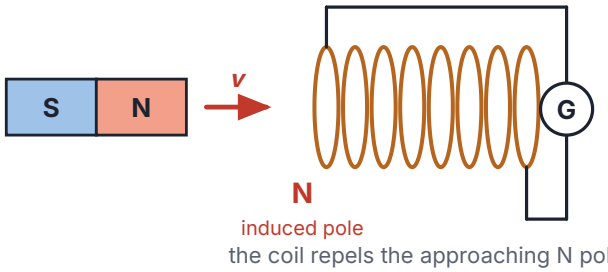


Figure 17.3: a magnet pushed into a coil connected to a galvanometer.

Faraday's law

the **magnitude of the induced e.m.f. is proportional to the rate of change of magnetic flux linkage**.

Lenz's law

the **induced current flows in a direction that opposes the change producing it**.

$$\varepsilon = -N\frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

the minus sign is Lenz's law. Average e.m.f.: $|\varepsilon| = N|\Delta\Phi|/\Delta t$

- $\Phi = BA\cos\theta$ can change in three ways: change B ($\varepsilon = -NA\,\mathrm{d}B/\mathrm{d}t$), change the area A ($\varepsilon = -NB\,\mathrm{d}A/\mathrm{d}t$), or change the angle θ (rotate the coil).
- If the coil is part of a closed circuit of resistance R, the induced current is $I = |\varepsilon|/R$.
- Lenz's law follows from the **conservation of energy**: if the induced current helped the change, energy would be created from nothing. Work must be done against the opposing force, and this work becomes electrical energy.

Using Lenz's law



flux increasing: B ind opposes B ext

flux decreasing: B ind is along B ext

Figure 17.4: the induced field opposes an increase of flux and supports a decrease.

Decide whether the flux through the loop is increasing or decreasing.

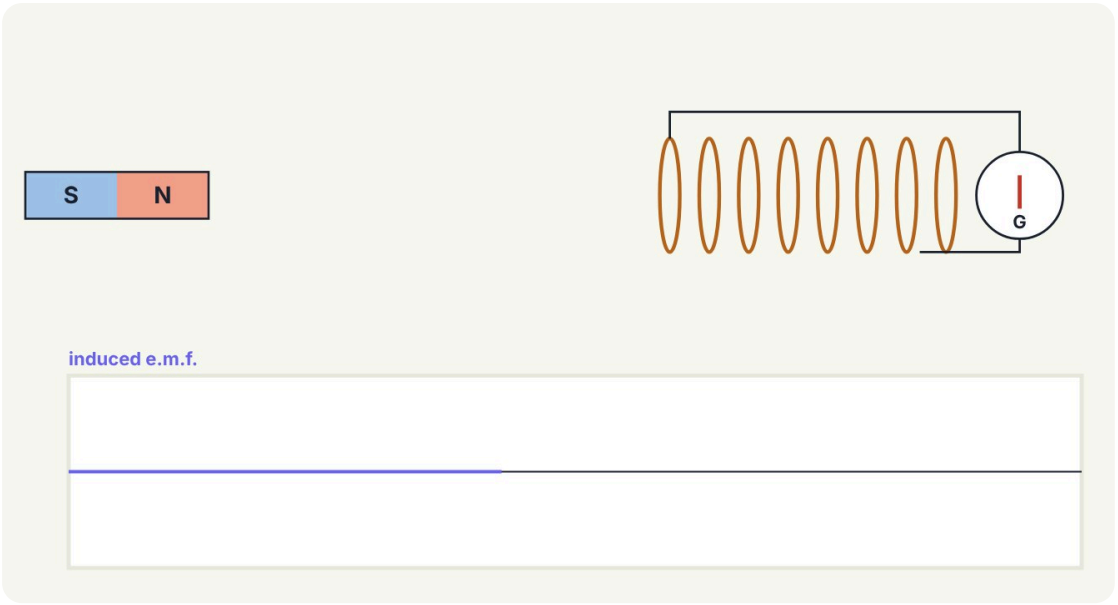
The induced current makes its own field **opposite** to the external field if the flux is increasing, and **along** it if the flux is decreasing.

Use the right-hand grip rule to turn the induced field into a current direction.

A north pole pushed towards a coil makes the near end of the coil a north pole (repelling it). Pulled away, the near end becomes a south pole (attracting it). Either way you must do work.

Simulation Magnet and coil

Move the magnet in and out of the coil with the slider (or press "Move it for me"). The galvanometer only deflects while the flux is changing; faster motion and more turns give a bigger e.m.f. The colour of the coil end shows the induced pole.



magnet	near end of the coil	e.m.f. (relative)	N turns
at rest: no e.m.f.	—	**0.00**	**10**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.2.1

A coil of 50 turns and diameter 8.0 cm is in a 1.8 T field perpendicular to its plane. The field falls to 0.60 T in 2.0 ms. Find the average induced e.m.f.

Solution

$A = \pi(0.040)^2 = 5.03 \times 10^{-3}\ \text{m}^2$

$$|\varepsilon| = NA\frac{|\Delta B|}{\Delta t} = (50)(5.03 \times 10^{-3})\frac{1.2}{2.0 \times 10^{-3}} = \mathbf{151\ V}$$

Example 17.2.2

An elastic circular loop in the plane of the paper is in a 0.75 T field into the paper. Its diameter shrinks from 20.0 cm to 6.0 cm in 0.50 s. Find (a) the direction of the induced current, (b) the average induced e.m.f., (c) the average current if the loop's resistance is 2.5 Ω.

Solution

(a) The flux into the paper decreases, so the induced current makes a field into the paper inside the loop: **clockwise**.

(b) $|\Delta A| = \pi(0.100^2 - 0.030^2) = 2.86 \times 10^{-2}\ \text{m}^2$; $|\varepsilon| = B\dfrac{|\Delta A|}{\Delta t} = (0.75)\dfrac{2.86 \times 10^{-2}}{0.50} = \mathbf{43\ mV}$

(c) $I = \dfrac{0.043}{2.5} = \mathbf{17\ mA}$

Example 17.2.3

A circular coil of radius 3.05 cm, 40 turns and resistance 3.55 Ω is perpendicular to a field of 1.25×10^{-2} T. The field increases to 0.450 T in 0.250 s. Find the induced current.

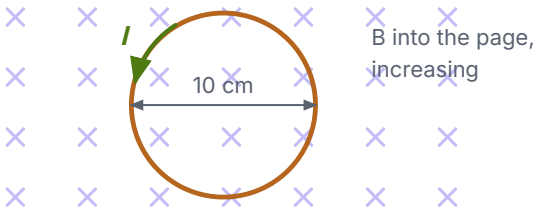
Solution

$|\varepsilon| = NA\dfrac{\Delta B}{\Delta t} = (40)\pi(0.0305)^2\dfrac{0.450 - 0.0125}{0.250} = 0.205\ \text{V}$

$I = \dfrac{0.205}{3.55} = \mathbf{0.058\ A}$

Example 17.2.4

A single circular loop of diameter 10 cm is connected in series with a 37 Ω resistor. The field through the loop is into the page and increasing at 0.050 T s^{-1}. Find the current and its direction.



Solution

$$|\varepsilon| = A\frac{dB}{dt} = \pi(0.050)^2(0.050) = 3.93 \times 10^{-4}\ \text{V}$$

$$I = \frac{3.93 \times 10^{-4}}{37} = \mathbf{1.1 \times 10^{-5}\ A}$$

The flux into the page increases, so the induced field inside the loop points out of the page: the current is **anticlockwise**.

E.m.f. induced in a straight conductor moving in a field

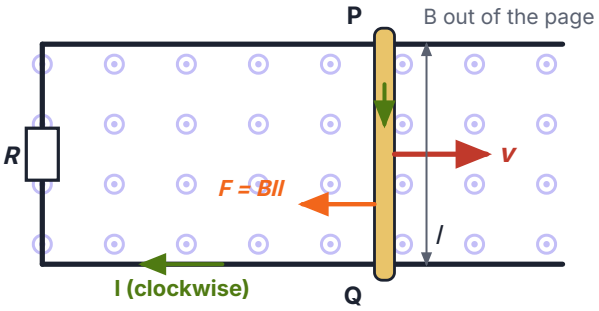


Figure 17.5: a rod of length l sliding at speed v along two rails in a uniform field.

In a time dt the rod moves $dx = v\,dt$, so the area of the circuit increases by $dA = l\,dx = lv\,dt$.

The flux increases by $d\Phi = B\,dA = Blv\,dt$.

Faraday's law: $|\varepsilon| = \frac{d\Phi}{dt} = Blv$. If **v** makes an angle θ with **B**, only the component v sin θ cuts the field lines:

$$\varepsilon = Blv \sin\theta$$

θ = angle between v and B (rod perpendicular to both)

- The same result comes from the magnetic force on the free electrons in the rod: they are pushed along the rod, one end becomes negative and the other positive.
- Direction of the induced current: **Fleming's right-hand rule** (thumb = motion, first finger = field, second finger = induced current). It always agrees with Lenz's law.
- The current in the rod feels a force BIl that **opposes the motion**; to keep the rod moving at constant speed an equal applied force is needed. The mechanical power Fv equals the electrical power I^2R.

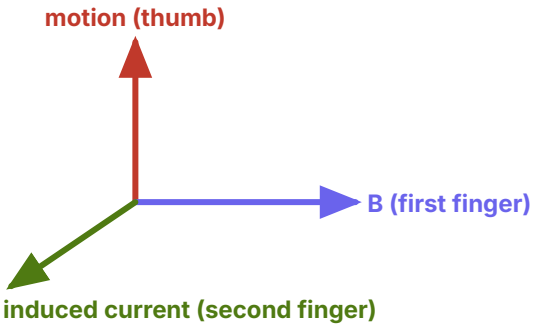
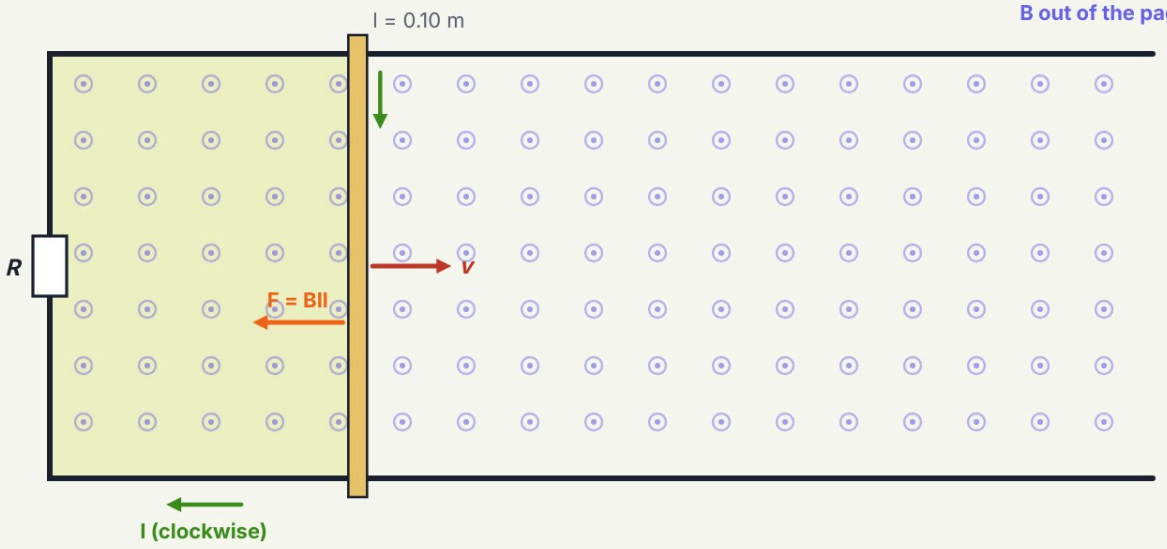


Figure 17.6: Fleming's right-hand rule (for generators: induced current).

Simulation

A rod sliding on rails

Push the rod with a steady speed. The area of the circuit and the flux grow, an e.m.f. Blv is induced and a current flows. The current makes a force on the rod that opposes the motion; the work done against it becomes heat in R.



ε = Blv	I = ε/R	force on the rod BIl (opposes v)	power I^2R = Fv
0.150 V	**5.00 A**	**0.300 N**	**0.750 W**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.2.5

In Figure 17.5, l = 0.10 m, v = 2.5 m s^{-1}, B = 0.60 T and the total resistance of the circuit is 0.030 Ω. Find (a) the induced e.m.f., (b) the current, (c) the force on the rod, (d) the power dissipated.

Solution

(a) ε = Blv = (0.60)(0.10)(2.5) = **0.15 V**

(b) $I = \dfrac{0.15}{0.030}$ = **5.0 A**

(c) F = BIl = (0.60)(5.0)(0.10) = **0.30 N**, opposing the motion

(d) P = I^2R = $(5.0)^2(0.030)$ = **0.75 W** (= Fv = (0.30)(2.5))

Example 17.2.6

A wire of length 0.20 m moves at 4.0 m s^{-1} at 40° to a field of 0.50 T. Find the induced e.m.f.

Solution

$\varepsilon = Blv \sin\theta = (0.50)(0.20)(4.0)\sin 40° =$ **0.26 V**

Example 17.2.7

A rod of length 0.400 m moves on rails in a field of 1.20 T and the induced e.m.f. is 3.60 V. Find (a) its speed, (b) the current if the total resistance is 0.900 Ω, (c) the force the field exerts on the rod.

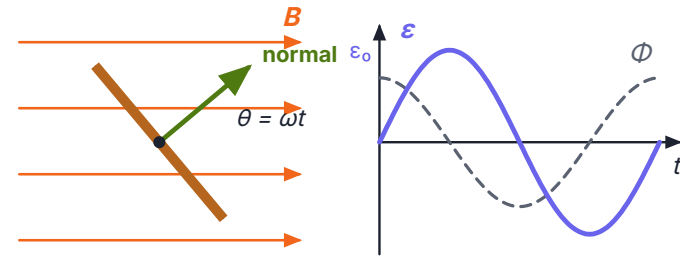
Solution

(a) $v = \dfrac{\varepsilon}{Bl} = \dfrac{3.60}{(1.20)(0.400)} =$ **7.50 m s^{-1}**

(b) $I = \dfrac{3.60}{0.900} =$ **4.00 A**

(c) $F = BIl = (1.20)(4.00)(0.400) =$ **1.92 N**, opposite to the velocity

E.m.f. induced in a rotating coil (a.c. generator)



coil seen edge-on, axis ⊥ page

Figure 17.7: a coil rotating at angular speed ω in a uniform field, and the e.m.f. it produces.

A coil of N turns, area A, rotates at constant angular speed ω about an axis perpendicular to a uniform field B. At time t the normal makes $\theta = \omega t$ with B.

Flux through each turn: $\Phi = BA\cos\omega t$.

$$\varepsilon = -N\frac{d\Phi}{dt} = -NBA\frac{d}{dt}(\cos \omega t) = NBA\omega \sin \omega t$$

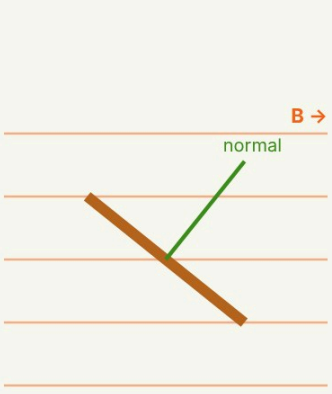
$$\varepsilon = NBA\omega \sin \omega t = \varepsilon_0 \sin \omega t$$

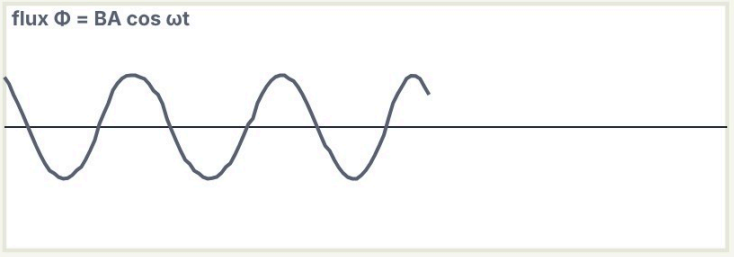
peak e.m.f. $\varepsilon_0 = NBA\omega$; $\omega = 2\pi f$

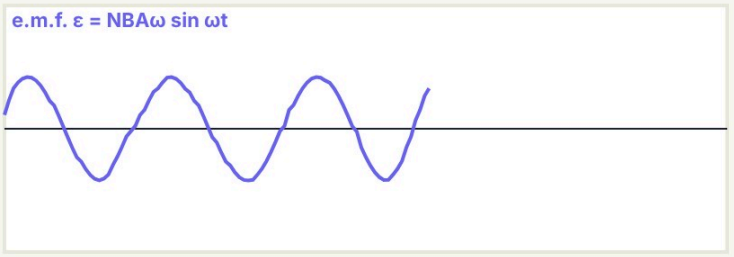
- The e.m.f. is **largest when the flux is zero** (plane of the coil parallel to B: the sides cut the field lines fastest) and **zero when the flux is largest** (plane perpendicular to B).
- Doubling the speed of rotation doubles both the peak e.m.f. and the frequency.

Simulation A.C. generator

Change the speed of rotation, the flux density and the number of turns. Compare the flux graph with the e.m.f. graph: the e.m.f. is largest where the flux changes fastest, which is where the flux is zero.







peak e.m.f. $\varepsilon_0 = NBA\omega$	e.m.f. now	angle $\theta = \omega t$	coil area A
1.57 V	**1.23 V**	**51°**	**0.010 m²**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.2.8

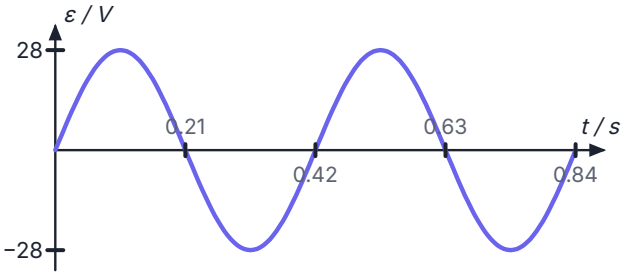
A generator coil has 100 turns of area 0.20 m^2 and turns at 60 rev s^{-1} in a field of 1.0×10^{-3} T. Find the peak e.m.f.

Solution

$\varepsilon_0 = NBA\omega = (100)(1.0 \times 10^{-3})(0.20)(2\pi \times 60) =$ **7.5 V**

Example 17.2.9

The graph shows the output of a generator whose coil has 150 turns of area 0.020 m^2. Find (a) the frequency, (b) the angular speed, (c) the flux density.



Solution

(a) Period $T = 0.42$ s, so $f = \dfrac{1}{0.42} =$ **2.4 Hz**

(b) $\omega = 2\pi f =$ **15 rad s^{-1}**

(c) $B = \dfrac{\varepsilon_0}{NA\omega} = \dfrac{28}{(150)(0.020)(14.96)} =$ **0.62 T**

Example 17.2.10

A generator coil has 500 turns, each of area 60 cm^2, and turns at 3600 rev min^{-1} in a 2.0 mT field. Find (a) the frequency of the e.m.f., (b) the peak e.m.f., (c) the e.m.f. when the plane of the coil makes 60° with the field.

Solution

(a) $f = \dfrac{3600}{60}$ = **60 Hz** ($\omega = 2\pi \times 60 = 377$ rad s^{-1})

(b) $\varepsilon_0 = NBA\omega = (500)(2.0 \times 10^{-3})(60 \times 10^{-4})(377)$ = **2.26 V**

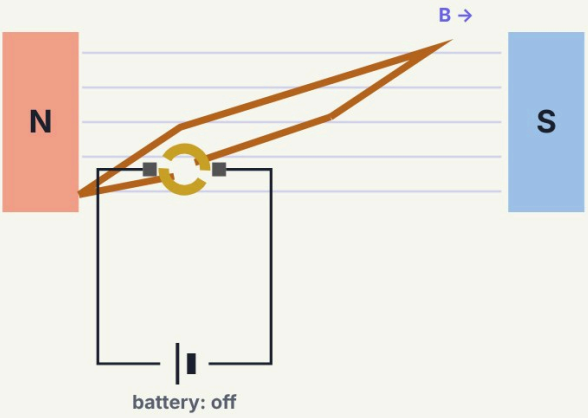
(c) The plane is at 60° to B, so the normal is at $\theta = 30°$ to B: $\varepsilon = \varepsilon_0 \sin 30°$ = **1.13 V**

Motor and dynamo: the same machine

- An **electric motor** is a coil in a magnetic field. A current in the coil makes forces BIl on its two long sides, one up and one down, which turn the coil (Chapter 16). A **split-ring commutator** reverses the current every half turn, so the coil keeps turning the same way.
- A **dynamo** (like a bicycle dynamo) is the same machine used the other way round: you turn the coil, the flux through it changes, and an e.m.f. $\varepsilon = NBA\omega \sin \omega t$ is induced. With **slip rings** it gives a.c.; with a commutator the output is always in one direction (d.c.).
- In a working motor the turning coil also acts as a dynamo: it induces a back e.m.f. that opposes the supply (Lenz's law), which is why a motor takes a large current when it starts and a smaller one at full speed.

Simulation Motor and dynamo

As a motor: switch on the current and watch the forces BIl on the two long sides turn the coil, with the commutator swapping the current every half turn. As a dynamo: turn the handle faster or slower and see the lamp and the e.m.f. graph; switch between slip rings (a.c.) and a commutator (d.c.).



turning effect	speed	commutator
no current, no force	**0.0 turns per second**	**reverses the current every half turn**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Exercise: Faraday's and Lenz's laws

1. A 200-turn coil of area 4.0 cm^2 is turned from a position where its plane is perpendicular to a 0.30 T field to one where its plane is parallel to it, in 0.050 s. Find the average e.m.f.

   Answer

   $\Delta\Phi = BA - 0$; $|\varepsilon| = NBA/\Delta t = (200)(0.30)(4.0 \times 10^{-4})/0.050 =$ **0.48 V**

2. Explain, using energy, why the induced current must oppose the change that produces it.

   Answer

   If it helped the change, the motion would speed up by itself and electrical energy would appear without any work being done, breaking the conservation of energy. Because it opposes the change, work must be done, and that work becomes the electrical energy.

3. An aircraft with a wingspan of 40 m flies horizontally at 250 m s^{-1} where the vertical component of the Earth's field is 4.0 × 10^{-5} T. Find the e.m.f. between its wing tips.

Answer

$\varepsilon = Blv = (4.0 \times 10^{-5})(40)(250) =$ 0.40 V

17.3 Self induction

Self induction

the **production of an induced e.m.f. in a coil because of a change of the current in the same coil**.

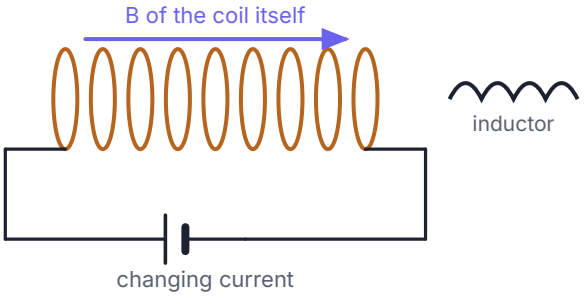


Figure 17.8: the coil's own changing field changes the flux linking it.

- The current in the coil produces a field and a flux through the coil itself. When the current changes, the flux changes and an e.m.f. is induced (a **back e.m.f.**).
- By Lenz's law it opposes the change: while the current is **rising** the induced e.m.f. opposes the current; while it is **falling** the induced e.m.f. acts in the direction of the current, trying to keep it flowing.
- A coil used for its self-inductance is called an **inductor**.

Self-inductance, L

the **ratio of the self-induced e.m.f. to the rate of change of current in the coil**: $L = -\varepsilon/(\mathrm{d}I/\mathrm{d}t)$.

Equivalently, the flux linkage per unit current, $L = N\Phi/I$.

$$\varepsilon = -L\frac{\mathrm{d}I}{\mathrm{d}t} \qquad LI = N\Phi$$

unit: henry (H) = Wb A^{-1} = V s A^{-1}. A coil has L = 1 H if a current changing at 1 A s^{-1} induces 1 V.

- $LI = N\Phi$ comes from Faraday's law: $-L\,\mathrm{d}I/\mathrm{d}t = -N\,\mathrm{d}\Phi/\mathrm{d}t$, integrated from zero current (and zero flux).
- L depends on the size and shape of the coil, the number of turns and the core material; **not on the current**.

Self-inductance of a solenoid

Field inside a long solenoid of N turns and length l: $B = \dfrac{\mu_0 N I}{l}$.

Flux through each turn: $\Phi = BA = \dfrac{\mu_0 N I A}{l}$.

$L = \dfrac{N\Phi}{I} = \dfrac{N}{I} \times \dfrac{\mu_0 N I A}{l}$:

$$L = \frac{\mu_0 N^2 A}{l}$$

with a core of relative permeability μ_r: $L = \mu_r \mu_0 N^2 A / l$

Example 17.3.1

The current in a 230 mH coil rises steadily from 20.0 mA to 28.0 mA in 140 ms. Find the induced e.m.f.

Solution

$\varepsilon = -L\dfrac{\Delta I}{\Delta t} = -(0.230)\dfrac{8.0 \times 10^{-3}}{0.140} =$ **-1.3×10^{-2} V** (it opposes the rising current)

Example 17.3.2

A solenoid of length 0.052 m and cross-sectional area 1.2×10^{-3} m^2 must have an inductance of 1.4 mH. How many turns does it need?

Solution

$N = \sqrt{\dfrac{Ll}{\mu_0 A}} = \sqrt{\dfrac{(1.4 \times 10^{-3})(0.052)}{(4\pi \times 10^{-7})(1.2 \times 10^{-3})}} =$ **220 turns**

Example 17.3.3

The current in a coil rises steadily from zero to 50.0 A in 10.0 s, inducing an e.m.f. of 45.0 V. Find (a) the self-inductance, (b) the flux linkage when the current is 50.0 A.

Solution

(a) $L = \dfrac{|\varepsilon|}{dI/dt} = \dfrac{45.0}{5.00} = \mathbf{9.00\ H}$

(b) $N\Phi = LI = (9.00)(50.0) = \mathbf{450\ Wb}$

Example 17.3.4

An air-cored solenoid of 450 turns, diameter 15.0 mm and length 12.0 cm carries 40.0 mA. Find (a) the field inside, (b) the flux through each turn, (c) the inductance.

Solution

(a) $B = \dfrac{\mu_0 NI}{l} = \dfrac{(4\pi \times 10^{-7})(450)(0.0400)}{0.120} = \mathbf{1.88 \times 10^{-4}\ T}$

(b) $\Phi = BA = (1.88 \times 10^{-4})\pi(7.50 \times 10^{-3})^2 = \mathbf{3.33 \times 10^{-8}\ Wb}$

(c) $L = \dfrac{N\Phi}{I} = \dfrac{(450)(3.33 \times 10^{-8})}{0.0400} = \mathbf{3.75 \times 10^{-4}\ H}$

17.4 Energy stored in an inductor

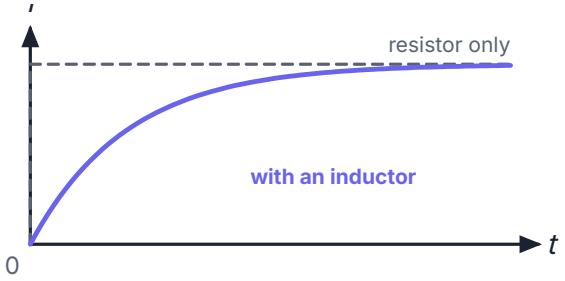


Figure 17.9: when the switch is closed, the back e.m.f. of an inductor makes the current rise gradually.

- While the current in an inductor is rising, a back e.m.f. opposes it. The source must do work to drive charge against this e.m.f.

- This work is not lost: it is **stored in the magnetic field** of the inductor. When the current is switched off, the energy is given back (this is why a spark can jump across a switch that breaks an inductive circuit).

Deriving $U = \frac{1}{2}LI^2$

Power supplied against the back e.m.f.: $P = I|\varepsilon| = IL\dfrac{\mathrm{d}I}{\mathrm{d}t}$

Energy in a time $\mathrm{d}t$: $\mathrm{d}U = P\,\mathrm{d}t = LI\,\mathrm{d}I$

Total energy as the current rises from 0 to I: $U = \int_0^I LI\,\mathrm{d}I = \frac{1}{2}LI^2$

$$U = \tfrac{1}{2}LI^2$$

for a solenoid: $U = \tfrac{1}{2}\dfrac{\mu_0 N^2 A}{l}I^2$

Simulation

Switching on an inductor

Close the switch and watch the current rise gradually while the back e.m.f. falls. Open it again and the stored energy $\frac{1}{2}LI^2$ is given back. Change L and R: a larger L/R makes the rise slower.



current I	back e.m.f. L dI/dt	energy stored $\frac{1}{2}LI^2$	supply e.m.f.
0.00 A	**0.00 V**	**0.00 J**	**6.0 V**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.4.1

How much energy is stored in a 0.085 H inductor carrying 2.5 A?

Solution

$U = \tfrac{1}{2}LI^2 = \tfrac{1}{2}(0.085)(2.5)^2 = \mathbf{0.27\ J}$

Example 17.4.2

A steady current of 2.5 A in a coil of 500 turns produces a flux of 1.4×10^{-4} Wb through each turn. Find (a) the average back e.m.f. if the current is stopped in 0.080 s, (b) the inductance, (c) the energy stored at 2.5 A.

Solution

(a) $|\varepsilon| = N\dfrac{\Delta\Phi}{\Delta t} = (500)\dfrac{1.4 \times 10^{-4}}{0.080} = \mathbf{0.88\ V}$

(b) $L = \dfrac{N\Phi}{I} = \dfrac{(500)(1.4 \times 10^{-4})}{2.5} = \mathbf{2.8 \times 10^{-2}\ H}$

(c) $U = \tfrac{1}{2}LI^2 = \tfrac{1}{2}(0.028)(2.5)^2 = \mathbf{8.8 \times 10^{-2}\ J}$

17.5 Mutual induction

Mutual induction

the **production of an induced e.m.f. in one coil because of a change of current in a neighbouring coil**.

- The changing current in the **primary** coil changes the flux linking the **secondary** coil, so an e.m.f. is induced in the secondary. (This is what Faraday saw when he switched the current on and off.)
- The transformer and the induction coil work by mutual induction.

Mutual inductance, M

the **ratio of the e.m.f. induced in one coil to the rate of change of current in the other coil**: $M = -\varepsilon_s/(\mathrm{d}I_p/\mathrm{d}t)$. Equivalently, the flux linkage of the secondary per unit current in the primary.

$$\varepsilon_s = -M\frac{dI_p}{dt} \qquad MI_p = N_s\Phi_s$$

unit: henry (H). M is the same whichever coil is called the primary.

Two coaxial solenoids

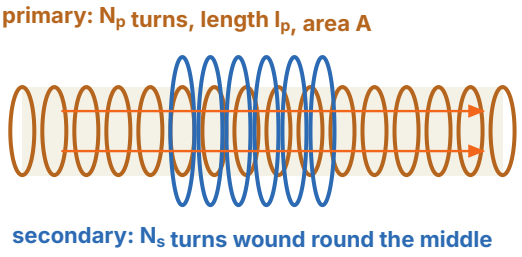


Figure 17.10: a short secondary coil wound round the middle of a long primary solenoid.

Field inside the primary: $B_p = \frac{\mu_0 N_p I_p}{l_p}$.

All of this flux passes through each turn of the secondary: $\Phi_s = B_p A = \frac{\mu_0 N_p I_p A}{l_p}$.

$M = \frac{N_s\Phi_s}{I_p}$:

$$M = \frac{\mu_0 N_p N_s A}{l_p}$$

A = cross-sectional area of the primary; l_p = length of the primary

Example 17.5.1

A primary solenoid of radius 2.0 cm and length 24 cm has 500 turns. A secondary coil of 80 turns is wound round its centre. Find (a) the mutual inductance, (b) the e.m.f. induced in the secondary when the primary current changes at 4.8 A s^{-1}.

Solution

(a) $M = \frac{\mu_0 N_p N_s A}{l_p} = \frac{(4\pi \times 10^{-7})(500)(80)\pi(0.020)^2}{0.24} = \mathbf{2.6 \times 10^{-4}\ H}$

(b) $|\varepsilon_s| = M\frac{dI_p}{dt} = (2.63 \times 10^{-4})(4.8) = \mathbf{1.3 \times 10^{-3}\ V}$

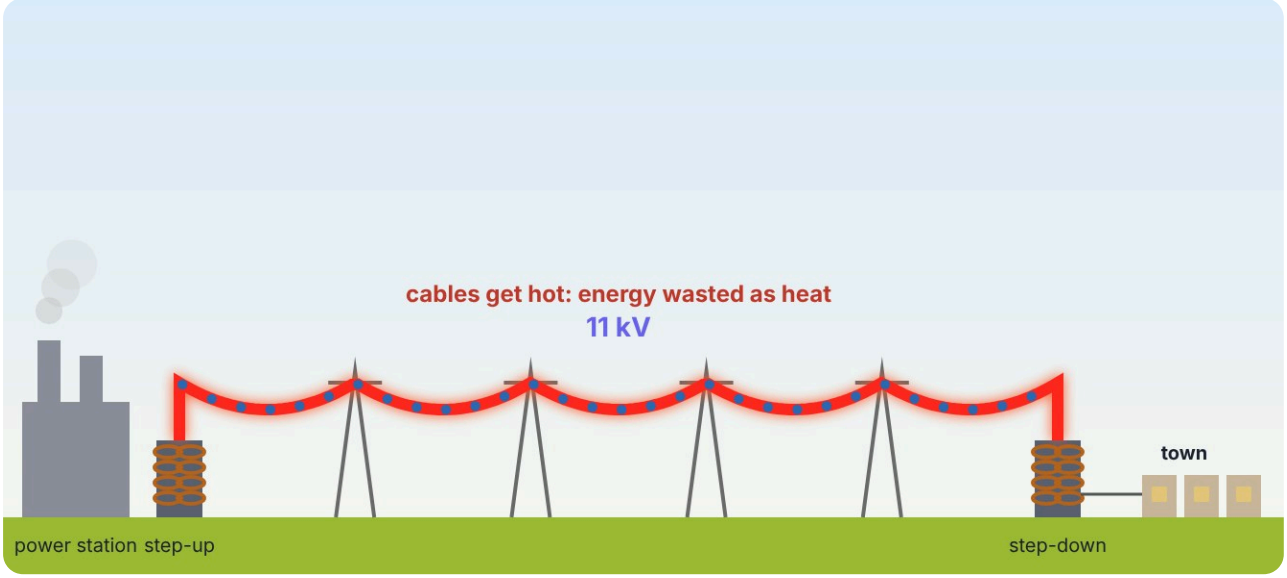
A real use: transformers and power lines

- A **transformer** is two coils on one iron core. An **alternating** current in the primary makes a changing flux in the core, which induces an e.m.f. in the secondary by **mutual induction**. (A steady d.c. would give no changing flux and no output.)
- For an ideal transformer: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ and, with no energy lost, $V_p I_p = V_s I_s$.
- Power stations send electricity at very **high voltage** (132–400 kV). For the same power $P = VI$, a higher voltage means a **smaller current**, and the heat wasted in the cables, I^2R, falls with the **square** of the current.
- Step-up transformers raise the voltage at the power station; step-down transformers near our homes bring it back to 240 V.

Simulation

Transformers and power lines

Send 20 MW from the power station to the town through cables of total resistance 4.0 Ω. Change the transmission voltage and watch the current, the heat lost in the cables and how bright the town is.



current in the cables I = P/V	power lost I^2R	fraction lost	step-up turns ratio Ns/Np (from 11 kV)
1818 A	**13.2 MW**	**66 %**	**1.0**

step-down turns ratio Ns/Np (to 240 V)
1 : 46

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.5.2

A power station sends 20 MW through cables of total resistance 4.0 Ω. Find the power lost in the cables if it is sent at (a) 11 kV, (b) 275 kV.

Solution

(a) $I = \dfrac{P}{V} = \dfrac{20 \times 10^6}{11 \times 10^3} = 1820$ A; $P_{\text{loss}} = I^2R = (1820)^2(4.0) =$ **13 MW** (two-thirds of the power!)

(b) $I = \dfrac{20 \times 10^6}{275 \times 10^3} = 72.7$ A; $P_{\text{loss}} = (72.7)^2(4.0) =$ **21 kW** (about 0.1%)

Exercise: inductance

1. A solenoid 25 cm long has 1000 turns and a cross-sectional area of 4.0 cm^2. Find (a) its self-inductance, (b) the energy stored when it carries 3.0 A.

   Answer

   (a) $L = \mu_0 N^2 A/l =$ 2.0 × 10^{-3} H (b) $U = \tfrac{1}{2}LI^2 =$ 9.0 × 10^{-3} J

2. The current in the primary of a pair of coils (M = 0.15 H) falls steadily from 4.0 A to zero in 0.020 s. Find the e.m.f. induced in the secondary.

   Answer

   $|\varepsilon| = M\Delta I/\Delta t = (0.15)(4.0/0.020) =$ 30 V

3. State two differences between self induction and mutual induction.

   Answer

   Self induction: the e.m.f. is induced in the same coil whose current changes, and it opposes the change of that current. Mutual induction: the e.m.f. is induced in a second coil, by a change of current in the first; it needs two coils linked by flux.

Chapter 18 · Semester 2

ALTERNATING CURRENT CIRCUITS

Notes by Cikgu Suhaili

9 worked examples 3 simulations Exercises with answers

18.1 Alternating current through a resistor

An **alternating current** (a.c.) changes its magnitude and direction periodically. The output of an a.c. generator is sinusoidal:

$$V = V_0 \sin \omega t \qquad I = I_0 \sin \omega t \qquad \omega = 2\pi f = \frac{2\pi}{T}$$

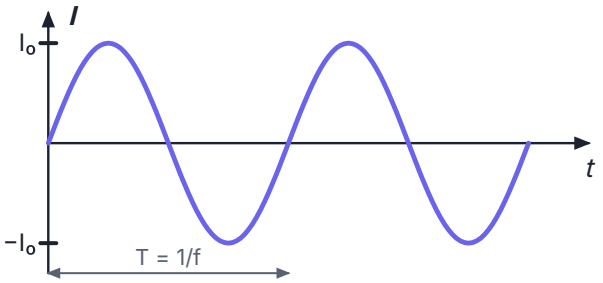


Figure 18.1: a sinusoidal alternating current with peak value I_o and period T.

- V, I: instantaneous values; V_0, I_0: peak values; ω: angular frequency (rad s^{-1}); f: frequency (Hz).

Phasors

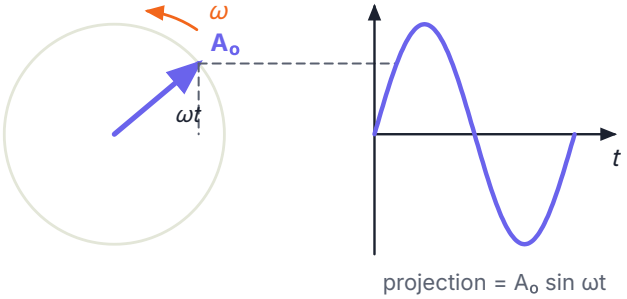


Figure 18.2: a phasor of length A_o rotating anticlockwise at ω. Its vertical projection is A_o sin ωt.

A **phasor** is a vector that rotates anticlockwise at the angular frequency ω. Its length is the peak value and its projection on the vertical axis gives the instantaneous value. A **phasor diagram** shows the phase difference between voltage and current.

Root mean square (r.m.s.) value

r.m.s. current, I_{rms}

the **value of the steady direct current that produces the same average power** in a resistor as the alternating current. (The r.m.s. voltage is defined in the same way.)

Power in a resistor at any instant: $P = I^2R$. The average power is $\langle I^2 \rangle R$.

A steady current I_{rms} gives the same power if $I_{rms}^2 R = \langle I^2 \rangle R$, so $I_{rms} = \sqrt{\langle I^2 \rangle}$: the **square root of the mean of the square** of the current.

For $I = I_0 \sin \omega t$: $I^2 = I_0^2 \sin^2 \omega t = \frac{1}{2} I_0^2 (1 - \cos 2\omega t)$. Over a whole cycle the mean of $\cos 2\omega t$ is zero, so $\langle I^2 \rangle = \frac{1}{2} I_0^2$.

$$I_{rms} = \frac{I_0}{\sqrt{2}} \qquad V_{rms} = \frac{V_0}{\sqrt{2}}$$

sinusoidal a.c. only. Mains 240 V means $V_{rms} = 240$ V, so $V_0 = 339$ V.

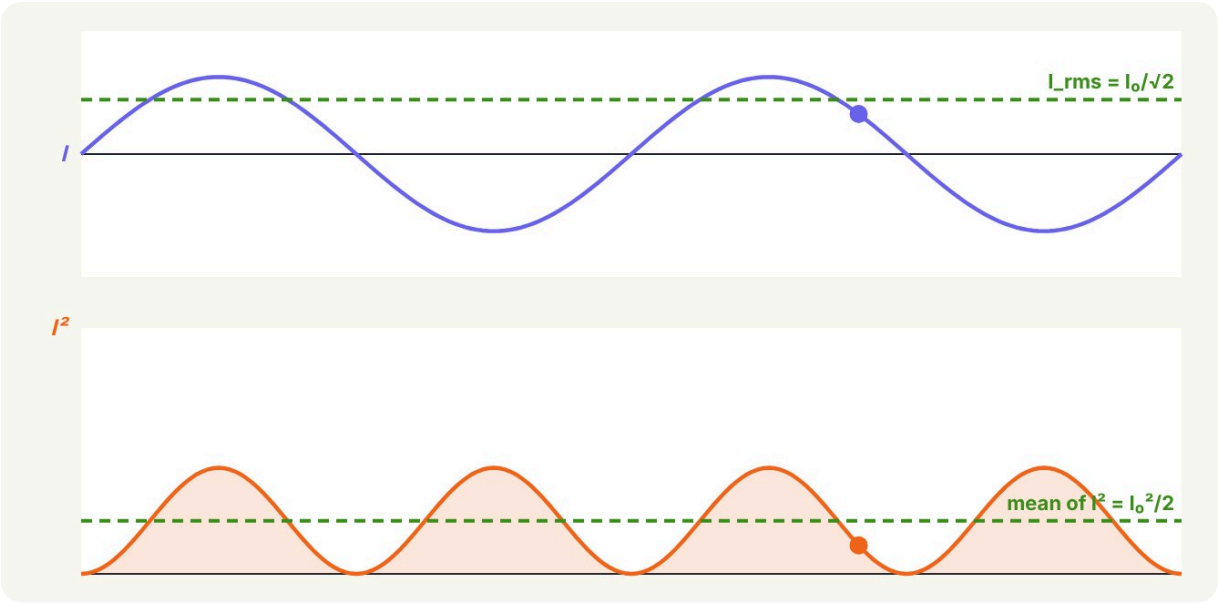
Teacher's note

Ammeters and voltmeters for a.c. read r.m.s. values. Unless a question says "peak", assume a quoted a.c. voltage or current is r.m.s.

Simulation

Where the r.m.s. value comes from

Watch the current, its square and the mean of the square. The steady current that gives the same heating is the square root of that mean: $I_0/\sqrt{2}$ for a sine wave.



I_0	mean of I^2	$I_{rms} = \sqrt{(\text{mean of } I^2)}$	mean of I itself
4.0 A	**8.00 A²**	**2.83 A**	**0 (no use for heating)**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 18 to use the simulation.

Current in a pure resistor

Supply voltage across the resistor: $V = V_0 \sin \omega t$.

At every instant $I = \dfrac{V}{R} = \dfrac{V_0}{R} \sin \omega t = I_0 \sin \omega t$, with $I_0 = V_0/R$.

The current and voltage reach their maximum and zero values at the same time: they are **in phase** (phase difference 0).

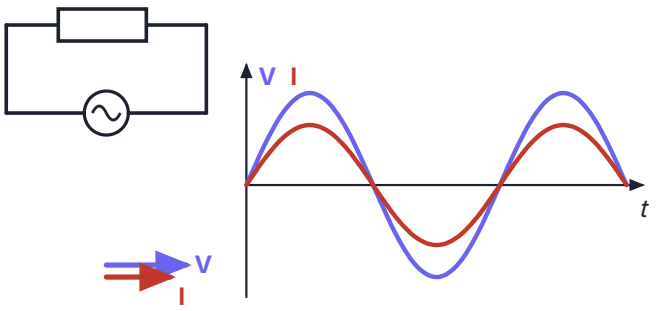


Figure 18.3: a pure resistor: circuit, phasor diagram and graphs.

Power in a pure resistor

$P = IV = I_0 V_0 \sin^2 \omega t = P_0 \sin^2 \omega t$, always positive: the resistor always takes energy and turns it into heat.

Mean of $\sin^2 \omega t = ½$, so:

$$\langle P \rangle = ½ I_0 V_0 = I_{rms} V_{rms} = I_{rms}^2 R = \frac{V_{rms}^2}{R}$$

average power = half the peak power $P_o = I_o V_o$

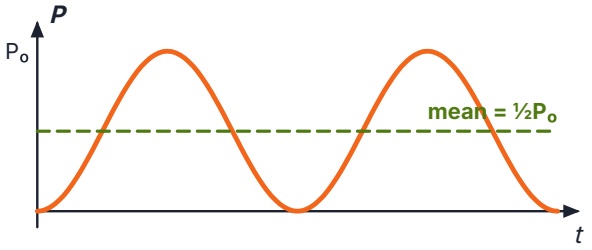


Figure 18.4: power in a resistor varies between 0 and P_o at twice the supply frequency.

Example 18.1.1

A 240 V (r.m.s.), 50 Hz supply is connected to a 60 Ω heater. Find (a) the peak voltage, (b) the r.m.s. and peak currents, (c) the mean power, (d) the peak power.

Solution

(a) $V_0 = \sqrt{2}(240) =$ **339 V**

(b) $I_{rms} = \frac{240}{60} =$ **4.0 A**; $I_0 = \sqrt{2}(4.0) =$ **5.7 A**

(c) $\langle P \rangle = I_{rms} V_{rms} = (4.0)(240) =$ **960 W**

(d) $P_0 = I_0 V_0 = 2\langle P \rangle =$ **1920 W**

Example 18.1.2

The current in a resistor is $I = 3.0 \sin 100\pi t$ A. Find (a) the frequency, (b) the r.m.s. current, (c) the current at $t = 2.5$ ms.

Solution

(a) $\omega = 100\pi$, $f = \frac{\omega}{2\pi} =$ **50 Hz**

(b) $I_{rms} = \frac{3.0}{\sqrt{2}} =$ **2.1 A**

(c) $I = 3.0 \sin(100\pi \times 0.0025) = 3.0 \sin(\pi/4) =$ **2.1 A**

18.2 Alternating current through an inductor

A **pure (ideal) inductor** has inductance but no resistance.

Deriving the current

The supply voltage is $V = V_0 \sin \omega t$. With no resistance, the applied voltage just balances the back e.m.f.: $V = L\frac{dI}{dt}$.

$$\frac{dI}{dt} = \frac{V_0}{L} \sin \omega t \Rightarrow I = -\frac{V_0}{\omega L} \cos \omega t$$

$I = I_0 \sin(\omega t - \frac{\pi}{2})$, with $I_0 = \frac{V_0}{\omega L}$.

In an inductor the **current lags the voltage by π/2** (a quarter of a cycle): the voltage reaches its peak a quarter-cycle before the current. (Equivalently, V leads I by π/2.)

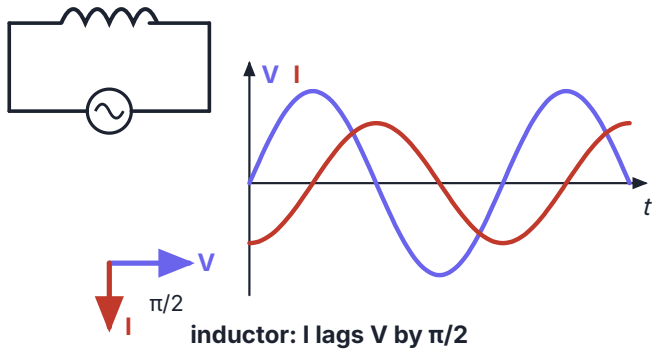


Figure 18.5: a pure inductor: circuit, phasor diagram and graphs.

Inductive reactance, X_L

the **opposition of an inductor to alternating current**: the ratio of the r.m.s. (or peak) voltage across it to the r.m.s. (or peak) current through it.

$$X_L = \frac{V_0}{I_0} = \omega L = 2\pi f L$$

unit: ohm. $X_L \propto f$: an inductor passes d.c. (f = 0) freely but opposes high frequencies.

Power in a pure inductor

$P = IV = I_0(-\cos \omega t) \times V_0 \sin \omega t = -\frac{1}{2} I_0 V_0 \sin 2\omega t$

This is positive for half the time (energy stored in the magnetic field) and negative for the other half (energy returned to the supply). Its mean over a cycle is zero.

$$\langle P \rangle = 0$$

a pure inductor (or capacitor) uses no energy on average

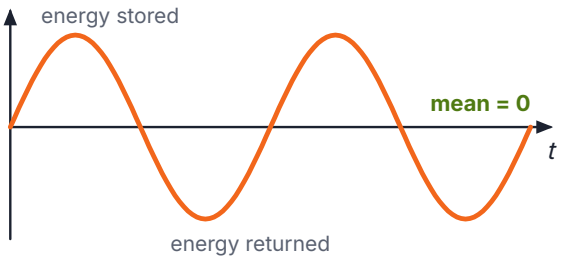


Figure 18.6: power in a pure inductor or capacitor: energy is stored and returned every half-cycle.

Example 18.2.1

A coil of inductance 0.50 H and negligible resistance is connected to a 120 V, 60 Hz supply. Find the r.m.s. current.

Solution

$X_L = 2\pi f L = 2\pi(60)(0.50) = 188\ \Omega$

$I_{rms} = \dfrac{120}{188} = \mathbf{0.64\ A}$

Example 18.2.2

A 240 V, 50 Hz supply drives a current of 3.0 A through a pure inductor. Find its inductance.

Solution

$X_L = \dfrac{240}{3.0} = 80\ \Omega$

$L = \dfrac{X_L}{2\pi f} = \dfrac{80}{2\pi(50)} = \mathbf{0.25\ H}$

18.3 Alternating current through a capacitor

A **pure (ideal) capacitor** has capacitance but no resistance or inductance.

Deriving the current

With $V = V_0 \sin \omega t$ across it, the charge on the capacitor is $Q = CV = CV_0 \sin \omega t$.

$$I = \frac{dQ}{dt} = \omega C V_0 \cos \omega t$$

$$I = I_0 \sin(\omega t + \frac{\pi}{2}), \text{ with } I_0 = \omega C V_0.$$

In a capacitor the **current leads the voltage by π/2**: the current is largest when the capacitor is uncharged (V = 0) and zero when it is fully charged.

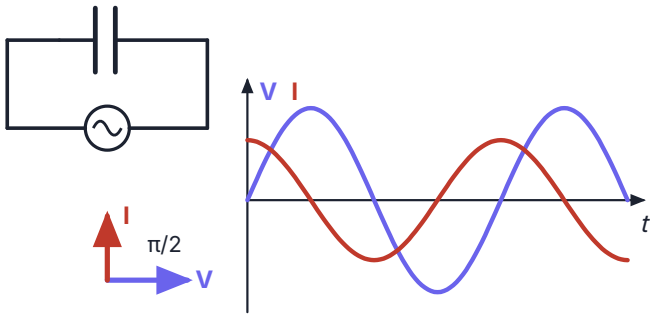


Figure 18.7: a pure capacitor: circuit, phasor diagram and graphs.

Teacher's note

A memory aid: **CIVIL**. In a **C**apacitor, **I** leads **V**; **V** leads **I** in an inductor (**L**).

Capacitive reactance, X_C

the **opposition of a capacitor to alternating current**: the ratio of the r.m.s. (or peak) voltage across it to the r.m.s. (or peak) current.

$$X_C = \frac{V_0}{I_0} = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

unit: ohm. $X_C \propto 1/f$: a capacitor blocks d.c. but passes high frequencies easily.

Power in a pure capacitor

$P = IV = I_0 \cos \omega t \times V_0 \sin \omega t = \tfrac{1}{2} I_0 V_0 \sin 2\omega t$.

Energy is stored in the electric field while the capacitor charges and returned while it discharges: $\langle P \rangle = 0$ (Figure 18.6).

Example 18.3.1

An 8.00 μF capacitor is connected to a 150 V (r.m.s.), 60.0 Hz supply. Find the capacitive reactance, the r.m.s. current and the peak current.

Solution

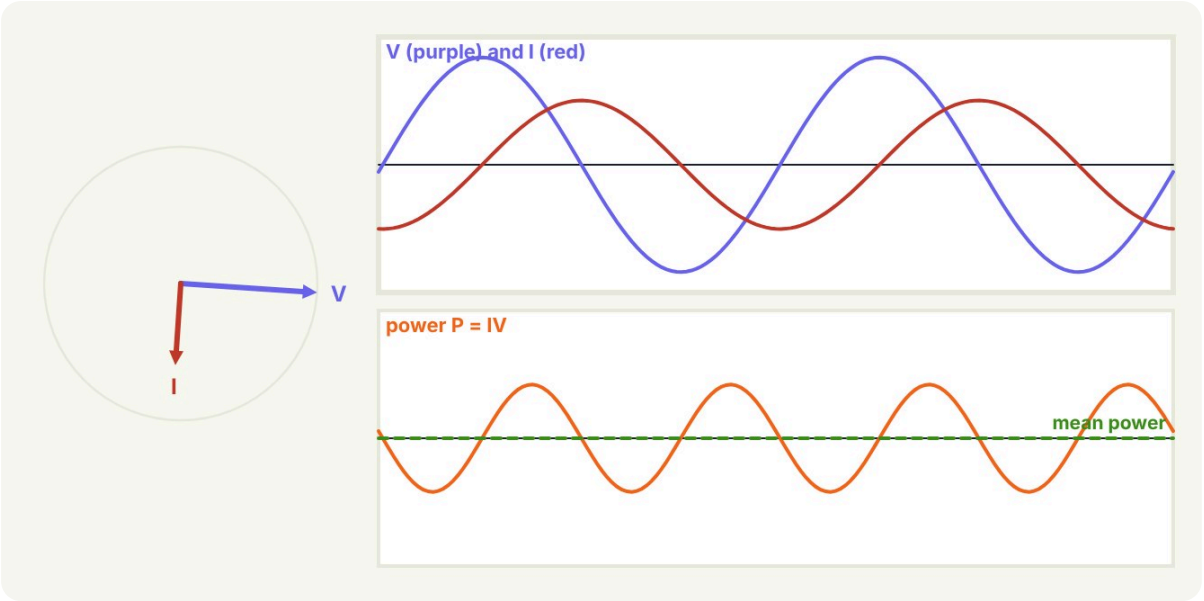
$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(60.0)(8.00 \times 10^{-6})} = \mathbf{332\ \Omega}$$

$$I_{rms} = \frac{150}{332} = \mathbf{0.452\ A};\ I_0 = \sqrt{2}\,(0.452) = \mathbf{0.639\ A}$$

Simulation

R, L or C on an a.c. supply

Choose a resistor, an inductor or a capacitor. The rotating phasors and the graphs show the phase of the current against the voltage; the power curve shows why only the resistor uses energy on average. Change the frequency and watch the reactance.



phase of I relative to V	reactance X_L = 2πfL	peak current (V_o = 10 V)	average power
I lags V by π/2	**100.5 Ω**	**99.5 mA**	**0 W**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 18 to use the simulation.

Exercise: single elements

1. At what frequency does a 2.0 μF capacitor have a reactance of 400 Ω?

   Answer

   $f = 1/(2\pi X_C C)$ = **199 Hz**

2. The frequency of the supply to a pure inductor is doubled while the r.m.s. voltage stays the same. What happens to the r.m.s. current?

   Answer

   $X_L = 2\pi f L$ **doubles, so the current is halved.**

3. Explain why the average power taken by a pure capacitor is zero.

Answer

In one quarter-cycle the capacitor charges and stores energy taken from the supply; in the next quarter-cycle it discharges and returns the same energy. Over a whole cycle the net energy taken is zero.

18.4 R-C, R-L and R-L-C circuits in series

In a series circuit the **same current** flows through every component, so the current is used as the reference phasor. The voltages across the components have different phases, so they add as **phasors** (vectors), not as numbers.

> **Impedance, Z**
>
> the **total opposition of a circuit to alternating current**: $Z = V_{rms}/I_{rms} = V_0/I_0$. It combines resistance and reactance; unit: ohm.

R-C series circuit



R-C: V lags I by φ **impedance triangle**

Figure 18.8: R-C series: V_R is in phase with I, V_C lags I by π/2.

$$V^2 = V_R^2 + V_C^2 = (IR)^2 + (IX_C)^2$$

$$Z = \frac{V}{I} = \sqrt{R^2+X_C^2}, \quad \tan\varphi = \frac{X_C}{R} = \frac{1}{\omega CR}$$, the current **leads** the voltage by φ.

R-L series circuit



Figure 18.9: R-L series: V_L leads I by π/2.

$V^2 = V_R^2 + V_L^2$

$Z = \sqrt{R^2 + X_L^2}$, $\tan\varphi = \frac{X_L}{R} = \frac{\omega L}{R}$, the voltage **leads** the current by φ.



Figure 18.10: how the impedance of R-C and R-L circuits changes with frequency.

R-L-C series circuit

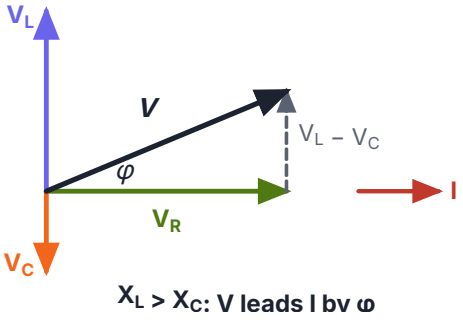


Figure 18.11: R-L-C series: V_L and V_C point in opposite directions, so they partly cancel.

V_L leads I by π/2 and V_C lags I by π/2, so they are in **opposite** directions on the phasor diagram. Their resultant is $(V_L - V_C)$.

$V^2 = V_R^2 + (V_L - V_C)^2 = I^2[R^2 + (X_L - X_C)^2]$

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \qquad \tan\varphi = \frac{X_L - X_C}{R}$$

$X_L > X_C$: V leads I (inductive). $X_L < X_C$: V lags I (capacitive).

Teacher's note

The r.m.s. voltages across the components do **not** add up to the supply voltage. Only the phasor sum does. A voltmeter across the inductor alone can read more than the supply!

Example 18.4.1

A 10 kΩ resistor and a 0.10 μF capacitor are in series with an a.c. supply of angular frequency 1.0×10^4 rad s^{-1}. The r.m.s. voltage across the resistor is 20 V. Find the r.m.s. voltage across the capacitor.

Solution

$$X_C = \frac{1}{\omega C} = \frac{1}{(1.0 \times 10^4)(0.10 \times 10^{-6})} = 1.0\ \text{k}\Omega$$

$$I = \frac{20}{10 \times 10^3} = 2.0\ \text{mA}, \text{ so } V_C = IX_C = \mathbf{2.0\ V}$$

Example 18.4.2

A series R-L-C circuit has $R = 25.0\ \Omega$, $C = 50.0\ \mu\text{F}$ and $L = 0.300$ H, and is driven by a 120 V, 60 Hz supply. Find (a) the impedance, (b) the r.m.s. current, (c) the phase angle between the voltage and the current.

Solution

$$X_L = 2\pi(60)(0.300) = 113.1\ \Omega;\ X_C = \frac{1}{2\pi(60)(50.0 \times 10^{-6})} = 53.1\ \Omega$$

(a) $Z = \sqrt{25.0^2 + (113.1 - 53.1)^2} = \mathbf{65.0\ \Omega}$

(b) $I_{rms} = \frac{120}{65.0} = \mathbf{1.85\ A}$

(c) $\tan\varphi = \frac{60.0}{25.0}$, $\varphi = \mathbf{67.4°}$; $X_L > X_C$, so the voltage leads the current.

Resonance in an R-L-C circuit

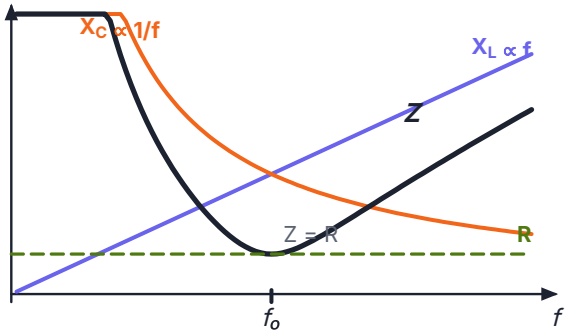


Figure 18.12: X_L, X_C, R and Z against frequency. Z is smallest, and equal to R, at f_o.

- At low frequency X_C is large; at high frequency X_L is large. In between there is a frequency where they are equal.
- When $X_L = X_C$, V_L and V_C cancel exactly: $Z = R$ (its **minimum**), the current is a **maximum** ($I = V/R$) and it is **in phase** with the supply voltage ($\varphi = 0$).

> **Resonance**
> in a series R-L-C circuit occurs when the **frequency of the supply makes the inductive reactance equal to the capacitive reactance**, so the impedance is a minimum (equal to R) and the current is a maximum.

Deriving the resonance frequency

Condition: $X_L = X_C \Rightarrow 2\pi f_0 L = \dfrac{1}{2\pi f_0 C}$

$f_0^2 = \dfrac{1}{4\pi^2 LC}$:

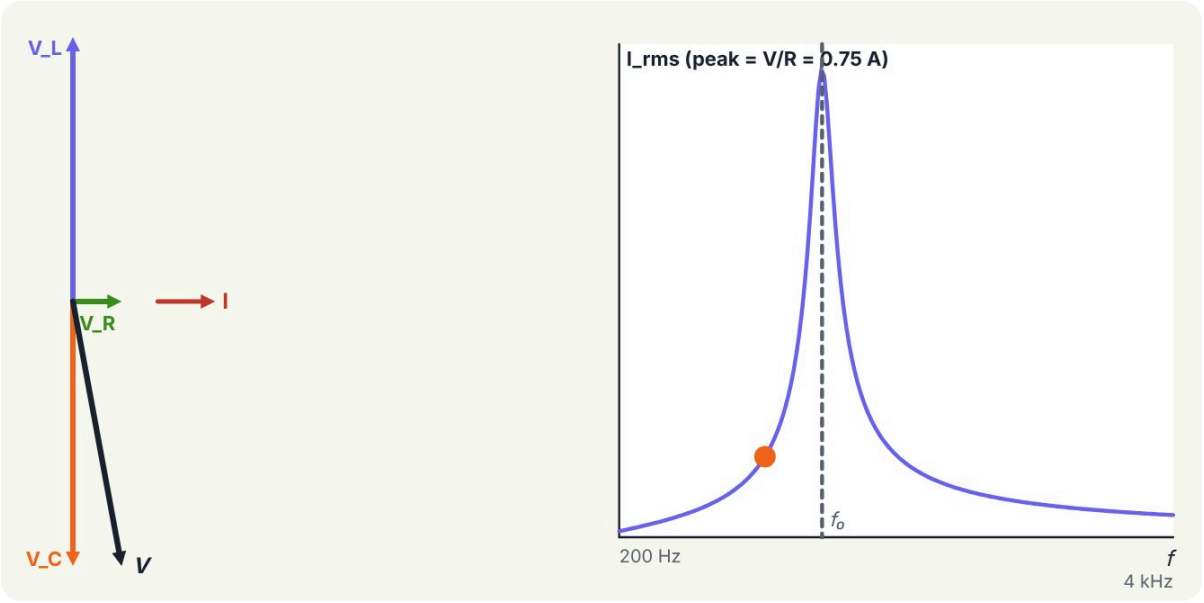
$$f_0 = \frac{1}{2\pi\sqrt{LC}} \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$

independent of R. A smaller R gives a taller, sharper current peak.

- Tuning a radio: the capacitance is changed until f_0 equals the frequency of the wanted station, so that station gives a large current.

Simulation Series R-L-C circuit and resonance

Choose R-C, R-L or R-L-C and change the frequency. The phasor diagram shows V_R, V_L, V_C and the supply V; the graph shows the current against frequency. Find the resonance: V_L and V_C cancel and the current peaks.



impedance Z	current I_rms	phase angle φ	X_L, X_C	V_R, V_L, V_C
1162 Ω	**0.129 A**	**80.1° (V lags I)**	**1508 Ω, 2653 Ω**	**26, 195, 342 V**

$f_o = 1/(2\pi\sqrt{LC})$ **1592 Hz**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 18 to use the simulation.

Example 18.4.3

A 50 Ω resistor, a 200 mH inductor and a 0.050 μF capacitor are in series with a 150 V (r.m.s.) supply of variable frequency. Find (a) the resonance frequency, (b) the r.m.s. voltage across each component at resonance, (c) the voltage across the inductor and capacitor together, (d) the equation of the supply voltage at resonance.

Solution

(a) $f_0 = \dfrac{1}{2\pi\sqrt{(0.200)(0.050 \times 10^{-6})}}$ = **1.59 kHz** ($\omega_0 = 1.0 \times 10^4$ rad s^{-1})

(b) $I = \dfrac{150}{50}$ = 3.0 A. $X_L = \omega_0 L$ = 2000 Ω = X_C. V_R = **150 V**, $V_L = V_C$ = (3.0)(2000) = **6.0 kV**

(c) V_L and V_C are equal and opposite: **0 V**

(d) $V_0 = \sqrt{2}\,(150)$ = 212 V: V **= 212 sin(1.0 × 10^4 t)** V

Example 18.4.4

A 200 Ω resistor, a 0.75 H inductor and a capacitor C are in series with a 250 V supply. Resonance occurs at 600 Hz. Find (a) X_L and X_C at resonance, (b) C, (c) the impedance at resonance, (d) the current at resonance.

Solution

(a) $X_L = 2\pi(600)(0.75)$ = **2.83 kΩ** = X_C

(b) $C = \dfrac{1}{2\pi f X_C} = \dfrac{1}{2\pi(600)(2827)}$ = **93.8 nF**

(c) $Z = R$ = **200 Ω**

(d) $I = \dfrac{250}{200}$ = **1.25 A**. Phasor diagram: V_R along I, V_L and V_C equal and opposite.

Exercise: series circuits

1. A 40 Ω resistor and a 0.10 H inductor are in series with a 100 V, 50 Hz supply. Find the current and the phase angle.

   Answer

   X_L = 31.4 Ω, Z = 50.9 Ω, I = **1.97 A**, φ = $\tan^{-1}(31.4/40)$ = **38.1° (V leads I)**

2. What capacitance must be put in series with a 25 mH inductor so that the circuit resonates at 1.0 kHz?

   Answer

   $C = 1/(4\pi^2 f_0^2 L)$ = **1.0 μF**

3. Sketch the phasor diagram for a series R-L-C circuit driven below its resonance frequency and state whether the current leads or lags the voltage.

   Answer

   Below f_o, $X_C > X_L$, so V_C is longer than V_L; the resultant V lies below the current phasor. The current leads the voltage (the circuit is capacitive).